The Consciousness Series

From Source Formation to Geometry, Gravity, and the Structural Horizon of Self-Description

Complete Collected Volume

Foundation Trilogy · Consciousness Trilogy

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Abstract

This volume collects the six papers comprising the *Consciousness Series*—a programme that investigates how source-side open-system structure, manifest carriers, geometry, gravitational consistency, and the structural limits of self-description fit into a single operator-algebraic chain.

The programme is organised in two parts:

1. **Part I: Foundation Trilogy** (Foundations I–III). The Foundation papers develop the source-side scaffolding of the programme. They argue that: (a) a single Type III_1 source datum can, under named conditional inputs, reach an admissible manifest carrier; (b) multiple source chains can admit a conditional shared manifest sector without any tensor-product assumption; and (c) a manifest calibration signal can be lifted back to a conditional one-step source-level feedback loop by exact semigroup composition.

2. **Part II: Consciousness Trilogy** (Papers I–III). The Consciousness papers analyze what follows once such an admissible manifest sector is available. They argue that: (a) geometry and modular flow are structurally inseparable in the stated Type III_1 regime; (b) higher-dimensional consistency of the modular datum yields a conditional leading-order relation in the CHM small-ball regime; and (c) the former local minimax drift-floor theorem for finite self-description is withdrawn, leaving the explanatory gap open.

Taken together, the two parts establish a six-paper programme: source formation, shared stabilization, one-step feedback, geometry–modular inseparability, local gravitational consistency analysis, and an open finite self-description horizon.

Keywords: open quantum systems, quantum dynamical semigroups, von Neumann algebras, Type III_1 factors, modular theory, same-root structure, geometry–modular inseparability, linearized gravity, self-description

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Part I

Foundation Trilogy: Source Formation, Shared Manifestation, and Feedback

Foundation I: From Source to Manifestation
A Conditional Same-Root Program for Geometry and Modular Flow

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Abstract

Paper I of the Consciousness Trilogy [1] begins from an admissible accessible algebra $(\mathcal{A}_c, \omega)$ and derives the inseparability of geometry and modular flow. The deeper source-side question is how one reaches such an object from the pre-observer substrate developed across Genesis, especially the HAFF and T-DOME papers. The present paper, Foundation I of the Consciousness Series, does not claim a completed unified-source theorem. Instead, it formulates an honest source-side foundation program and pushes its currently closed parts into paper form.

We first state the correct conditional target: if a persistent Nakajima–Zwanzig projection exists in the source datum, and if the foreground-to-algebra transfer can be carried out, then the chain

$$S_{\text{source}} \to \mathcal{K}_* \to \mathfrak{F} \to (\mathcal{A}_c, \omega) \to (\text{Geom}, \text{Mod})$$

is forced. Within that program, we prove the packaging result that budget violation in the persistent record process forces symmetry-broken foreground compression, thereby rewriting the T-DOME I–II chain as a clean Foundation I proposition. We then prove a first restricted PS2 closure in a product-classical, decoherence-dominated pointer regime: pointer coarse-graining yields fragment-local omission losses, semigroup decay yields a uniformly bounded coherence remainder, and a weight gap larger than the resulting remainder forces exact top-k^* pointer selection. In a first finite-dimensional faithful pointer-support regime, the required coherence constants become explicit in Hilbert–Schmidt norm. We then define a pointer-aligned realization of the foreground frame inside the ambient algebra, construct the associated pointer-generated von Neumann algebra, and replace the earlier open-ended transfer candidate by a minimal modular-decoherence stable hull $\Phi^{\text{st}}_{\mathfrak{F}}(\mathfrak{F})$. This stable-hull transfer is well-defined, yields a von Neumann subalgebra, and carries a faithful restricted state.

Finally, under the exact fixed-point form of the HAFF accessibility condition, we prove the inclusion-level comparison

$$\Phi^{\text{st}}_{\mathfrak{F}}(\mathfrak{F}) \subseteq \mathcal{A}_c^{\text{HAFF}},$$

while leaving equality as an open strong-form problem. The upshot is deliberately calibrated: Genesis already supplies substantial antecedent support for Foundation I, but the

completed deductive source-side foundation still depends on two genuinely open interfaces, namely persistent projection selection and the model-independent derivation of PS2 beyond the restricted product-classical pointer regime treated here.

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1 Introduction

Paper I of the Consciousness Trilogy [1] starts at the manifest layer: one assumes an admissible accessible algebra $(\mathcal{A}_c, \omega)$, proves that geometry and modular flow are not independently deformable, and then studies the consequences of that inseparability. That was the correct place for the first trilogy paper to begin, but it leaves a source-foundation question unresolved. What, in the broader Genesis programme, plays the role of the source-side antecedent to $(\mathcal{A}_c, \omega)$?

The present paper, Foundation I of the Consciousness Series, addresses exactly that question. It does not claim that Genesis already contains a completed theorem deriving the trilogy from a single source object. What it does claim is narrower and more precise: the relevant source-side foundation can now be decomposed cleanly, its nearly closed parts can be written as formal propositions, and its actual bottlenecks can be localized without conflating vision and theorem.

The philosophical motivation is correspondingly limited. Paper I showed that geometry and modular flow are not independently deformable once one is already at the manifest pair $(\mathcal{A}_c, \omega)$. Foundation I asks the prior structural question: why should those two readings belong together at all? In the present paper, the same-root idea means only that one seeks a source-side chain from which geometric and modular aspects later arise together. This paper treats that mathematical bridge from source data to accessible algebra; it does not attempt to settle consciousness ontology or the broader philosophy of mind.

The correct three-layer structure is

$$S_{\text{source}} \longrightarrow (\mathcal{A}_c, \omega) \longrightarrow (\text{Geom}, \text{Mod}),$$

with an important internal refinement:

$$S_{\text{source}} \longrightarrow \mathcal{O}_A \longrightarrow \mathcal{O}_B \longrightarrow (\mathcal{A}_c, \omega).$$

Here S_{source} is the pre-observer source datum, $\mathcal{O}_A$ is a persistent projected record process with non-Markovian memory, and $\mathcal{O}_B$ is the symmetry-broken foreground frame produced when full-rank persistence exceeds the available processing budget.

Four claims structure the paper.

1. We state the honest *conditional* Foundation I target. The two genuine open points are persistent projection selection and the model-independent foreground-alignment step PS2 that makes the transfer construction canonical; the existence of an exact fixed-point pointer sector is treated only as an exact-regime PS1 idealization.

2. We isolate a finite-coordinate coding regime in which a binding rank budget and an irreducible symmetry action yield a symmetry-breaking foreground choice. This is a conditional model result, not an import of a universal T-DOME necessity theorem.

3. We prove a first restricted PS2 closure package: in a product-classical, decoherence-dominated pointer regime, pointer coarse-graining yields fragment-local omission losses, semigroup decay yields a uniformly bounded coherence remainder, and a weight gap larger than the resulting remainder forces exact top-k^* pointer selection.

4. We give the paper's main new operator-algebraic construction: a stable-hull realization of the foreground sector. Concretely, we construct a minimal modular-decoherence stable hull, prove that it is a von Neumann subalgebra with faithful restricted state, and prove the first comparison theorem $\mathcal{A}_{\mathfrak{F}} \subseteq \mathcal{A}_{\text{fix}}$ under an explicit fixed-set algebra-closure condition.

This is enough to fix the current calibration. Genesis provides substantial antecedent support for Foundation I. It does not yet provide a finished deductive source-side foundation. No equality between the foreground-generated exact fixed sector, probe-accessible effects, and the environmental record algebra is claimed here.

The paper is organized as follows. Section 2 defines the source datum and the stage outputs. Section 3 states the target conditional theorem and isolates the source-selection problem. Section 4 proves the conditional finite-coordinate compression proposition. Section 5 defines pointer alignment, foreground realization, and the PS2 target. Section 6 proves a first restricted PS2 closure package in the product-classical, decoherence-dominated pointer regime, and records a bookkeeping lemma fixing the diagonal/off-sector split needed for the later model-independent PS2 programme. Section 7 proves the first inclusion-level comparison with the typed exact fixed sector and states a partial Lemma C. Section 8 records the remaining open problems. Appendix A summarizes the external Genesis inputs used in the paper in self-contained mathematical form, and Appendix B collects the named conditional interfaces that remain open.

2 Source Datum and Object Ledger

Definition 2.1 (Source datum)**.** A *source datum* is a quadruple

$$S_{\text{source}} := (\mathcal{H}_U, \mathcal{A}_U, \Omega, \{\mathcal{E}_t\}_{t\geq 0}),$$

where:

1. $(\mathcal{H}_U, \mathcal{A}_U, \Omega)$ is a Type III_1 von Neumann algebra in standard form, with Ω cyclic and separating for $\mathcal{A}_U$;

2. $\{\mathcal{E}_t\}_{t\geq 0}$ is a pointwise σ-weakly continuous semigroup of normal completely positive unital maps on $\mathcal{A}_U$.

Remark 2.2 (Heisenberg-picture continuity convention). The semigroup $\{\mathcal{E}_t\}$ acts on observables, so the continuity assumption in Definition 2 is pointwise σ-weak continuity on $\mathcal{A}_U$, not norm continuity on the ambient algebra. This is the natural continuity regime for the Heisenberg-picture QDS language used throughout the paper.

The ambient standard-form side is the HAFF input [3]; the projected open-system side is expressed in Nakajima–Zwanzig language [8, 11, 7].

Definition 2.3 (Stage outputs)**.** Given a source datum S_{source}, we use the following ledger.

1. **Stage A output**
$$\mathcal{O}_A := (\mathcal{P}, \omega_{\mathcal{P},t}, \mathcal{K}_*(t,s)),$$
where $\mathcal{P} : \mathcal{A}_U \to \mathcal{A}_U$ is a Nakajima–Zwanzig projection, $\omega_{\mathcal{P},t} := \mathcal{P}_*(\omega_t)$ is the projected normal state on the predual, and $\mathcal{K}_*(t,s)$ is the dual memory kernel.

2. **Stage B output**
$$\mathcal{O}_B := (\mathcal{P}, \omega_{\mathcal{P},t}, \mathcal{K}_*(t,s), \mathfrak{F}, \pi_{\text{ego}}),$$
where $\mathfrak{F}$ is a k^*-dimensional foreground frame and π_{ego} is the ego prior carried forward from the broken phase of T-DOME II and into the self-calibration analysis of T-DOME III [5, 6].

3. **Stage C output**

$$\mathcal{O}_C := (\mathcal{A}_c, \omega), \qquad \omega = \langle \Omega, \cdot\, \Omega \rangle|_{\mathcal{A}_c}.$$

Remark 2.4 (Stage D). Once an admissible $(\mathcal{A}_c, \omega)$ is available, the Paper I chain from accessible algebra to jointly constrained geometry and modular flow is already in place [1]. The present note is therefore concerned only with Stages A–C.

3 Target Conditional Theorem and the Source-Selection Problem

The honest Foundation I target is conditional and should be stated that way from the start.

Target conditional theorem. Let

$$S_{\text{source}} = (\mathcal{H}_U, \mathcal{A}_U, \Omega, \{\mathcal{E}_t\}_{t \geq 0})$$

be a source datum. Assume:

1. there exists a persistent Nakajima–Zwanzig projection $\mathcal{P}$ whose projected predual dynamics carries a non-Markovian kernel $\mathcal{K}_*(t, s)$;
2. a specified finite-coordinate coding model has a binding rank budget and satisfies the optimization and group-action hypotheses of Proposition 4.1, producing a compressed foreground frame $\mathfrak{F}$;
3. there exists a foreground embedding

$$\iota_{\mathfrak{F}} : V_{\text{fg}}(\mathfrak{F}) \hookrightarrow \mathcal{A}_U^{\text{sa}}$$

   from which the transfer construction can be carried out and compared, at the inclusion level, with the separately typed exact fixed sector;
4. the resulting $(\mathcal{A}_c, \omega)$ satisfies the admissibility hypotheses needed by Paper I.

Then the chain

$$S_{\text{source}} \to \mathcal{O}_A \to \mathcal{O}_B \to (\mathcal{A}_c, \omega) \to (\text{Geom}, \text{Mod})$$

is defined conditional on assumptions (1)–(4). No arrow in this display is asserted to follow from the bare source datum alone.

Remark 3.1 (What is actually open). The conditional theorem has several independent burdens. They include existence and selection of a persistent projection $\mathcal{P}$, the model-specific coding and frame-selection package in Proposition 4.1, and the transfer from the foreground frame to the operator-algebraic stable hull. Paper I shell generation is a further interface input rather than a Foundation I consequence. These burdens must not be compressed into a single claimed necessity chain.

Lemma 3.2 (Conditional source-to-kernel interface)**.** *Let S_{source} be a source datum. Suppose there exists a Nakajima–Zwanzig projection $\mathcal{P}$ such that the projected predual dynamics admits a non-Markovian kernel $\mathcal{K}_*(t, s)$ and yields a persistent record process. Then*

$$\mathcal{O}_A = (\mathcal{P}, \omega_{\mathcal{P},t}, \mathcal{K}_*(t, s))$$

satisfies the input conditions of Stage B.

Proof. This is the conditional form of the source-side interface. The assumed projection already provides the Stage A output. Within the T-DOME I modeling framework [4], the persistence assumption is analyzed in a non-Markovian Nakajima–Zwanzig regime; the present paper imports that framing rather than claiming it as a general open-quantum-systems theorem. The point is not to solve the selection problem constructively, but to separate that problem from the downstream thermodynamic and algebraic consequences. □

Remark 3.3 (Constructive version remains open). Lemma 3.2 is intentionally conditional. The missing constructive question is why a given source datum selects a persistent projection at all. Nothing in the present note should be read as claiming that this observer-emergence problem has been solved.

4 Conditional Foreground Compression

We now isolate a finite-coordinate model in which the foreground choice is a valid consequence of explicit coding assumptions. The revised T-DOME results do not establish that memory is necessary for persistence, that every processing budget is saturated, or that every compression problem must break a symmetry. The proposition below therefore states the extra assumptions directly.

Proposition 4.1 (Conditional finite-coordinate foreground compression)**.** *Let*

$$\mathcal{O}_A = (\mathcal{P}, \omega_{\mathcal{P},t}, \mathcal{K}_*(t,s))$$

be a persistent record process arising from a source datum S_{source}*, carrying a specified real coordinate space* V *of dimension* D*. Let* $h_\mu > 0$ *be the stipulated processing rate per retained coordinate. Let admissible encoders be orthogonal projections* Q *on* V*, with processing cost* $h_\mu \operatorname{rank} Q$*, and assume:*

1. *the full-rank encoder is infeasible,*

$$h_\mu D > \dot{\mathcal{C}}_{\text{budget}};$$

2. *the survival-distortion functional attains its minimum on the feasible projections and is strictly improved whenever a feasible projection can be enlarged by one retained coordinate;*

3. *a group* G *acts irreducibly and orthogonally on* V*, the survival-distortion functional is* G*-invariant, and* $0 < \lfloor \dot{\mathcal{C}}_{\text{budget}}/h_\mu \rfloor < D$*.*

Then every minimizing encoder has rank

$$k^* = \left\lfloor \frac{\dot{\mathcal{C}}_{\text{budget}}}{h_\mu} \right\rfloor < D.$$

Its range defines a k^**-component foreground frame* $\mathfrak{F}$*. No such proper nonzero projection is* G*-invariant, so choosing a minimizer selects a frame and breaks the stipulated* G*-symmetry.*

Proof. Feasibility gives $\operatorname{rank} Q \leq \lfloor \dot{\mathcal{C}}_{\text{budget}}/h_\mu \rfloor$. By strict improvement under every feasible rank-one enlargement, no minimizer can have smaller rank. Hence every minimizer has rank

$$k^* = \Big\lfloor \frac{\dot{\mathcal{C}}_{\text{budget}}}{h_\mu} \Big\rfloor.$$

The range of a minimizing projection supplies the foreground subspace. If that projection commuted with the irreducible G-representation, its range and kernel would be invariant subspaces. Irreducibility would then force the projection to be either 0 or 1, contrary to $0 < k^* < D$. Thus each minimizing choice breaks the stipulated symmetry. This proof uses only the stated coding model; T-DOME I–II motivate the variables but supply no universal necessity premise here [4, 5]. □

Remark 4.2 (Status of Proposition 4.1). The proposition is an FD/I model result. Its rank charge, strict improvement, and irreducible group action are explicit inputs. In particular, it does not say that persistence requires memory, that a general optimum saturates a budget, or that SSB is unavoidable outside this encoder class.

5 Foreground Realization inside the Ambient Algebra

Lemma C requires a genuinely new object: a way to realize the foreground frame inside the ambient algebra.

Definition 5.1 (Exact and finite-window pointer alignment)**.** A self-adjoint observable $O \in \mathcal{A}_U^{\text{sa}}$ is called *pointer-aligned* with respect to the decoherence semigroup $\{\mathcal{E}_t\}_{t\geq 0}$ if it is exactly fixed:

$$\mathcal{E}_t(O) = O \qquad \forall\, t \geq 0.$$

For an explicitly approximate finite-window statement, define

$$d_\tau(O) := \sup_{0\leq t\leq \tau} \|\mathcal{E}_t(O) - O\|.$$

We call O (ε, τ)-pointer-aligned when $d_\tau(O) \leq \varepsilon$.

Remark 5.2 (Why the former asymptotic definition was not weaker). For a bounded semigroup, the former condition $\lim_{t\to\infty} \|\mathcal{E}_t(O) - O\| = 0$ already implies exact fixation. Indeed, for fixed $s \geq 0$, boundedness and the semigroup law give

$$\mathcal{E}_s(O) = \mathcal{E}_s\Big(\lim_{t\to\infty} \mathcal{E}_t(O)\Big) = \lim_{t\to\infty} \mathcal{E}_{s+t}(O) = O.$$

The finite-window quantity d_τ is used only when an honestly approximate notion is intended. Exact fixation remains an idealization of approximate einselection [9, 10].

Let

$$V_{\text{fg}}(\mathfrak{F}) = \operatorname{span}\{f_1, \ldots, f_{k^*}\}$$

be an orthonormal foreground basis selected by the T-DOME II rate-distortion optimum. Guided by the pointer basis selected in the broken phase [5], choose self-adjoint ambient observables

$$O_1, \ldots, O_{k^*} \in \mathcal{A}_U^{\text{sa}}$$

and define

$$\iota_{\mathfrak{F}}(f_i) := O_i, \qquad \iota_{\mathfrak{F}}\Big(\sum_{i=1}^{k^*} a_i f_i\Big) := \sum_{i=1}^{k^*} a_i O_i.$$

Write

$$\mathfrak{S}_{\mathfrak{F}} := \mathrm{span}\{1, O_1, \ldots, O_{k^*}\} \subset \mathcal{A}_U$$

for the resulting operator system and

$$\mathcal{A}_{\mathfrak{F}} := W^*(\mathfrak{S}_{\mathfrak{F}})$$

for its von Neumann envelope.

Proposition 5.3 (Pointer realization of the foreground sector)**.** *Let $\mathfrak{F}$ be a T-DOME foreground frame with*

$$V_{\mathrm{fg}}(\mathfrak{F}) = \mathrm{span}\{f_1, \ldots, f_{k^*}\}.$$

Assume there exists a family of self-adjoint ambient observables

$$O_1, \ldots, O_{k^*} \in \mathcal{A}_U^{\mathrm{sa}}$$

such that:

1. *$O_1, \ldots, O_{k^*}$ are linearly independent;*
2. *each O_i is pointer-aligned under $\{\mathcal{E}_t\}$;*
3. *the ordering of their survival relevance matches the ordering selected by the T-DOME foreground optimization.*

Then the linear extension $f_i \mapsto O_i$ defines an injective map

$$\iota_{\mathfrak{F}} : V_{\mathrm{fg}}(\mathfrak{F}) \hookrightarrow \mathcal{A}_U^{\mathrm{sa}}.$$

Moreover,

$$\mathfrak{S}_{\mathfrak{F}} = \mathrm{span}\{1, O_1, \ldots, O_{k^*}\}$$

is a finite-dimensional operator system, and

$$\mathcal{A}_{\mathfrak{F}} = W^*(\mathfrak{S}_{\mathfrak{F}})$$

is a von Neumann subalgebra of $\mathcal{A}_U$.

Proof. Linear independence of $O_1, \ldots, O_{k^*}$ makes the linear extension of $f_i \mapsto O_i$ injective. Since the O_i are self-adjoint, $\mathfrak{S}_{\mathfrak{F}}$ is self-adjoint and contains the identity, hence is an operator system. By definition, $W^*(\mathfrak{S}_{\mathfrak{F}})$ is the von Neumann algebra generated by that operator system inside $\mathcal{A}_U$. Therefore $\mathcal{A}_{\mathfrak{F}}$ is a von Neumann subalgebra of $\mathcal{A}_U$. □

The next refinement is to replace the bare existence assumption on the ambient observables O_i by a more structured compatibility package that points directly back to T-DOME I and II.

The PS1–PS3 package. The structured foreground-realization input naturally separates into three pieces:

(PS1) in the exact dephasing-dominated regime, the decoherence semigroup admits a commuting exact fixed-point pointer sector inside $\mathcal{A}_U$;

(PS2) the T-DOME sufficient-statistic coordinates agree with the coordinates induced by that pointer sector;

(PS3) the T-DOME rate-distortion optimum selects an ordered subset of pointer channels by survival relevance.

In this decomposition, PS1 is an exact-regime idealization of the decoherence-side input (not a consequence of generic decoherence or quantum Darwinism alone, but of the noiseless-sector limit in which pointer projections are exactly fixed rather than approximately einselected), whereas PS2 is the genuine model-independent bridge bottleneck and PS3 is downstream from it.

Proposition 5.4 (Conditional pointer-sector realization)**.** *Assume:*

(PS1) in the exact dephasing-dominated regime, the decoherence semigroup admits a commuting family of nonzero exact fixed-point projections $\{P_1, \dots, P_D\} \subset \mathrm{Fix}(\mathcal{E})$ *generating a classical pointer sector*

$$\mathcal{A}_{\mathrm{ptr}} := W^*(P_1, \dots, P_D) \subseteq \mathcal{A}_U;$$

(PS2) the sufficient-statistic record coordinates used in T-DOME II are the coordinates induced by this pointer sector through the T-DOME I coarse-graining map to the classical sector;

(PS3) the T-DOME II rate-distortion optimum selects an ordered subset $I^* = \{i_1, \dots, i_{k^*}\} \subseteq \{1, \dots, D\}$ *by survival relevance.*

Let

$$V_{\mathrm{fg}}(\mathfrak{F}) = \mathrm{span}\{f_1, \dots, f_{k^*}\}$$

be ordered compatibly with I^**, and define*

$$\iota_{\mathfrak{F}}(f_j) := P_{i_j}.$$

Then:

1. $\iota_{\mathfrak{F}} : V_{\mathrm{fg}}(\mathfrak{F}) \hookrightarrow \mathcal{A}_U^{\mathrm{sa}}$ *is injective;*
2. *the image lies in the exact decoherence fixed-point algebra;*
3. *the induced pointer-generated algebra satisfies*

$$\mathcal{A}_{\mathfrak{F}} \subseteq \mathcal{A}_{\mathrm{ptr}}.$$

Proof. By assumption (1), each P_i is self-adjoint and satisfies $\mathcal{E}_t(P_i) = P_i$ for all $t \geq 0$. Distinct nonzero orthogonal projections are linearly independent, so the assignment $f_j \mapsto P_{i_j}$ extends to an injective linear map $\iota_{\mathfrak{F}}$. Exact fixedness gives the second claim. Since the P_{i_j} lie in the commuting pointer sector $\mathcal{A}_{\mathrm{ptr}}$, the von Neumann algebra generated by $1, P_{i_1}, \dots, P_{i_{k^*}}$ is contained in $\mathcal{A}_{\mathrm{ptr}}$. □

Remark 5.5 (Status of PS1). PS1 is the least problematic part of the package. In decoherence theory and quantum Darwinism, robust classical records are carried by a commuting pointer sector selected by the environment [9, 10]. In the exact fixed-point regime used here, this is precisely the statement that the decoherence semigroup admits a commuting family of exact fixed-point projections generating a classical pointer algebra. HAFF Paper D realizes this exactly in the dephasing-dominated finite-dimensional model, where the diagonal fixed-point algebra is the pointer basis algebra [3]. The real model-independent unresolved content is therefore PS2, with PS3 following once PS2 is established.

The PS2 alignment problem. Proposition 5.4 sharpens the existence assumption on $\iota_{\mathfrak{F}}$, but it does not yet derive the compatibility package from the broken-phase output itself in full generality. Section 6 closes a first restricted version in a product-classical pointer regime. The real unresolved content is therefore the model-independent form of the middle condition: the claim that the T-DOME sufficient-statistic coordinates and the pointer-sector coordinates are the same optimization problem written in two languages. That is the full PS2 problem.

Target lemma (PS2 alignment; open). Let

$$\mathcal{A}_{\text{ptr}} = W^*(P_1, \ldots, P_D)$$

be the pointer sector from Proposition 5.4, and define the pointer-coordinate process

$$p_i(t) := \omega_t(P_i), \qquad i = 1, \ldots, D.$$

For an ordered subset

$$I = \{i_1, \ldots, i_k\} \subseteq \{1, \ldots, D\},$$

write $\mathcal{F}_I$ for the pointer-adapted frame retaining those channels. A natural target theorem is the following.

Assume:

1. the T-DOME sufficient statistics $\{c_i(t)\}_{i=1}^D$ agree, after reindexing, with the pointer-coordinate process $\{p_i(t)\}_{i=1}^D$;
2. there exist nonnegative weights $w_1, \ldots, w_D$ and a constant D_0 such that for every pointer-adapted frame $\mathcal{F}_I$,
$$D(\mathcal{F}_I) = D_0 + \sum_{i \notin I} w_i;$$
3. admissible frames satisfy $|I| = k^* = \lfloor \mathcal{C}_{\text{budget}}/h_\mu \rfloor$.

Then every survival-distortion minimizer retains exactly the k^* channels of largest weight, up to ties. In particular, the T-DOME foreground directions coincide with the top-ranked pointer directions, and the ordered subset I^* of Proposition 5.4 is selected canonically.

Proof strategy. This target should be attacked by an exchange argument. If an admissible subset I omits some channel a and retains some channel b with $w_a > w_b$, then replacing b by a changes the distortion by

$$D(\mathcal{F}_{I\setminus\{b\}\cup\{a\}}) - D(\mathcal{F}_I) = -w_a + w_b < 0,$$

contradicting optimality. Once the distortion has been reduced to such a channelwise omission law, alignment is immediate. The real analytic burden is therefore to derive the weights w_i from the T-DOME survival functional in a way compatible with the pointer-sector coarse-graining.

First realistic route: approximate PS2b in the decoherence-dominated regime. The exact separability law above is likely too strong as a first target in the fully correlated setting. A more plausible theorem program is to prove approximate separability once decoherence has suppressed the cross-channel terms strongly enough that the pointer sector is approximately classical.

Introduce a decoherence-depth parameter τ_{dec}, and let $\varepsilon_{\mathrm{dec}}(\tau_{\mathrm{dec}}) \to 0$ measure the size of the residual cross-channel distortion after coarse-graining beyond that scale. The natural asymptotic target is:

$$D(\mathcal{F}_I) = D_0 + \sum_{i \notin I} w_i + R_I, \qquad |R_I| \leq \varepsilon_{\mathrm{dec}}(\tau_{\mathrm{dec}}).$$

Under this bound, the same exchange argument gives

$$D(\mathcal{F}_{I\setminus\{b\}\cup\{a\}}) - D(\mathcal{F}_I) \leq -(w_a - w_b) + 2\varepsilon_{\mathrm{dec}}(\tau_{\mathrm{dec}}).$$

Therefore any exchange across a weight gap larger than $2\varepsilon_{\mathrm{dec}}(\tau_{\mathrm{dec}})$ strictly improves the objective. In particular, if the threshold gap

$$\Delta_{k^*} := w_{(k^*)} - w_{(k^*+1)}$$

exceeds $2\varepsilon_{\mathrm{dec}}(\tau_{\mathrm{dec}})$, then the top-$k^*$ pointer subset is forced, up to ties above the gap.

This is likely the right first Bridge 1 theorem target. It matches the physics of decoherence, which suppresses off-diagonal couplings rather than eliminating them exactly in full generality. Section 6 closes this target in a first restricted product-classical regime. The real analytic task is thus to prove that the T-DOME survival distortion admits such a uniform remainder estimate in the decoherence-dominated regime.

Analytic source of $\varepsilon_{\mathrm{dec}}$. The remainder estimate cannot come from abstract operator algebra alone. It has to be extracted from the concrete T-DOME I survival identity

$$\beta\,\mathcal{S} = -\Delta I(S{:}E) - \Delta D_{\mathrm{KL}} - \beta\,\Delta\langle H_{\mathrm{ctrl}}\rangle.$$

In the pointer basis, the natural next step is to decompose the record process and the induced state into diagonal and off-diagonal parts,

$$X(t) = X^{\mathrm{diag}}(t) + X^{\mathrm{off}}(t), \qquad \rho(t) = \rho_{\mathrm{diag}}(t) + \rho_{\mathrm{off}}(t),$$

where the diagonal part is generated by the commuting pointer projections and the off-diagonal part carries the residual coherences.

The target analytic lemma is then the following: in the decoherence-dominated regime, the off-diagonal contribution to the survival identity is uniformly bounded by a function $\varepsilon_{\mathrm{dec}}(\tau_{\mathrm{dec}}) \to 0$, so that for every pointer-adapted frame $\mathcal{F}_I$,

$$D(\mathcal{F}_I) = D_0 + \sum_{i \notin I} w_i + R_I, \qquad |R_I| \leq \varepsilon_{\mathrm{dec}}(\tau_{\mathrm{dec}}).$$

The likely proof route is:

1. use the decoherence semigroup to obtain exponential or semigroup-gap suppression of the off-pointer sector;
2. propagate that suppression through the three terms in the survival identity;
3. identify the diagonal leading term as an additive pointer-channel omission loss.

This identifies the analytic content that Section 6 closes in a first restricted regime.

6 A Restricted PS2 Result in the Decoherence-Dominated Pointer Regime

The full PS2 alignment problem remains open. Still, a genuine theorem can already be proved in a restricted regime. The point of the present section is not to claim a model-independent PS2 theorem, but to show that in a product-classical, decoherence-dominated pointer regime, the structural reduction to fragment-local omission losses and the estimate reduction to a uniformly small coherence remainder can both be closed explicitly, after which approximate separability and top-k^* selection follow by a completely explicit argument.

Proposition 6.1 (Restricted approximate PS2b in the product-classical pointer regime)**.** *Fix $k \in \{1, \dots, D-1\}$, and let $\mathcal{F}_I$ range over all pointer-adapted frames with $|I| = k$. Assume the T-DOME survival distortion uses the squared loss*

$$\ell(x) = x^2.$$

Suppose that for each admissible I there exist random variables

$$L_i = L_i(\xi), \qquad i = 1, \dots, D,$$

and a coherence remainder $C_I = C_I(\xi)$ such that

$$\mathcal{S}_{\text{full}}(\xi) - \mathcal{S}_{\mathcal{F}_I}(\xi) = \sum_{i \notin I} L_i(\xi) + C_I(\xi)$$

for almost every environmental realization ξ. Assume further:

1. ***Pairwise diagonal orthogonality:***

$$\mathbb{E}_\xi[L_i L_j] = 0 \qquad \textit{for } i \neq j;$$

2. ***Uniform coherence bound:***

$$\|C_I\|_{L^\infty(\xi)} \leq \eta_{\text{dec}}(\tau_{\text{dec}}) \qquad \textit{for every admissible } I;$$

3. ***Finite diagonal size:***

$$\Lambda_k := \sup_{|I|=k} \sum_{i \notin I} \|L_i\|_{L^2(\xi)} < \infty.$$

Define

$$w_i := \|L_i\|^2_{L^2(\xi)}$$

and

$$\varepsilon_{\text{dec}}(\tau_{\text{dec}}) := 2\Lambda_k\, \eta_{\text{dec}}(\tau_{\text{dec}}) + \eta_{\text{dec}}(\tau_{\text{dec}})^2.$$

Then for every admissible I,

$$D(\mathcal{F}_I) = \sum_{i \notin I} w_i + R_I, \qquad |R_I| \leq \varepsilon_{\text{dec}}(\tau_{\text{dec}}).$$

Consequently, if

$$\Delta_k := w_{(k)} - w_{(k+1)} > 2\varepsilon_{\text{dec}}(\tau_{\text{dec}}),$$

then every minimizer of $D(\mathcal{F}_I)$ retains exactly the top-k pointer channels, up to ties above the gap.

Proof. By definition of the distortion and the squared loss,

$$D(\mathcal{F}_I) = \mathbb{E}_\xi \left[\left(\sum_{i \notin I} L_i + C_I \right)^2 \right].$$

Expanding the square gives

$$D(\mathcal{F}_I) = \sum_{i \notin I} \mathbb{E}_\xi[L_i^2] + 2 \sum_{\substack{i,j \notin I \\ i<j}} \mathbb{E}_\xi[L_i L_j] + 2 \sum_{i \notin I} \mathbb{E}_\xi[L_i C_I] + \mathbb{E}_\xi[C_I^2].$$

The pairwise orthogonality assumption cancels the middle sum, so

$$D(\mathcal{F}_I) = \sum_{i \notin I} w_i + R_I,$$

where

$$R_I := 2 \sum_{i \notin I} \mathbb{E}_\xi[L_i C_I] + \mathbb{E}_\xi[C_I^2].$$

Because the environmental expectation is taken against a probability measure, $\|C_I\|_{L^2(\xi)} \leq \|C_I\|_{L^\infty(\xi)}$. Hence

$$|\mathbb{E}_\xi[L_i C_I]| \leq \|L_i\|_{L^2(\xi)} \|C_I\|_{L^2(\xi)} \leq \|L_i\|_{L^2(\xi)} \eta_{\text{dec}}(\tau_{\text{dec}}),$$

and therefore

$$|R_I| \leq 2\eta_{\text{dec}}(\tau_{\text{dec}}) \sum_{i \notin I} \|L_i\|_{L^2(\xi)} + \eta_{\text{dec}}(\tau_{\text{dec}})^2 \leq \varepsilon_{\text{dec}}(\tau_{\text{dec}}).$$

This proves the approximate separability law.

Now let I be admissible, and suppose $a \notin I$, $b \in I$ with $w_a > w_b$. Writing

$$I' := I \setminus \{b\} \cup \{a\},$$

the additive part changes by

$$\sum_{i \notin I'} w_i - \sum_{i \notin I} w_i = -w_a + w_b.$$

The remainder terms satisfy

$$R_{I'} - R_I \leq |R_{I'}| + |R_I| \leq 2\varepsilon_{\text{dec}}(\tau_{\text{dec}}),$$

so

$$D(\mathcal{F}_{I'}) - D(\mathcal{F}_I) \leq -(w_a - w_b) + 2\varepsilon_{\text{dec}}(\tau_{\text{dec}}).$$

Therefore any exchange across a weight gap strictly larger than $2\varepsilon_{\text{dec}}(\tau_{\text{dec}})$ lowers the distortion. If $\Delta_k > 2\varepsilon_{\text{dec}}(\tau_{\text{dec}})$, no minimizer can omit a channel above the threshold while retaining one below it. Hence every minimizer is exactly the top-k pointer subset, up to ties above the gap. □

Lemma 6.2 (Product-classical coarse-graining yields fragment-local omission losses)**.** *Assume that after applying the pointer coarse-graining map Φ, the diagonal classical sector is described by D local channel records*

$$\xi = (\xi_1, \ldots, \xi_D), \qquad \xi_i = (z_i, e_i),$$

with full diagonal law

$$P^{\text{full}}(\xi) = \prod_{i=1}^{D} P_i(z_i, e_i).$$

Let $\mathcal{F}_I$ be an admissible pointer-adapted frame. Assume its diagonal projected law is obtained by retaining the full local channel law on I and replacing each omitted channel by a fixed reference law:

$$P^I(\xi) = \prod_{i \in I} P_i(z_i, e_i) \prod_{i \notin I} Q_i(z_i, e_i).$$

Assume further:

1. *the diagonal survival identity is represented pointwise by*

$$\beta\, \mathcal{S}^{\text{diag}}(\xi) = -\imath(\xi) - d(\xi) - \beta\, \delta h(\xi),$$

   where $\imath$ is the pointwise mutual-information density, d is the pointwise reference-divergence density, and δh is the pointwise control contribution;

2. *the reference divergence is taken with respect to a product reference law*

$$\tau(e) = \prod_{i=1}^{D} \tau_i(e_i);$$

3. *the diagonal control contribution is additive:*

$$\delta h^{\text{full}}(\xi) = \sum_{i=1}^{D} \delta h_i^{\text{full}}(\xi_i),$$

$$\delta h^{I}(\xi) = \sum_{i \in I} \delta h_i^{\text{full}}(\xi_i) + \sum_{i \notin I} \delta h_i^{\text{ref}}(\xi_i).$$

For each channel i, define the local pointwise mutual-information density

$$\imath_i^{\text{full}}(\xi_i) := \log \frac{P_i(z_i, e_i)}{P_i^Z(z_i)\, P_i^E(e_i)}, \qquad \imath_i^{\text{ref}}(\xi_i) := \log \frac{Q_i(z_i, e_i)}{Q_i^Z(z_i)\, Q_i^E(e_i)},$$

and the local reference-divergence density

$$d_i^{\text{full}}(e_i) := \log \frac{P_i^E(e_i)}{\tau_i(e_i)}, \qquad d_i^{\text{ref}}(e_i) := \log \frac{Q_i^E(e_i)}{\tau_i(e_i)}.$$

Then the diagonal survival deficit is fragment-local:

$$\beta\big(\mathcal{S}_{\text{full}}^{\text{diag}}(\xi) - \mathcal{S}_{\mathcal{F}_I}^{\text{diag}}(\xi)\big) = \sum_{i \notin I} \lambda_i(\xi_i),$$

where

$$\lambda_i(\xi_i) := -\big(\imath_i^{\text{full}}(\xi_i) - \imath_i^{\text{ref}}(\xi_i)\big) - \big(d_i^{\text{full}}(e_i) - d_i^{\text{ref}}(e_i)\big) - \beta\big(\delta h_i^{\text{full}}(\xi_i) - \delta h_i^{\text{ref}}(\xi_i)\big).$$

Consequently, the centered variables

$$L_i(\xi_i) := \lambda_i(\xi_i) - \mathbb{E}_\xi[\lambda_i(\xi_i)]$$

are fragment-local omission losses in the sense needed by Lemma 6.4.

Proof. Because the full diagonal law factorizes,

$$P^{\text{full}}(\xi) = \prod_{i=1}^{D} P_i(z_i, e_i), \quad P_Z^{\text{full}}(z) = \prod_{i=1}^{D} P_i^Z(z_i), \quad P_E^{\text{full}}(e) = \prod_{i=1}^{D} P_i^E(e_i).$$

Therefore the full pointwise mutual-information density is additive:

$$\imath^{\text{full}}(\xi) := \log \frac{P^{\text{full}}(\xi)}{P_Z^{\text{full}}(z)\, P_E^{\text{full}}(e)} = \sum_{i=1}^{D} \imath_i^{\text{full}}(\xi_i).$$

The same argument for the projected law gives

$$\imath^I(\xi) = \sum_{i \in I} \imath_i^{\text{full}}(\xi_i) + \sum_{i \notin I} \imath_i^{\text{ref}}(\xi_i).$$

Since the reference law τ is also a product law,

$$d^{\text{full}}(e) := \log \frac{P_E^{\text{full}}(e)}{\tau(e)} = \sum_{i=1}^{D} d_i^{\text{full}}(e_i),$$

and similarly

$$d^I(e) = \sum_{i \in I} d_i^{\text{full}}(e_i) + \sum_{i \notin I} d_i^{\text{ref}}(e_i).$$

The control contribution is additive by assumption:

$$\delta h^{\text{full}}(\xi) = \sum_{i=1}^{D} \delta h_i^{\text{full}}(\xi_i),$$

$$\delta h^I(\xi) = \sum_{i \in I} \delta h_i^{\text{full}}(\xi_i) + \sum_{i \notin I} \delta h_i^{\text{ref}}(\xi_i).$$

Substituting these three decompositions into the pointwise diagonal survival identity gives

$$\begin{aligned}
&\beta\big(\mathcal{S}_{\text{full}}^{\text{diag}}(\xi) - \mathcal{S}_{\mathcal{F}_I}^{\text{diag}}(\xi)\big) \\
&= -\big(\imath^{\text{full}}(\xi) - \imath^I(\xi)\big) - \big(d^{\text{full}}(e) - d^I(e)\big) - \beta\big(\delta h^{\text{full}}(\xi) - \delta h^I(\xi)\big) \\
&= \sum_{i \notin I} \Big[-\big(\imath_i^{\text{full}}(\xi_i) - \imath_i^{\text{ref}}(\xi_i)\big) - \big(d_i^{\text{full}}(e_i) - d_i^{\text{ref}}(e_i)\big) - \beta\big(\delta h_i^{\text{full}}(\xi_i) - \delta h_i^{\text{ref}}(\xi_i)\big) \Big].
\end{aligned}$$

This is exactly the stated fragment-local decomposition. Centering each λ_i produces the centered fragment-local omission losses L_i. □

Remark 6.3 (What Lemma 6.2 closes). Lemma 6.2 is the first theorem-level closure of the structure line PS2a. It does not derive the product-classical pointer regime from the broken-phase output; it shows that once the pointer coarse-graining reaches that regime, fragment-local omission losses follow automatically by additivity of information densities, reference-divergence densities, and diagonal control terms.

Lemma 6.4 (Orthogonality from fragment-local centered channel losses)**.** *Assume the environmental realization admits a pointer-channel factorization*

$$\xi = (\xi_1, \dots, \xi_D)$$

with independent coordinates. Suppose the diagonal omission losses of Proposition 6.1 admit a centered fragment-local realization

$$L_i(\xi) = \lambda_i(\xi_i) - \mathbb{E}_\xi[\lambda_i(\xi_i)], \qquad i = 1, \dots, D,$$

for measurable functions $\lambda_i(\xi_i) \in L^2(\xi_i)$. Then

$$\mathbb{E}_\xi[L_i L_j] = 0 \qquad \textit{for } i \neq j.$$

In particular, the pairwise orthogonality hypothesis of Proposition 6.1 is automatic under independent pointer channels together with centered fragment-local omission losses.

Proof. Because L_i depends only on ξ_i and L_j depends only on ξ_j, the independence of ξ_i and ξ_j implies

$$\mathbb{E}_\xi[L_i L_j] = \mathbb{E}_\xi[L_i]\, \mathbb{E}_\xi[L_j].$$

The centered form of L_i and L_j gives

$$\mathbb{E}_\xi[L_i] = 0 = \mathbb{E}_\xi[L_j],$$

hence $\mathbb{E}_\xi[L_i L_j] = 0$. □

Remark 6.5 (Why the orthogonality hypothesis is natural in the pointer regime). Lemma 6.4 explains why the first hypothesis of Proposition 6.1 is not an arbitrary add-on. On the T-DOME side, the computational-ceiling analysis already treats the active channels as independent rate carriers. On the HAFF/Quantum Darwinism side, accessible classical records are carried by independent environmental fragments. Once the diagonal omission loss localizes to fragment-wise centered channel contributions after coarse-graining, pairwise orthogonality becomes a probability identity rather than an extra dynamical postulate.

What remains open is exactly the next step: deriving that fragment-local centered realization, together with the uniform coherence bound, from the full T-DOME survival identity in a model-independent way.

Lemma 6.6 (Semigroup-decay coherence estimate)**.** *Let*

$$\mathbb{E}_{\text{ptr}} : \mathcal{A}_U \to \mathcal{A}_{\text{ptr}}$$

be a normal conditional expectation onto the pointer algebra, and write

$$Q_{\text{off}} := \text{id} - \mathbb{E}_{\text{ptr}}.$$

For each admissible pointer-adapted frame $\mathcal{F}_I$ and environmental realization ξ, let X_ξ denote the relevant state or record datum entering the survival identity, and decompose the coherence remainder as

$$C_I(\xi) = C_I^{(I)}(\xi) + C_I^{(D)}(\xi) + C_I^{(H)}(\xi),$$

where the three terms correspond respectively to the off-diagonal contributions of

$$-\Delta I(S{:}E), \qquad -\Delta D_{\text{KL}}, \qquad -\beta\, \Delta\langle H_{\text{ctrl}}\rangle.$$

Assume:

1. ***Off-sector semigroup decay:*** *there exists a non-increasing function* $g : [0, \infty) \to [0, \infty)$ *with* $g(\tau) \to 0$ *such that*

$$\sup_{t \geq \tau} \|Q_{\text{off}} \mathcal{E}_t(X)\| \leq g(\tau) \, \|Q_{\text{off}} X\|$$

for every relevant datum X*;*

2. ***Uniform off-sector size:***

$$\sup_{\xi} \|Q_{\text{off}} X_\xi\| \leq B_{\text{off}} < \infty;$$

3. ***Termwise continuity bounds:*** *there exist constants* $L_I, L_D, L_H \geq 0$ *such that for every admissible* I *and every* ξ*,*

$$|C_I^{(I)}(\xi)| \leq L_I \sup_{t \geq \tau_{\text{dec}}} \|Q_{\text{off}} \mathcal{E}_t(X_\xi)\|,$$

$$|C_I^{(D)}(\xi)| \leq L_D \sup_{t \geq \tau_{\text{dec}}} \|Q_{\text{off}} \mathcal{E}_t(X_\xi)\|,$$

$$|C_I^{(H)}(\xi)| \leq L_H \sup_{t \geq \tau_{\text{dec}}} \|Q_{\text{off}} \mathcal{E}_t(X_\xi)\|.$$

Then for every admissible I*,*

$$\|C_I\|_{L^\infty(\xi)} \leq \eta_{\text{dec}}(\tau_{\text{dec}}) := (L_I + L_D + L_H) \, B_{\text{off}} \, g(\tau_{\text{dec}}).$$

In particular, the uniform coherence bound required in Proposition 6.1 holds with this choice of η_{dec}*.*

Proof. By the triangle inequality,

$$|C_I(\xi)| \leq |C_I^{(I)}(\xi)| + |C_I^{(D)}(\xi)| + |C_I^{(H)}(\xi)|.$$

Applying the termwise continuity bounds gives

$$|C_I(\xi)| \leq (L_I + L_D + L_H) \sup_{t \geq \tau_{\text{dec}}} \|Q_{\text{off}} \mathcal{E}_t(X_\xi)\|.$$

The off-sector semigroup decay assumption then yields

$$|C_I(\xi)| \leq (L_I + L_D + L_H) \, g(\tau_{\text{dec}}) \, \|Q_{\text{off}} X_\xi\|.$$

Using the uniform off-sector bound,

$$|C_I(\xi)| \leq (L_I + L_D + L_H) \, B_{\text{off}} \, g(\tau_{\text{dec}}).$$

Taking the essential supremum over ξ gives the stated L^∞ bound. □

Remark 6.7 (What Lemma 6.6 closes). Lemma 6.6 is the abstract translation step from standard decoherence-semigroup suppression of the off-pointer sector to the exact η_{dec} object used in Proposition 6.1. It does not yet identify the best norm or derive the constants L_I, L_D, L_H from the full T-DOME survival identity; those remain the true analytic content.

In concrete pointer-diagonal control models, one often has $L_H = 0$, so only the mutual-information and relative-entropy terms contribute to the coherence remainder.

Corollary 6.8 (Hilbert–Schmidt continuity package on a uniformly faithful pointer support)**.** *Assume that, after pointer coarse-graining, the relevant joint system–environment states are supported on a finite-dimensional space*

$$\mathcal{H}_S \otimes \mathcal{H}_E$$

with

$$d_S := \dim \mathcal{H}_S, \qquad d_E := \dim \mathcal{H}_E.$$

Assume there exists $m_* > 0$ such that every relevant joint state ρ_{SE}, every relevant marginal state ρ_S, ρ_E, and the fixed reference state τ_E entering the T-DOME ΔD_{KL} term satisfy

$$\lambda_{\min}(\rho_{SE}), \lambda_{\min}(\rho_S), \lambda_{\min}(\rho_E), \lambda_{\min}(\tau_E) \geq m_*.$$

Write $\|\cdot\|_2$ for the Hilbert–Schmidt norm.
Then:

1. *for any two relevant joint states ρ_{SE}, σ_{SE},*

$$\big|I(\rho_{SE}) - I(\sigma_{SE})\big| \leq 3\sqrt{d_S d_E}\,\big(|\log m_*| + 1\big)\,\|\rho_{SE} - \sigma_{SE}\|_2;$$

2. *for any two relevant environment states ρ_E, σ_E,*

$$\big|D_{\mathrm{KL}}(\rho_E\|\tau_E) - D_{\mathrm{KL}}(\sigma_E\|\tau_E)\big| \leq \sqrt{d_E}\,\big(2|\log m_*| + 1\big)\,\|\rho_E - \sigma_E\|_2;$$

3. *for any self-adjoint control Hamiltonian $H_{\mathrm{ctrl}} \in \mathcal{B}(\mathcal{H}_S)$,*

$$\big|\mathrm{Tr}[H_{\mathrm{ctrl}}(\rho_S - \sigma_S)]\big| \leq \|H_{\mathrm{ctrl}}\|_2\,\|\rho_S - \sigma_S\|_2.$$

Consequently, in this restricted finite-dimensional faithful regime, one may take Hilbert–Schmidt continuity constants

$$L_I^{\mathrm{HS}} := 3\sqrt{d_S d_E}\,\big(|\log m_*| + 1\big),$$

$$L_D^{\mathrm{HS}} := \sqrt{d_E}\,\big(2|\log m_*| + 1\big),$$

$$L_H^{\mathrm{HS}} := \|H_{\mathrm{ctrl}}\|_2$$

in Lemma 6.6, after translating the relevant off-sector perturbations into the corresponding Hilbert–Schmidt norms.

Proof. Set

$$S(\rho) := -\,\mathrm{Tr}(\rho \log \rho).$$

On the convex set of density operators with spectrum contained in $[m_*, 1]$, the Fréchet derivative is

$$DS_\rho(H) = -\,\mathrm{Tr}\big(H(\log\rho + \mathbf{1})\big).$$

Hence

$$|DS_\rho(H)| \leq \|H\|_2\,\|\log\rho + \mathbf{1}\|_2 \leq \sqrt{d}\,\big(|\log m_*| + 1\big)\|H\|_2$$

for $d = \dim \mathrm{supp}\,\rho$. Applying the mean-value theorem along the line segment between two states gives

$$|S(\rho) - S(\sigma)| \leq \sqrt{d}\,\big(|\log m_*| + 1\big)\|\rho - \sigma\|_2.$$

For the mutual-information term, use

$$I(\rho_{SE}) = S(\rho_S) + S(\rho_E) - S(\rho_{SE}).$$

The partial trace satisfies the Hilbert–Schmidt bound

$$\|\operatorname{Tr}_E A\|_2 \le \sqrt{d_E}\,\|A\|_2, \qquad \|\operatorname{Tr}_S A\|_2 \le \sqrt{d_S}\,\|A\|_2.$$

Therefore

$$|S(\rho_S) - S(\sigma_S)| \le \sqrt{d_S}\,\big(|\log m_*| + 1\big)\|\rho_S - \sigma_S\|_2 \le \sqrt{d_S d_E}\,\big(|\log m_*| + 1\big)\|\rho_{SE} - \sigma_{SE}\|_2,$$

and similarly for the environment marginal. For the joint entropy term,

$$|S(\rho_{SE}) - S(\sigma_{SE})| \le \sqrt{d_S d_E}\,\big(|\log m_*| + 1\big)\|\rho_{SE} - \sigma_{SE}\|_2.$$

Summing the three contributions proves part (1).

For the relative-entropy term with fixed faithful reference τ_E, write

$$D_{\mathrm{KL}}(\rho_E\|\tau_E) = \operatorname{Tr}(\rho_E \log \rho_E) - \operatorname{Tr}(\rho_E \log \tau_E).$$

Hence

$$\begin{aligned}&\big|D_{\mathrm{KL}}(\rho_E\|\tau_E) - D_{\mathrm{KL}}(\sigma_E\|\tau_E)\big| \\ &\le \big|S(\rho_E) - S(\sigma_E)\big| + \big|\operatorname{Tr}\big((\rho_E - \sigma_E)\log \tau_E\big)\big|.\end{aligned}$$

The entropy term is bounded by

$$\sqrt{d_E}\,\big(|\log m_*| + 1\big)\|\rho_E - \sigma_E\|_2,$$

while

$$\|\log \tau_E\|_2 \le \sqrt{d_E}\,|\log m_*|.$$

Cauchy–Schwarz therefore gives part (2).

Part (3) is immediate from the Hilbert–Schmidt Cauchy–Schwarz inequality. □

Remark 6.9 (Status of Corollary 6.8). Corollary 6.8 does not close the full estimate track in general Type III form. It closes the first finite-dimensional faithful pointer-support regime, where the Hilbert–Schmidt norm is natural and all three continuity constants become explicit. For the restricted Bridge 1 target developed in the present note, that is enough: the estimate line is closed at theorem level. The further question of whether an equally clean package survives beyond that regime, or must be reformulated in a predual norm, belongs to a later strengthening rather than to the current closure claim.

Corollary 6.10 (Restricted PS2 closure in the product-classical faithful pointer regime)**.** *Assume:*

1. *the hypotheses of Lemma 6.2;*
2. *the independent pointer-channel factorization hypothesis of Lemma 6.4;*
3. *the off-sector semigroup decay and uniform off-sector size hypotheses of Lemma 6.6;*
4. *the finite-dimensional uniformly faithful pointer-support hypotheses of Corollary 6.8;*

5. *the centered fragment-local omission losses from Lemma 6.2 satisfy*

$$\Lambda_k := \sup_{|I|=k} \sum_{i \notin I} \|L_i\|_{L^2(\xi)} < \infty.$$

Define

$$w_i := \|L_i\|^2_{L^2(\xi)},$$
$$\eta^{\mathrm{HS}}_{\mathrm{dec}}(\tau_{\mathrm{dec}}) := \left(L^{\mathrm{HS}}_I + L^{\mathrm{HS}}_D + L^{\mathrm{HS}}_H\right) B_{\mathrm{off}}\, g(\tau_{\mathrm{dec}}),$$

and

$$\varepsilon^{\mathrm{HS}}_{\mathrm{dec}}(\tau_{\mathrm{dec}}) := 2\Lambda_k\, \eta^{\mathrm{HS}}_{\mathrm{dec}}(\tau_{\mathrm{dec}}) + \left(\eta^{\mathrm{HS}}_{\mathrm{dec}}(\tau_{\mathrm{dec}})\right)^2.$$

Then every admissible pointer-adapted frame $\mathcal{F}_I$ satisfies

$$D(\mathcal{F}_I) = \sum_{i \notin I} w_i + R_I, \qquad |R_I| \le \varepsilon^{\mathrm{HS}}_{\mathrm{dec}}(\tau_{\mathrm{dec}}).$$

Consequently, if

$$\Delta_k := w_{(k)} - w_{(k+1)} > 2\varepsilon^{\mathrm{HS}}_{\mathrm{dec}}(\tau_{\mathrm{dec}}),$$

then every minimizer of $D(\mathcal{F}_I)$ retains exactly the top-k pointer channels, up to ties above the gap.

Proof. Lemma 6.2 produces centered fragment-local omission losses L_i. Lemma 6.4 upgrades the independent pointer-channel factorization to the pairwise orthogonality required in Proposition 6.1. By Corollary 6.8, the continuity constants appearing in Lemma 6.6 may be chosen as $L^{\mathrm{HS}}_I, L^{\mathrm{HS}}_D, L^{\mathrm{HS}}_H$, so that Lemma 6.6 yields the uniform coherence bound

$$\|C_I\|_{L^\infty(\xi)} \le \eta^{\mathrm{HS}}_{\mathrm{dec}}(\tau_{\mathrm{dec}}).$$

The finiteness hypothesis gives the remaining size bound $\Lambda_k < \infty$. Proposition 6.1 therefore applies and gives exactly the stated approximate separability law and top-k selection criterion. □

Corollary 6.11 (Pure-dephasing envelope)**.** *Assume the hypotheses of Proposition 6.1, and suppose in addition that the coherence remainder is controlled by the omitted pointer-channel coherences in the form*

$$|C_I(\xi, t)| \le \sum_{i \notin I} a_i\, |p_i(t)| \qquad \textit{for all } t \ge \tau_{\mathrm{dec}},$$

where $a_i \ge 0$ are constants and each $p_i(t)$ is the exact Lorentz–Drude pure-dephasing decoherence function from T-DOME I [4, 7],

$$\gamma_i < 4\lambda_i, \qquad \Omega_i := \tfrac{1}{2}\sqrt{4\lambda_i\gamma_i - \gamma_i^2} \in \mathbb{R}_{>0},$$

and

$$p_i(t) = e^{-\gamma_i t/2}\left[\cos(\Omega_i t) + \frac{\gamma_i}{2\Omega_i}\sin(\Omega_i t)\right].$$

Define

$$R_i := \sqrt{1 + \left(\frac{\gamma_i}{2\Omega_i}\right)^2}, \qquad A_k := \max_{|I|=k} \sum_{i \notin I} a_i R_i.$$

Then one may take

$$\eta_{\mathrm{dec}}(\tau_{\mathrm{dec}}) = A_k\, e^{-\gamma_* \tau_{\mathrm{dec}}/2}, \qquad \gamma_* := \min_{1\le i\le D} \gamma_i,$$

and hence

$$\varepsilon_{\mathrm{dec}}(\tau_{\mathrm{dec}}) \le 2\Lambda_k A_k\, e^{-\gamma_* \tau_{\mathrm{dec}}/2} + A_k^2\, e^{-\gamma_* \tau_{\mathrm{dec}}}.$$

In particular, the distortion remainder decays exponentially with the decoherence depth.

Proof. For each channel,

$$|p_i(t)| \le e^{-\gamma_i t/2} \sqrt{1 + \left(\frac{\gamma_i}{2\Omega_i}\right)^2} = R_i\, e^{-\gamma_i t/2} \le R_i\, e^{-\gamma_* t/2}$$

because $|a\cos x + b \sin x| \le \sqrt{a^2+b^2}$. Therefore

$$|C_I(\xi,t)| \le \sum_{i\notin I} a_i R_i\, e^{-\gamma_* t/2} \le A_k\, e^{-\gamma_* \tau_{\mathrm{dec}}/2}$$

for all $t \ge \tau_{\mathrm{dec}}$. This identifies a valid choice of $\eta_{\mathrm{dec}}(\tau_{\mathrm{dec}})$. The stated bound on $\varepsilon_{\mathrm{dec}}(\tau_{\mathrm{dec}})$ is then immediate from Proposition 6.1. □

A bookkeeping split for the model-independent PS2 programme. The restricted results above close the PS2 package only after explicit regime assumptions have been imposed. For the later model-independent programme, one further lemma is useful already now: it fixes the exact place where the off-sector estimates enter the survival deficit.

For $X \in \{\mathrm{full}, \mathcal{F}_I\}$, write the survival identity as

$$\beta\, \mathcal{S}_X = -\Delta I_X - \Delta D_X - \beta\, \Delta H_X,$$

where

$$\Delta D_X := \Delta D_{\mathrm{KL},X}, \qquad \Delta H_X := \Delta \langle H_{\mathrm{ctrl}} \rangle_X.$$

Assume each of the three terms admits a pointer-diagonal/off-sector decomposition,

$$\Delta I_X := \Delta I_X^{\mathrm{diag}} + \Delta I_X^{\mathrm{off}},$$

$$\Delta D_X := \Delta D_X^{\mathrm{diag}} + \Delta D_X^{\mathrm{off}},$$

$$\Delta H_X := \Delta H_X^{\mathrm{diag}} + \Delta H_X^{\mathrm{off}}.$$

Define the pointer-diagonal survival functional and off-sector remainder by

$$\beta\, \mathcal{S}_X^{\mathrm{diag}} := -\Delta I_X^{\mathrm{diag}} - \Delta D_X^{\mathrm{diag}} - \beta\, \Delta H_X^{\mathrm{diag}},$$

$$R_X^{\mathrm{off}} := -\Delta I_X^{\mathrm{off}} - \Delta D_X^{\mathrm{off}} - \beta\, \Delta H_X^{\mathrm{off}}.$$

Lemma 6.12 (Diagonal/off-sector split of the survival deficit)**.** *Let $\mathbb{E}_{\mathrm{ptr}} : \mathcal{A}_U \to \mathcal{A}_{\mathrm{ptr}}$ be the conditional expectation onto the pointer algebra selected by the broken phase, and assume the above termwise decompositions are computed with respect to $\mathbb{E}_{\mathrm{ptr}}$. Then for every admissible pointer-adapted frame $\mathcal{F}_I$,*

$$\beta\big(\mathcal{S}_{\mathrm{full}} - \mathcal{S}_{\mathcal{F}_I}\big) = \beta\big(\mathcal{S}_{\mathrm{full}}^{\mathrm{diag}} - \mathcal{S}_{\mathcal{F}_I}^{\mathrm{diag}}\big) + R_I^{\mathrm{off}},$$

where

$$R_I^{\mathrm{off}} := R_{\mathrm{full}}^{\mathrm{off}} - R_{\mathcal{F}_I}^{\mathrm{off}}.$$

Equivalently,

$$\begin{aligned} R_I^{\mathrm{off}} = &- \big(\Delta I_{\mathrm{full}}^{\mathrm{off}} - \Delta I_{\mathcal{F}_I}^{\mathrm{off}}\big) \\ &- \big(\Delta D_{\mathrm{full}}^{\mathrm{off}} - \Delta D_{\mathcal{F}_I}^{\mathrm{off}}\big) \\ &- \beta\big(\Delta H_{\mathrm{full}}^{\mathrm{off}} - \Delta H_{\mathcal{F}_I}^{\mathrm{off}}\big). \end{aligned}$$

Proof. Write the survival identity separately for the full process and the frame-adapted process $\mathcal{F}_I$, insert the termwise diagonal/off-sector decompositions into both identities, and subtract. The pointer-diagonal contributions group into

$$\beta\big(\mathcal{S}_{\mathrm{full}}^{\mathrm{diag}} - \mathcal{S}_{\mathcal{F}_I}^{\mathrm{diag}}\big),$$

while the off-sector contributions group into

$$R_{\mathrm{full}}^{\mathrm{off}} - R_{\mathcal{F}_I}^{\mathrm{off}}.$$

This is exactly the stated identity. □

Remark 6.13 (What this does and does not close). Section 6 is a restricted theorem-level closure of PS2, not a general one. What is now closed is the implication

$$\begin{aligned} \text{uniformly small coherence remainder} &\Longrightarrow \text{approximate separability} \\ &\Longrightarrow \text{top-}k \text{ pointer selection.} \end{aligned}$$

Corollary 6.10 packages this as a single restricted selection theorem in the product-classical and finite-dimensional faithful pointer regime, while Corollary 6.11 gives a model-specific exponential envelope. Lemma 6.12 does not enlarge that closure claim; it only fixes the bookkeeping split showing that the already-controlled semigroup-decay terms belong to the off-sector remainder. What remains open is the model-independent derivation of the regime assumptions, and in particular control of the diagonal product defect, from the full T-DOME survival identity and the broken-phase output.

A conditional model-independent foreground-background package. The bookkeeping split also makes it possible to state an honest conditional version of the next model-independent target. For budget-feasible I with $|I| = k^*$, define the diagonal-tail survival deficit by

$$\Delta_I^{\mathrm{diag}}(\xi, t) := \beta\big(\mathcal{S}_{\mathrm{full}}^{\mathrm{diag}}(\xi, t) - \mathcal{S}_{\mathcal{F}_I}^{\mathrm{diag}}(\xi, t)\big),$$

the corresponding tail distortion by

$$D_\tau^{\mathrm{diag}}(I) := \sup_{t \geq \tau} \mathbb{E}_{P_t^{\mathrm{diag}}}\Big[\big(\Delta_I^{\mathrm{diag}}(\xi, t)\big)^2\Big],$$

and for an exchange pair

$$\mathfrak{P} := I^* \times (I^*)^{\mathrm{c}}, \qquad N_{\mathrm{ex}} := |\mathfrak{P}| = k^*(D - k^*),$$

define

$$\begin{aligned} G_{a\leftrightarrow b}^{\mathrm{diag}}(\xi, t) &:= \beta\big(\mathcal{S}_{\mathcal{F}_{I_{a\leftrightarrow b}}}^{\mathrm{diag}}(\xi, t) - \mathcal{S}_{\mathcal{F}_{I^*}}^{\mathrm{diag}}(\xi, t)\big), \\ \gamma_{\mathrm{fb}}^{\mathrm{diag}}(\xi, t) &:= \beta\big(\mathcal{S}_{\mathrm{full}}^{\mathrm{diag}}(\xi, t) - \mathcal{S}_{\mathrm{ff}}^{\mathrm{diag}}(u, t) - \mathcal{S}_{\mathrm{bb}}^{\mathrm{diag}}(v, t)\big), \\ \overline{\Gamma}_{\mathrm{fb}}^{\mathrm{diag}}(t) &:= \mathbb{E}_{P_t^{\mathrm{diag}}}\big[\gamma_{\mathrm{fb}}^{\mathrm{diag}}(\xi, t)\big]. \end{aligned}$$

Proposition 6.14 (Conditional diagonal foreground-background coupling suppression)**.** *Let I^* be the broken-phase foreground set. Assume:*

1. ***Approximate tail optimality.*** *There exists $r_{\mathrm{opt}}(\tau) \to 0$ such that*

$$D_\tau^{\mathrm{diag}}(I^*) \le D_\tau^{\mathrm{diag}}(I) + r_{\mathrm{opt}}(\tau) \qquad \textit{for all } |I| = k^*.$$

2. ***Exchange cover and tail persistence.*** *There exist $c_{\mathrm{cov}} > 0$, $M_\gamma < \infty$, $e_{\mathrm{cov}}(\tau) \to 0$, and $r_{\mathrm{pers}}(\tau) \to 0$ such that:*

$$G_{a\leftrightarrow b}^{\mathrm{diag}}(\xi,t)\,\gamma_{\mathrm{fb}}^{\mathrm{diag}}(\xi,t) \ge 0 \qquad \textit{for all } (a,b) \in \mathfrak{P},\ t \ge \tau,$$

$$\frac{1}{N_{\mathrm{ex}}} \sum_{(a,b)\in\mathfrak{P}} \left|G_{a\leftrightarrow b}^{\mathrm{diag}}(\xi,t)\right| \ge c_{\mathrm{cov}} \left|\gamma_{\mathrm{fb}}^{\mathrm{diag}}(\xi,t)\right| - e_{\mathrm{cov}}(\tau) \quad \textit{for a.e. } \xi,\ t \ge \tau,$$

$$\sup_{t\ge\tau} \mathbb{E}\big[|\gamma_{\mathrm{fb}}^{\mathrm{diag}}(\cdot,t)|\big] \le M_\gamma,$$

and for

$$\Gamma_{a\leftrightarrow b}^{\mathrm{ex}}(t) := \mathbb{E}\big[G_{a\leftrightarrow b}^{\mathrm{diag}}(\cdot,t)\,\gamma_{\mathrm{fb}}^{\mathrm{diag}}(\cdot,t)\big]$$

one has

$$\sup_{(a,b)\in\mathfrak{P}} \sup_{s,t\ge\tau} \left|\Gamma_{a\leftrightarrow b}^{\mathrm{ex}}(t) - \Gamma_{a\leftrightarrow b}^{\mathrm{ex}}(s)\right| \le r_{\mathrm{pers}}(\tau).$$

3. ***Tail-passive background survival.*** *There exist $M_{\mathrm{ex}}(\tau) < \infty$ and $\eta_{\mathrm{bb}}(\tau) \to 0$ such that*

$$\sup_{(a,b)\in\mathfrak{P}} \sup_{t\ge\tau} \left\|G_{a\leftrightarrow b}^{\mathrm{diag}}(\cdot,t)\right\|_{L^1} \le M_{\mathrm{ex}}(\tau),$$

$$\sup_{t\ge\tau} \left\|\mathcal{S}_{\mathrm{bb}}^{\mathrm{diag}}(\cdot,t)\right\|_{L^\infty} \le \eta_{\mathrm{bb}}(\tau).$$

4. ***Exchange efficiency gap.*** *There exist $M_G(\tau) < \infty$, $m_\gamma(\tau) > 0$, and a constant $\delta_Q > 0$ such that*

$$\sup_{(a,b)\in\mathfrak{P}} \sup_{t\ge\tau} \left\|G_{a\leftrightarrow b}^{\mathrm{diag}}(\cdot,t)\right\|_{L^\infty} \le M_G(\tau),$$

$$\left|\gamma_{\mathrm{fb}}^{\mathrm{diag}}(\xi,t)\right| \ge m_\gamma(\tau) \quad \textit{whenever } G_{a\leftrightarrow b}^{\mathrm{diag}}(\xi,t) \ne 0,$$

and the efficiency ratio

$$\kappa_Q(\tau) := \frac{M_G(\tau)}{m_\gamma(\tau)}$$

satisfies

$$\kappa_Q(\tau) \le 2 - \delta_Q \qquad \textit{for all sufficiently large } \tau.$$

Then there exists a function $\varepsilon_\Gamma(\tau) \to 0$ such that

$$\sup_{t\ge\tau} \left|\overline{\Gamma}_{\mathrm{fb}}^{\mathrm{diag}}(t)\right| \le \varepsilon_\Gamma(\tau)$$

for all sufficiently large τ. One may take

$$\varepsilon_\Gamma(\tau) := \left[\frac{N_{\mathrm{ex}}}{\delta_Q c_{\mathrm{cov}}} \left(r_{\mathrm{opt}}(\tau) + \frac{2e_{\mathrm{cov}}(\tau)}{N_{\mathrm{ex}}} M_\gamma + 2r_{\mathrm{pers}}(\tau) + 2\beta M_{\mathrm{ex}}(\tau)\eta_{\mathrm{bb}}(\tau)\right)\right]^{1/2}.$$

Proof. The exact exchange expansion gives, for every exchange pair,

$$D_\tau^{\mathrm{diag}}(I_{a\leftrightarrow b}) - D_\tau^{\mathrm{diag}}(I^*) \le Q_{a\leftrightarrow b}^\tau + 2B_{a\leftrightarrow b}^\tau - 2A_{a\leftrightarrow b}^\tau,$$

where

$$Q_{a\leftrightarrow b}^\tau := \sup_{t\ge\tau} \mathbb{E}\Big[\big(G_{a\leftrightarrow b}^{\mathrm{diag}}(\cdot,t)\big)^2\Big],$$

$$B_{a\leftrightarrow b}^\tau := \sup_{t\ge\tau} \Big|\beta\,\mathbb{E}\Big[G_{a\leftrightarrow b}^{\mathrm{diag}}(\cdot,t)\,\mathcal{S}_{\mathrm{bb}}^{\mathrm{diag}}(\cdot,t)\Big]\Big|,$$

$$A_{a\leftrightarrow b}^\tau := \inf_{t\ge\tau} \Gamma_{a\leftrightarrow b}^{\mathrm{ex}}(t).$$

If

$$\sup_{t\ge\tau} \big|\overline{\Gamma}_{\mathrm{fb}}^{\mathrm{diag}}(t)\big| > \varepsilon,$$

then the exchange-cover and tail-persistence assumptions imply the existence of $(a,b) \in \mathfrak{P}$ with

$$A_{a\leftrightarrow b}^\tau \ge \frac{c_{\mathrm{cov}}}{N_{\mathrm{ex}}}\,\varepsilon^2 - \frac{e_{\mathrm{cov}}(\tau)}{N_{\mathrm{ex}}} M_\gamma - r_{\mathrm{pers}}(\tau).$$

The tail-passive background assumption gives

$$2B_{a\leftrightarrow b}^\tau \le 2\beta M_{\mathrm{ex}}(\tau)\eta_{\mathrm{bb}}(\tau).$$

The exchange-efficiency gap implies

$$Q_{a\leftrightarrow b}^\tau \le (2-\delta_Q)\,\sup_{t\ge\tau}\Gamma_{a\leftrightarrow b}^{\mathrm{ex}}(t) \le (2-\delta_Q)\big(A_{a\leftrightarrow b}^\tau + r_{\mathrm{pers}}(\tau)\big).$$

Substituting these estimates into the exchange bound gives

$$D_\tau^{\mathrm{diag}}(I_{a\leftrightarrow b}) - D_\tau^{\mathrm{diag}}(I^*) \le -\frac{\delta_Q c_{\mathrm{cov}}}{N_{\mathrm{ex}}}\,\varepsilon^2 + \frac{2e_{\mathrm{cov}}(\tau)}{N_{\mathrm{ex}}} M_\gamma + 2r_{\mathrm{pers}}(\tau) + 2\beta M_{\mathrm{ex}}(\tau)\eta_{\mathrm{bb}}(\tau).$$

Approximate tail optimality of I^* yields

$$-r_{\mathrm{opt}}(\tau) \le D_\tau^{\mathrm{diag}}(I_{a\leftrightarrow b}) - D_\tau^{\mathrm{diag}}(I^*).$$

Combining the two inequalities gives the stated bound on ε, hence on $\sup_{t\ge\tau}|\overline{\Gamma}_{\mathrm{fb}}^{\mathrm{diag}}(t)|$. □

Remark 6.15 (Compatibility and physical nonvacuity of Proposition 6.14). An engineered finite classical Markov witness constructed in the audit layer realizes the four tail packages simultaneously. It establishes their abstract mathematical compatibility, so the antecedent is not formally contradictory. It is not a natural source-datum model and is not used as a physical example here. No known physical source/control model currently derives all four packages jointly, and no known example also realizes the Type III modular shell interface needed downstream. Thus Proposition 6.14 remains a conditional result; its natural physical nonvacuity is open.

Corollary 6.16 (Conditional foreground-background mutual-information decay)**.** *Assume the hypotheses of Proposition 6.14, and in addition suppose that the pointwise diagonal foreground-background defect admits the block-local decomposition*

$$\gamma_{\mathrm{fb}}^{\mathrm{diag}}(\xi,t) = -\imath_{\mathrm{fb}}(\xi,t) + e_{\mathrm{fb}}^D(\xi,t) + e_{\mathrm{fb}}^H(\xi,t),$$

where

$$\imath_{\text{fb}}(\xi,t) := \log \frac{P_t^{\text{diag}}(u,v)}{P_t^{\text{diag}}(u)P_t^{\text{diag}}(v)}$$

and the averaged locality defects satisfy

$$\sup_{t\geq\tau}\big|\overline{E}^{D}_{\text{fb}}(t)\big| \leq \varepsilon_D(\tau), \qquad \sup_{t\geq\tau}\big|\overline{E}^{H}_{\text{fb}}(t)\big| \leq \varepsilon_H(\tau),$$

with $\varepsilon_D(\tau), \varepsilon_H(\tau) \to 0$. *Then*

$$\sup_{t\geq\tau} I_t(u:v) \leq \varepsilon_\Gamma(\tau) + \varepsilon_D(\tau) + \varepsilon_H(\tau),$$

hence $I_t(u:v) \to 0$ *uniformly on the tail.*

Proof. Averaging the assumed pointwise decomposition gives

$$\overline{\Gamma}^{\text{diag}}_{\text{fb}}(t) = -I_t(u:v) + \overline{E}^{D}_{\text{fb}}(t) + \overline{E}^{H}_{\text{fb}}(t).$$

Taking absolute values and applying Proposition 6.14 gives the stated estimate. □

Corollary 6.17 (Conditional full diagonal product-defect closure)**.** *Fix orderings*

$$B_{\text{fg}} = (i_1,\dots,i_{k^*}), \qquad B_{\text{bg}} = (j_1,\dots,j_{D-k^*})$$

of the foreground and background indices. For an ordered block $B = (\ell_1,\dots,\ell_r)$*, write*

$$X_{B,\leq m} := (\xi_{\ell_1},\dots,\xi_{\ell_m}), \qquad I_{B,m}(t) := I_t\big(X_{B,\leq m} : \xi_{\ell_{m+1}}\big).$$

Assume the hypotheses of Corollary 6.16, and in addition suppose that for each ordered block $B \in \{B_{\text{fg}}, B_{\text{bg}}\}$ *and each* $m = 1,\dots,|B|-1$ *there is a pointwise diagonal prefix defect*

$$\gamma^{\text{diag}}_{B,m}(x_{B,\leq m+1},t) = -\imath_{B,m}(x_{B,\leq m+1},t) + e^{D}_{B,m}(x_{B,\leq m+1},t) + e^{H}_{B,m}(x_{B,\leq m+1},t),$$

where

$$\imath_{B,m}(x_{B,\leq m+1},t) := \log \frac{P^{\leq m+1}_{B,t}(x_{B,\leq m+1})}{P^{\leq m}_{B,t}(x_{B,\leq m})\,P_{\ell_{m+1},t}(x_{\ell_{m+1}})},$$

and the averaged defects satisfy

$$\sup_{t\geq\tau}\big|\overline{\Gamma}^{\text{diag}}_{B,m}(t)\big| \leq \varepsilon^{\Gamma}_{B,m}(\tau),$$

$$\sup_{t\geq\tau}\big|\mathbb{E}\big[e^{D}_{B,m}(\cdot,t)\big]\big| \leq \varepsilon^{D}_{B,m}(\tau), \qquad \sup_{t\geq\tau}\big|\mathbb{E}\big[e^{H}_{B,m}(\cdot,t)\big]\big| \leq \varepsilon^{H}_{B,m}(\tau),$$

with all these functions tending to zero as $\tau \to \infty$. *Define*

$$\varepsilon^{\text{seq}}_{\text{fg}}(\tau) := \sum_{m=1}^{k^*-1}\Big(\varepsilon^{\Gamma}_{\text{fg},m}(\tau) + \varepsilon^{D}_{\text{fg},m}(\tau) + \varepsilon^{H}_{\text{fg},m}(\tau)\Big),$$

$$\varepsilon^{\text{seq}}_{\text{bg}}(\tau) := \sum_{m=1}^{D-k^*-1}\Big(\varepsilon^{\Gamma}_{\text{bg},m}(\tau) + \varepsilon^{D}_{\text{bg},m}(\tau) + \varepsilon^{H}_{\text{bg},m}(\tau)\Big),$$

with the convention that an empty sum is zero. Then

$$\sup_{t\geq\tau} D_{\text{KL}}\left(P_t^{\text{diag}}\middle\|\prod_{i=1}^{D}P_{i,t}\right) \leq \varepsilon_\Gamma(\tau) + \varepsilon_D(\tau) + \varepsilon_H(\tau) + \varepsilon^{\text{seq}}_{\text{fg}}(\tau) + \varepsilon^{\text{seq}}_{\text{bg}}(\tau).$$

Proof. The exact block decomposition gives

$$D_{\mathrm{KL}}\left(P_t^{\mathrm{diag}}\middle\|\prod_{i=1}^{D}P_{i,t}\right)=I_t(u:v)+D_{\mathrm{KL}}\left(P_t^{\mathrm{diag}}(u)\middle\|\prod_{i\in I^*}P_{i,t}\right)+D_{\mathrm{KL}}\left(P_t^{\mathrm{diag}}(v)\middle\|\prod_{j\notin I^*}P_{j,t}\right).$$

For any ordered block $B=(\ell_1,\dots,\ell_r)$, the KL chain rule gives

$$D_{\mathrm{KL}}\left(P_{B,t}\middle\|\prod_{q=1}^{r}P_{\ell_q,t}\right)=\sum_{m=1}^{r-1}I_{B,m}(t).$$

Averaging the assumed prefix decomposition yields

$$\overline{\Gamma}^{\mathrm{diag}}_{B,m}(t)=-I_{B,m}(t)+\mathbb{E}\big[e^{D}_{B,m}(\cdot,t)\big]+\mathbb{E}\big[e^{H}_{B,m}(\cdot,t)\big],$$

hence

$$I_{B,m}(t)\le\left|\overline{\Gamma}^{\mathrm{diag}}_{B,m}(t)\right|+\left|\mathbb{E}\big[e^{D}_{B,m}(\cdot,t)\big]\right|+\left|\mathbb{E}\big[e^{H}_{B,m}(\cdot,t)\big]\right|.$$

Summing over m gives the stated bound for each block. Combining those two blockwise bounds with Corollary 6.16 proves the claim. □

7 Stable-Hull Transfer and Comparison with the Exact Fixed Sector

Let $\omega(\cdot)=\langle\Omega,\cdot\,\Omega\rangle$ on $\mathcal{A}_U$, and let $\sigma_s^{\omega,\mathrm{tot}}$ denote the ambient modular flow. Define

$$\mathfrak{L}^{\mathrm{st}}_{\mathfrak{F}}:=\left\{\begin{array}{l}\mathcal{M}\subseteq\mathcal{A}_U:\mathcal{M}\text{ is a von Neumann subalgebra,}\\ \qquad\mathcal{A}_{\mathfrak{F}}\subseteq\mathcal{M},\\ \qquad\sigma_s^{\omega,\mathrm{tot}}(\mathcal{M})=\mathcal{M}\quad\forall s\in\mathbb{R},\\ \qquad\mathcal{E}_t(\mathcal{M})\subseteq\mathcal{M}\quad\forall t\ge0\end{array}\right\}.$$

This class is nonempty because $\mathcal{A}_U\in\mathfrak{L}^{\mathrm{st}}_{\mathfrak{F}}$.

Three distinct structural inputs. For a general normal UCP semigroup, the linear fixed set

$$\mathrm{Fix}(\mathcal{E}):=\{A\in\mathcal{A}_U:\mathcal{E}_t(A)=A\ \forall t\ge0\}$$

is an operator system but need not be a von Neumann algebra under the ambient product. Whenever the comparison results below use

$$\mathcal{A}_{\mathrm{fix}}:=\mathrm{Fix}(\mathcal{E})$$

as an algebra, they assume separately:

(FSA) *fixed-set algebra closure:* $\mathcal{A}_{\mathrm{fix}}$ is a von Neumann subalgebra of $\mathcal{A}_U$;

(MOD) *Type III modular invariance:* $\sigma_s^{\omega,\mathrm{tot}}(\mathcal{A}_{\mathrm{fix}})=\mathcal{A}_{\mathrm{fix}}$ for every $s\in\mathbb{R}$;

(EXP) *expectedness when required:* there is a faithful normal ω-preserving conditional expectation $E_{\mathrm{fix}}:\mathcal{A}_U\to\mathcal{A}_{\mathrm{fix}}$.

FSA is a multiplicative-closure condition and is not supplied by UCP fixedness alone. MOD is a separate dynamical condition. In the present faithful standard-form setting, MOD permits the Takesaki expectation, but the mere existence of a faithful invariant state does not by itself establish MOD or EXP [12, 13]. The minimal stable-hull intersection below does not require EXP; an ω-preserving projection onto the exact fixed sector does.

Definition 7.1 (Stable-hull transfer map)**.** The *stable-hull transfer* of the foreground frame $\mathfrak{F}$ is

$$\Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F}) := \bigcap_{\mathcal{M}\in\mathfrak{L}^{\mathrm{st}}_{\mathfrak{F}}} \mathcal{M}.$$

Proposition 7.2 (Preliminary inclusion in the exact fixed sector)**.** *Assume FSA. Thus $\mathcal{A}_{\mathrm{fix}} = \mathrm{Fix}(\mathcal{E})$ is a von Neumann subalgebra, without identifying it with probe accessibility or with an environmental record algebra. Let $\mathfrak{F}$ be a T-DOME foreground frame, and let $O_1, \dots, O_{k^*}$ be exact pointer-aligned observables realizing $\iota_{\mathfrak{F}}$ as in Proposition 5.3. Then*

$$\mathcal{A}_{\mathfrak{F}} = W^*(1, O_1, \dots, O_{k^*}) \subseteq \mathcal{A}_{\mathrm{fix}}.$$

Proof. Exact pointer alignment gives

$$O_i \in \mathrm{Fix}(\mathcal{E}) = \mathcal{A}_{\mathrm{fix}} \qquad \text{for each } i.$$

Since $\mathcal{A}_{\mathrm{fix}}$ is a von Neumann algebra by FSA, it contains the von Neumann algebra generated by $1, O_1, \dots, O_{k^*}$. Hence

$$W^*(1, O_1, \dots, O_{k^*}) \subseteq \mathcal{A}_{\mathrm{fix}}.$$

□

Proposition 7.3 (Minimal stable-hull transfer)**.** *Assume the hypotheses of Proposition 5.3. Then*

$$\Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F}) = \bigcap_{\mathcal{M}\in\mathfrak{L}^{\mathrm{st}}_{\mathfrak{F}}} \mathcal{M}$$

is a well-defined von Neumann subalgebra of $\mathcal{A}_U$ satisfying:

1. $\mathcal{A}_{\mathfrak{F}} \subseteq \Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F})$*;*
2. $$\sigma^{\omega,\mathrm{tot}}_s\big(\Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F})\big) = \Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F}) \qquad \forall s \in \mathbb{R};$$
3. $$\mathcal{E}_t\big(\Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F})\big) \subseteq \Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F}) \qquad \forall t \geq 0;$$
4. *it is minimal with these properties: if $\mathcal{N} \subseteq \mathcal{A}_U$ is any von Neumann subalgebra containing $\mathcal{A}_{\mathfrak{F}}$ and stable under both ambient dynamics, then*
$$\Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F}) \subseteq \mathcal{N}.$$

Proof. The class $\mathfrak{L}^{\mathrm{st}}_{\mathfrak{F}}$ is nonempty because $\mathcal{A}_U$ belongs to it. An arbitrary intersection of von Neumann subalgebras is again a von Neumann subalgebra, so the intersection is well-defined and is itself a von Neumann subalgebra of $\mathcal{A}_U$. Since every element of $\mathfrak{L}^{\mathrm{st}}_{\mathfrak{F}}$ contains $\mathcal{A}_{\mathfrak{F}}$, so does the intersection.

If $A \in \Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F})$, then $A \in \mathcal{M}$ for every $\mathcal{M} \in \mathfrak{L}^{\mathrm{st}}_{\mathfrak{F}}$. By definition of the class, $\sigma^{\omega,\mathrm{tot}}_s(A) \in \mathcal{M}$ for every $s \in \mathbb{R}$, and $\mathcal{E}_t(A) \in \mathcal{M}$ for every $t \geq 0$. Therefore both $\sigma^{\omega,\mathrm{tot}}_s(A)$ and $\mathcal{E}_t(A)$ belong to the intersection. This proves $\sigma^{\omega,\mathrm{tot}}_s(\Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F})) \subseteq \Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F})$. Applying the same argument with $-s$ gives the reverse inclusion, hence equality.

Finally, if $\mathcal{N}$ is any von Neumann subalgebra with the same containment and stability properties, then $\mathcal{N} \in \mathfrak{L}^{\mathrm{st}}_{\mathfrak{F}}$. Since $\Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F})$ is the intersection of all members of that class, it is contained in $\mathcal{N}$. This is exactly the desired minimality. □

Proposition 7.4 (Stable-hull transfer lies in a modular exact fixed sector)**.** *Assume:*

1. *the hypotheses of Proposition 5.3;*
2. *FSA and the exact pointer-invariance condition of Proposition 7.2;*
3. *MOD holds for* $\mathcal{A}_{\mathrm{fix}}$.

Then

$$\Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F}) \subseteq \mathcal{A}_{\mathrm{fix}}.$$

Proof. By Proposition 7.2, $\mathcal{A}_{\mathfrak{F}} \subseteq \mathcal{A}_{\mathrm{fix}}$. FSA makes $\mathcal{A}_{\mathrm{fix}}$ a von Neumann algebra, MOD makes it invariant under the ambient modular flow, and exact fixedness implies forward stability under the decoherence semigroup. Hence $\mathcal{A}_{\mathrm{fix}} \in \mathfrak{L}^{\mathrm{st}}_{\mathfrak{F}}$. Since $\Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F})$ is the intersection of all members of that class, it is contained in $\mathcal{A}_{\mathrm{fix}}$. □

Corollary 7.5 (Partial Lemma C)**.** *Assume the hypotheses of Proposition 7.3. Then* $\Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F})$ *is a von Neumann subalgebra of* $\mathcal{A}_U$*, and the restricted vector state*

$$\omega^{\mathrm{st}}_{\mathfrak{F}} := \langle \Omega, \cdot\, \Omega \rangle|_{\Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F})}$$

is faithful. If, in addition, the hypotheses of Proposition 7.4 hold, then

$$\Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F}) \subseteq \mathcal{A}_{\mathrm{fix}}.$$

Proof. The subalgebra statement is exactly Proposition 7.3. Because Ω is separating for the ambient algebra $\mathcal{A}_U$, it is also separating for every von Neumann subalgebra of $\mathcal{A}_U$; hence the restricted vector state is faithful on $\Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F})$. The inclusion claim is Proposition 7.4. □

Anchored shell existence around the stable hull. The stable-hull output already anchors a Foundation I shell problem. Write

$$\mathcal{A}_* := \Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F}).$$

For $X \in \mathrm{Ball}(\mathcal{A}_U)$ and $q \geq 0$, define the $\mathcal{A}_*$-relative defect

$$\delta^{\mathcal{A}_*,\Omega}_q(X) := \mathrm{dist}^{\sharp}_{\Omega}\big(\mathcal{E}_q(X),\, \mathrm{Ball}(\mathcal{A}_*)\big), \qquad \|Y\|^{\sharp}_{\Omega} := \big(\|Y\Omega\|^2 + \|Y^*\Omega\|^2\big)^{1/2}.$$

For $\varepsilon \geq 0$ and $\tau > 0$, let

$$\mathfrak{R}^{\mathcal{A}_*,\Omega}_{\varepsilon,\tau} := \left\{ \mathcal{M} \;\middle|\; \begin{array}{l} \mathcal{A}_* \subseteq \mathcal{M} \subseteq \mathcal{A}_U,\ \mathcal{M} \text{ is a unital vN subalgebra,} \\ \delta^{\mathcal{A}_*,\Omega}_q(X) \leq \varepsilon,\ \forall X \in \mathrm{Ball}(\mathcal{M}),\ \forall q \in \mathbb{Q} \cap [0,\tau] \end{array} \right\}.$$

Proposition 7.6 (Anchored relative-shell existence around the stable hull)**.** *Assume the hypotheses of Corollary 7.5. Then for every* $\varepsilon \geq 0$ *and* $\tau > 0$*:*

1. *the class $\mathfrak{R}_{\varepsilon,\tau}^{\mathcal{A}_*,\Omega}$ is nonempty;*

2. *every chain in $\mathfrak{R}_{\varepsilon,\tau}^{\mathcal{A}_*,\Omega}$ has an upper bound;*

3. *therefore maximal elements exist.*

Proof. By Corollary 7.5, $\mathcal{A}_*$ is a von Neumann subalgebra of $\mathcal{A}_U$ and is forward-stable under the decoherence semigroup. Hence $\mathcal{A}_*$ itself belongs to $\mathfrak{R}_{\varepsilon,\tau}^{\mathcal{A}_*,\Omega}$: for $X \in \mathrm{Ball}(\mathcal{A}_*)$, forward stability gives $\mathcal{E}_q(X) \in \mathcal{A}_*$, and contractivity gives $\mathcal{E}_q(X) \in \mathrm{Ball}(\mathcal{A}_*)$, so $\delta_q^{\mathcal{A}_*,\Omega}(X) = 0$. Thus the class is nonempty.

Let $\{\mathcal{M}_i\}_{i\in I}$ be a chain in $\mathfrak{R}_{\varepsilon,\tau}^{\mathcal{A}_*,\Omega}$, and set

$$\mathcal{M} := W^*\Big(\bigcup_{i\in I} \mathcal{M}_i\Big).$$

Then $\mathcal{A}_* \subseteq \mathcal{M} \subseteq \mathcal{A}_U$, and $\mathcal{M}$ is a unital von Neumann subalgebra. By Kaplansky density, every $X \in \mathrm{Ball}(\mathcal{M})$ is a bounded strong* limit of a net from the unit ball of the algebraic union $\bigcup_i \mathcal{M}_i$. Since the chain is totally ordered, each approximant lies in some admissible member of the chain, so its $\mathcal{A}_*$-relative defect is bounded by ε for all $q \in \mathbb{Q} \cap [0,\tau]$. Passing to the limit in the $\Omega^\sharp$-seminorm yields the same bound for X, hence $\mathcal{M} \in \mathfrak{R}_{\varepsilon,\tau}^{\mathcal{A}_*,\Omega}$. Therefore every chain has an upper bound. Zorn's lemma now gives a maximal element. □

Remark 7.7 (Status of anchored shells). Proposition 7.6 does not yet construct the full shell family required on the Paper I side. It closes only the anchored existence problem around the theorem-level Foundation I output $\mathcal{A}_*$. What remains open is the upgrade to a canonical, nontrivial, and calibrated shell family with the correct modular transport law.

Remark 7.8 (Why equality is not the right first target). In general, $\mathcal{A}_{\mathrm{fix}}$ may be much larger than the minimal modular-decoherence stable hull of finitely many foreground pointer observables. The honest first target is therefore the inclusion

$$\Phi_{\mathfrak{F}}^{\mathrm{st}}(\mathfrak{F}) \subseteq \mathcal{A}_{\mathrm{fix}},$$

not the equality

$$\Phi_{\mathfrak{F}}^{\mathrm{st}}(\mathfrak{F}) \stackrel{?}{=} \mathcal{A}_{\mathrm{fix}}.$$

The equality problem should remain open until one proves that the foreground realization together with the stable-hull transfer generates the full exact fixed sector. A further typed interface would still be needed to relate either object to probe-accessible effects or to the environmental record algebra.

Remark 7.9 (Relation to Paper III). One should not identify the foreground complexity k^* directly with the statistical complexity parameter n of Paper III [2]. The safer statement is only that $\Phi_{\mathfrak{F}}^{\mathrm{st}}(\mathfrak{F})$ provides a natural algebraic candidate for a later bridge to the finite-observer sector studied there.

8 What Remains Open

The bottlenecks are now sharper than they were before this note.

1. **Persistent projection selection.** Lemma 3.2 is only conditional. The constructive question of which projection emerges from the source datum remains open.

2. **Foreground realization and model-independent PS2.** The stable-hull route closes the transfer-map well-definedness problem once $\mathcal{A}_{\mathfrak{F}}$ exists. Within the restricted product-classical and finite-dimensional faithful pointer regime, the PS2 package is now theorem-backed: Lemma 6.2 closes the structural reduction to fragment-local omission losses, Lemma 6.4 closes orthogonality, and Corollary 6.10 packages the restricted selection theorem. Lemma 6.12 now fixes the bookkeeping split of the survival deficit into a pointer-diagonal term and an off-sector remainder. In addition, Proposition 6.14 and Corollary 6.16 now show that the foreground-background part of the model-independent programme closes conditionally once four tail inputs are granted: approximate tail optimality, exchange-cover plus persistence, tail-passive background survival, and an exchange-efficiency gap, together with the block-locality defects for the KL and control terms. Corollary 6.17 then reduces the remaining diagonal product-defect problem to finitely many ordered within-block prefix couplings. The remaining open issue is therefore to derive those tail hypotheses together with the prefix-level coupling and locality bounds model-independently from the broken-phase output and the full T-DOME survival identity rather than assume them. Beyond the restricted regime, the unresolved analytic obstacles are explicit: cross-channel correlations need not decouple, pointer spectra need not remain finite or discrete, and the Hilbert–Schmidt control used here is not available in general Type III infinite-dimensional settings, where a predual reformulation may be necessary. The pointer-sector existence input PS1 is treated here only as an exact-regime idealization; what remains genuinely open is the model-independent alignment step PS2 and the induced ordered-selection step PS3.

3. **Comparison with typed HAFF objects.** The present paper proves inclusion only into the exact fixed sector under FSA and MOD. Equality with that sector remains open, and no canonical equivalence with probe-accessible effects or the environmental record algebra is asserted.

4. **Bridge to Paper I admissibility.** To feed directly into Paper I, one must still verify that the algebra produced by the lower bridge satisfies the additional admissibility hypotheses used there, beyond the already-closed faithfulness condition. More precisely, the remaining interface is to supply the independent dynamical modular-decoherence scaling axiom (C) together with an operational shell-generation/closure interface $\{\mathcal{A}_*(\tau)\}_{\tau>0}$ around the stable-hull output $\mathcal{A}_* := \Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F})$, and one-scale cyclicity of the corresponding shell-annulus algebra
$$\mathcal{D}_\tau := \mathcal{A}_*(\tau) \cap \mathcal{A}_*'.$$
In Paper I language these are exactly the remaining inputs $(C)+(Amb)+(CN_\exists)$. Foundation I does not derive the shell-generation rule or its covariance. The latter is to be derived in Paper I from the independently stated dynamical axiom (C), once membership and closure are supplied. The shell-family side has only partially internalized: Proposition 7.6 closes anchored relative-shell existence around $\mathcal{A}_*$. What remains open is the canonical and calibrated nontrivial shell family. The naive scalar-threshold route now has an exact blocker: fixed thresholds are compatible with modular transport but not with exact core calibration, vanishing thresholds give exact core calibration but not automatic modular transport, and the only scalar profile with both properties is the trivial zero-threshold profile, which collapses the shells back to $\mathcal{A}_*$. More sharply, even the fixed-threshold branch now reduces to finding a scale-invariant nontriviality window between core collapse and ambient collapse, together with uniqueness of maximal shells inside that window; exact calibration is a separate issue beyond that.

These open problems are real, but they are no longer vague. Foundation I is now separated into one conditional source-selection problem, one theorem-level thermodynamic packaging result, one theorem-level stable-hull transfer construction, and one remaining foreground-realization/comparison problem.

9 Conclusion

The purpose of this paper has been calibration, not inflation. Foundation I is neither absent nor complete. It is partially available in theorem-level form, partially available as an explicit construction problem, and partially still conditional.

The main positive result is that the T-DOME I–II chain can already be written as a formal Foundation I proposition, and that the first restricted PS2 bottleneck can now be closed at theorem level in a first restricted regime. In the product-classical, decoherence-dominated pointer regime, pointer coarse-graining yields fragment-local omission losses, independent fragment records force orthogonality, semigroup decay plus explicit Hilbert–Schmidt continuity constants bound the coherence remainder, and a sufficiently large weight gap forces exact top-k^* pointer selection. On the algebraic side, the first operator-algebraic realization of the foreground sector can now be extended to a well-defined minimal modular-decoherence stable hull, yielding a faithful von Neumann subalgebra together with the inclusion

$$\Phi_{\mathfrak{F}}^{\mathrm{st}}(\mathfrak{F}) \subseteq \mathcal{A}_c^{\mathrm{HAFF}}.$$

At the level of future PS2 work, the paper now also fixes the bookkeeping split of the survival deficit into a pointer-diagonal term and an off-sector remainder, and it closes the first foreground-background decoupling step conditionally: if the tail distortion localizes, the exchange-weighted defect persists, the background diagonal survival becomes tail-passive, the exchange efficiency ratio stays below the squared-loss threshold, and the remaining KL/control locality defects are small, then the broken phase forces tail-small diagonal foreground-background coupling and hence tail-small foreground-background mutual information. Corollary 6.17 then shows that the unresolved product-defect part is no longer a monolithic statement: it is a finite family of ordered within-block prefix decoupling problems. The remaining model-independent problem is therefore no longer vague: it is to derive those tail hypotheses and the ordered prefix-level coupling/locality bounds, and thereby to control the full diagonal product defect as a theorem-level output of the broken phase. On the Paper I side, the admissibility interface is now sharpened in the same way: faithfulness is already closed, ambient scaling covariance would transfer the dynamical axiom (C), and the remaining regularity burden is exactly the existence of a Foundation I shell family and one cyclic shell-annulus scale. Even the shell-family side is no longer a black box: anchored relative shells around $\mathcal{A}_*$ exist, but their upgrade to a canonical nontrivial shell family currently stops at the scalar-threshold trilemma between transport, exact core calibration, and nontriviality. Within the fixed-threshold route, this has already reduced further to a scale-invariant nontriviality window and uniqueness problem, plus a separate calibration theorem. The main negative result is equally important: nothing in the present note licenses the stronger claim that the source datum has already been shown to generate the full accessible algebra uniquely, nor that PS2 has been derived model-independently from the broken-phase output. Those remain the unfinished parts of the unified-source programme.

In that sense, the current status is simple. The thread can be strung, but only honestly: conditional at the source end, inclusion-first at the algebra end, and theorem-backed only where the intermediate steps have actually been closed.

A External Genesis Inputs (Self-Contained Statements)

The following list records the external Genesis-side inputs used in the paper in a self-contained form. The point is not to reproduce their full proofs, but to state exactly what is imported from HAFF and T-DOME.

1. **HAFF ambient datum.** The ambient setting is a Type III_1 von Neumann algebra in standard form $(\mathcal{H}_U, \mathcal{A}_U, \Omega)$, with Ω cyclic and separating for $\mathcal{A}_U$, together with a pointwise σ-weakly continuous semigroup of normal completely positive unital maps $\{\mathcal{E}_t\}_{t\geq 0}$ on $\mathcal{A}_U$. This is the source datum formalized in Section 2.

2. **T-DOME I projected record dynamics.** Given a persistent Nakajima–Zwanzig projection $\mathcal{P} : \mathcal{A}_U \to \mathcal{A}_U$, the projected predual dynamics is analyzed in a non-Markovian kernel form

$$\partial_t \omega_{\mathcal{P},t} = \int_0^t \mathcal{K}_*(t,s)\, \omega_{\mathcal{P},s}\, ds \; + \; \text{inhomogeneous term},$$

   or equivalently in the dual memory-kernel formulation used in Section 2. Foundation I uses only the existence of such a kernel and a persistent record process, not an explicit closed-form solution.

3. **T-DOME I survival identity and distortion functional.** The survival bookkeeping imported into the PS2 analysis is the identity

$$\beta\, \mathcal{S} = -\Delta I(S{:}E) - \Delta D_{\mathrm{KL}} - \beta\, \Delta\langle H_{\mathrm{ctrl}}\rangle,$$

   with frame comparisons measured in the squared-loss distortion

$$D(\mathcal{F}_I) := \mathbb{E}_\xi\Big[\big(\mathcal{S}_{\mathrm{full}}(\xi) - \mathcal{S}_{\mathcal{F}_I}(\xi)\big)^2\Big].$$

   The diagonal/off-sector split and all restricted PS2 estimates are built from this identity rather than assumed independently.

4. **Finite-coordinate compression model motivated by T-DOME II.** T-DOME II motivates a coordinate-coding problem, but its universal computational-ceiling and SSB-necessity claims are not imported. Proposition 4.1 instead assumes a rate charge, strict improvement up to the feasible rank, and an irreducible group action. Only in that supplied FD/I model does $k^* = \lfloor \dot{\mathcal{C}}_{\mathrm{budget}}/h_\mu \rfloor$ and the selection of a proper foreground subspace follow.

5. **Decoherence-side pointer sector (exact-regime idealization).** In the exact dephasing-dominated regime, robust classical records are carried by a commuting pointer sector. Foundation I uses this as the PS1 input: a commuting family of exact fixed-point projections for $\{\mathcal{E}_t\}$ generating a classical pointer algebra inside $\mathcal{A}_U$. This is an idealization of the approximate einselection and redundant-record structure described in the quantum Darwinism literature [9, 10], not a consequence of generic decoherence alone.

6. **Exact fixed sector, not a universal accessibility object.** The inclusion comparison uses $\mathcal{A}_{\mathrm{fix}} = \mathrm{Fix}(\mathcal{E})$ only under FSA and MOD. This object is an exact noiseless/fixed sector. It is not identified with the probe-accessible operator system or the environmental pointer/record algebra.

7. **T-DOME III / Paper III calibration distinction.** Foundation I uses T-DOME III and Paper III only as contextual support for what is *not* claimed: the foreground complexity k^* is not identified with Paper III's statistical complexity parameter n, and the present paper does not attempt a direct derivation of Paper III's self-description bounds from the source-side chain.

B Conditional Input Ledger

For ease of reference, the remaining conditional interfaces are listed here in one place.

1. **Source selection.** Lemma 3.2 is conditional on the existence of a persistent Nakajima–Zwanzig projection producing the projected record kernel. This is the observer-emergence bottleneck. No general semigroup-gap, CP-divisibility, or kernel-decay criterion is asserted here: each would require a specified persistence functional and can fail to capture the non-Markovian record regime. A rigorous sufficient criterion remains open rather than being installed for downstream use without proof.

2. **Pointer alignment package.** The foreground realization problem separates into:

   (PS1) existence of a decoherence-selected exact fixed-point pointer sector (exact dephasing-dominated regime idealization, not a consequence of generic decoherence);

   (PS2) identification of the T-DOME sufficient-statistic coordinates with the pointer-sector coordinates;

   (PS3) ordered pointer-channel selection by survival relevance.

   PS1 is an exact-regime idealization of the decoherence-side input; the genuine model-independent open problem is PS2, with PS3 downstream from it.

3. **Exact fixed-sector structural inputs.** The main text keeps the following conditions separate because none is a consequence of the bare Type III source datum:

   (FSA) $\mathrm{Fix}(\mathcal{E})$ is closed under the ambient product and is a von Neumann subalgebra;

   (MOD) that fixed algebra is invariant under the ambient modular flow;

   (EXP) when an expected subalgebra is required, an ω-preserving faithful normal conditional expectation exists;

   (PTR) the separately specified environmental pointer algebra is contained in the exact fixed algebra in the idealized noiseless regime.

   Faithfulness of ω alone does not establish MOD. In the faithful standard-form regime, Takesaki's theorem relates MOD to an ω-preserving expectation only after FSA supplies the subalgebra to which the theorem applies [12, 13].

4. **Restricted PS2 regime assumptions.** The theorem-level restricted PS2 closure is proved only in the product-classical, finite-dimensional faithful pointer regime stated in Section 6. In particular, the restricted closure does not yet treat general cross-channel correlations, continuous pointer spectra, or the genuinely infinite-dimensional Type III estimate problem.

5. **Conditional foreground-background decoupling inputs.** Proposition 6.14 requires the following named inputs on the tail regime: approximate tail optimality, exchange-cover plus persistence, tail-passive background survival, and an exchange-efficiency gap, together with the KL/control locality defects used in Corollary 6.16.

6. **Abstract compatibility of the four tail packages.** An engineered finite classical Markov witness realizes the four tail packages simultaneously. This is an FD/I audit witness establishing joint mathematical compatibility, not a physical derivation and not a Type III shell model.

7. **Natural physical nonvacuity remains open.** No natural source/control model is known to derive all four packages jointly, and no known model simultaneously realizes those packages and the modular shell interface. Proposition 6.14 therefore remains conditional despite the compatibility witness.

8. **Conditional full product-defect inputs.** Corollary 6.17 adds the ordered within-block prefix-chain analogues of the same decoupling/locality inputs. The full diagonal product-defect problem is therefore reduced to finitely many named prefix couplings rather than left as one undifferentiated gap.

9. **Paper I admissibility inputs.** The lower output feeds into Paper I only conditionally on three further interfaces: the independent dynamical covariance axiom (C), an operational shell membership/closure rule around the stable hull, and one-scale shell-annulus cyclicity. Proposition 7.6 closes only an anchored existence problem. It does not derive the Paper I shell interface. In particular, covariance is not placed into that interface; Paper I must derive shell scaling from (C).

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Foundation II: Shared Manifestation
A Conditional Two-Source Program for Shared Accessible Sectors

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Abstract

Foundation I of the Consciousness Series [1] constructs a conditional chain from a single source datum $S_{\text{source}} = (\mathcal{H}_U, \mathcal{A}_U, \Omega, \{\mathcal{E}_t\})$ to an admissible accessible pair $(\mathcal{A}_c, \omega)$, which then feeds into the inseparability theorem of Paper I [2]. That chain is single-source: it produces one observer's manifest sector. The present paper, Foundation II, asks the next structural question: when two source chains share the same ambient substrate, under what conditions does a non-trivial *shared* manifest sector emerge?

We define a multi-source datum as a family of source data sharing a common ambient triple $(\mathcal{H}_U, \mathcal{A}_U, \Omega)$, each equipped with its own decoherence semigroup. No global tensor-product decomposition is assumed; the multi-source structure is encoded entirely through subalgebra inclusions, following the non-canonicality of tensor-product factorizations in the Type III_1 setting established in [5]. We then introduce joint-stability conditions under which a candidate shared sector $\mathcal{A}_{\text{sh}}$ is simultaneously stable for both sources.

The paper's main result is a conditional two-source foothold, not a general multi-source closure theorem. Under named joint-stability and compatible-projection conditions, we construct a shared stable hull $\Phi^{\text{st}}_{\text{sh}}(\mathfrak{F}_{\text{sh}})$ from the overlap of the two foreground frames, prove that it is a von Neumann subalgebra carrying a faithful restricted state, and show that, under an additional admissibility input, the resulting pair $(\Phi^{\text{st}}_{\text{sh}}(\mathfrak{F}_{\text{sh}}), \omega^{\text{st}}_{\text{sh}})$ lies in the theorem domain of Paper I. The shared sector therefore inherits the geometry–modular-flow inseparability of the single-source case.

After the main theorem, we analyze a regime-dependent common/private structure: under a shared/private regime assumption, the shared sector is the locus on the overlap algebra where the two manifest states agree, while observer-dependent discrepancy can only appear outside that shared sector. Finally, we state a conditional N-source extension under pairwise-threshold and exchangeability assumptions, and record the open problems that remain, including the quantitative relationship between consensus strength and geometric stability.

This paper treats only the mathematical bridge from multi-source data to shared accessible sectors. It does not claim a general N-source theorem, does not provide a quantitative consensus–stability bound, and does not derive the candidate shared sector from bare multi-source dynamics alone. It also does not attempt to settle questions of intersubjective ontology.

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1 Introduction

Foundation I of the Consciousness Series [1] addressed the single-source problem. Starting from a source datum

$$S_{\text{source}} = (\mathcal{H}_U, \mathcal{A}_U, \Omega, \{\mathcal{E}_t\}_{t\geq 0}),$$

it isolated a conditional chain

$$S_{\text{source}} \to \mathcal{O}_A \to \mathcal{O}_B \to (\mathcal{A}_c, \omega),$$

where $\mathcal{O}_A$ is a persistent projected record process, $\mathcal{O}_B$ is a foreground frame supplied under explicit finite-coordinate coding assumptions, and $(\mathcal{A}_c, \omega)$ is the manifest accessible pair that then feeds into the geometry–modular-flow inseparability theorem of Paper I [2]. That was the correct scope for the first foundation paper: it explained how one observer-facing manifest sector could arise from one source chain.

The present paper asks the next structural question. If two source chains share the same ambient substrate, under what conditions does a non-trivial *shared* manifest sector arise? This is the minimum multi-source problem behind any later discussion of a shared world. The paper does not attempt to settle the full intersubjective problem in one step. It does not claim a general N-source theorem, does not derive a quantitative consensus-strength/stability law, and does not aim to settle questions of ontology. Its narrower aim is to identify a theorem-level foothold for *shared manifestation existence*.

That calibration matters. In the current programme there are two separate bridges beyond the already-written Consciousness Trilogy results. The first concerns whether entangled or correlated source chains can support a common manifest sector at all. The second concerns how the strength of that commonality controls the quantitative stability of the resulting geometry. Foundation II addresses only the first. The second remains a separate Bridge 3 problem and should not be smuggled into the present paper under vague language about "consensus" or "robustness".

The paper therefore adopts the smallest honest target. Its main theorem is proved in the two-source case. The theorem does not directly state that a full shared geometry has been derived. Its first conclusion is weaker and mathematically safer: under named compatibility and joint-stability inputs, one can construct a *shared admissible manifest sector*

$$(\mathcal{A}_{\text{sh}}, \omega_{\text{sh}}) \subseteq (\mathcal{A}_c^{(1)}, \omega_1) \cap (\mathcal{A}_c^{(2)}, \omega_2),$$

which is stable under the relevant source-side dynamics and can then be fed into the already-existing Paper I machinery. Geometry enters only as a corollary once that admissible shared sector is in hand. The main result should therefore be read as a conditional inclusion theorem and theorem-level foothold, not as a full emergence theorem for shared worlds.

At the heuristic level, this shared-sector question is related to the quantum Darwinism and spectrum-broadcast-structure literature, where the objective part of a quantum description is identified with redundantly accessible pointer information carried across multiple environment fragments [17, 18, 19, 20]. The present paper does not translate those frameworks into a theorem about Type III_1 ambient algebras. It only records the operator-algebraic analogue needed here: a shared foreground overlap together with a shared stable hull.

Three design choices govern the whole paper.

1. **Two sources first.** The paper starts at $N = 2$, because the two-source case is the minimum non-trivial setting in which one can ask whether a shared manifest sector exists at all. General N-source statements are postponed to a conditional extension section.

2. **No tensor-product starting point.** The multi-source datum is formulated inside a common ambient algebra rather than from a canonical tensor decomposition. This follows the HAFF viewpoint that physically relevant subsystem structure is selected by stability and accessibility rather than fixed in advance [5, 6].

3. **Common/private decomposition is secondary.** After the main theorem, we analyze how each observer's full manifest algebra may be written, under a regime assumption, as a shared sector plus a private complement. That decomposition is treated as a structural consequence, not as a competing main theorem.

The paper's mathematical contributions are correspondingly limited and explicit.

1. We define a *multi-source datum* on a common standard-form von Neumann algebra, with Type III_1 as the primary regime, together with independent faithful normal source states and source-specific decoherence semigroups.

2. We separate the new inputs into three named joint-stability conditions: shared-sector invariance, shared-state existence with faithfulness, and projection/kernel compatibility.

3. In the two-source case, we define the shared foreground overlap and the associated shared stable hull, and prove the abstract minimality properties of that hull.

4. Under an explicit candidate shared-sector input, we prove an inclusion theorem showing that the shared stable hull lies inside the common manifest carrier, and therefore yields a shared admissible manifest sector.

5. Under a shared/private regime assumption, we analyze the resulting common/private structure by identifying the shared sector as the locus of agreement on the overlap algebra and by localizing overlap-level discrepancy outside that sector.

The role of conditional language is exactly the same as in Foundation I. Wherever a piece can be internalized cleanly, we internalize it. Wherever it cannot, we isolate it and keep it explicit. The present theorem does not derive the existence of a candidate shared sector from bare multi-source dynamics or entanglement alone. It starts from such a sector as named input and analyzes the stable closure that follows from it. The main open points here are not hidden: model-independent joint stability, compatible Nakajima–Zwanzig projections, and the well-definedness of common/private decomposition in the Type III_1 setting.

The paper is organized as follows. Section 2 defines the multi-source datum and the joint-stability package. Section 3 states and proves the two-source main theorem. Section 4 analyzes the common/private structure under a regime assumption. Section 5 gives a conditional N-source extension. Section 6 records the open problems. Section 7 collects the named conditional inputs. Appendix A summarizes the imported Genesis and Consciousness results used by the paper.

2 Multi-Source Datum and Joint-Stability Inputs

The first task is to define the source-side object language without overloading it with later conditional interface data.

Definition 2.1 (Multi-source datum)**.** A *multi-source datum* is a tuple

$$S_{\text{multi}} := (\mathcal{H}_U, \mathcal{A}_U, \Omega, \{\varpi_i\}_{i=1}^N, \{\mathcal{E}_t^{(i)}\}_{i=1}^N),$$

where:

1. $(\mathcal{H}_U, \mathcal{A}_U, \Omega)$ is a von Neumann algebra in standard form, with Ω cyclic and separating for $\mathcal{A}_U$; the programme's primary infinite-dimensional regime is a Type III_1 factor, while explicitly marked model checks may be finite Type I;

2. $\varpi_1, \ldots, \varpi_N$ are independently specified faithful normal states on $\mathcal{A}_U$; no agreement of their restrictions to later overlap algebras is part of the datum;

3. for each $i \in \{1, \ldots, N\}$, $\{\mathcal{E}_t^{(i)}\}_{t\geq 0}$ is a pointwise σ-weakly continuous semigroup of normal completely positive unital maps on $\mathcal{A}_U$.

Remark 2.2 (Heisenberg-picture convention). The semigroups $\{\mathcal{E}_t^{(i)}\}$ act on observables, so the paper uses the Heisenberg-picture convention throughout. The continuity assumption in Definition 2.1 is therefore pointwise σ-weak continuity on $\mathcal{A}_U$, not norm continuity.

Remark 2.3 (A deliberately thin datum). The multi-source datum contains only the common ambient standard-form triple, the independent source states, and the family of source-side dynamical semigroups. It does *not* include Nakajima–Zwanzig projections, foreground frames, accessible algebras, or any private/shared decomposition. Those are not primitive data; they are interface or output objects that may or may not be obtainable from the datum under additional assumptions.

Remark 2.4 (Why no tensor-product factorization). The present paper does not assume a global tensor-product decomposition

$$\mathcal{H}_U \cong \bigotimes_{i=1}^N \mathcal{H}_i \qquad \text{or} \qquad \mathcal{A}_U \cong \bar{\otimes}_{i=1}^N \mathcal{A}_i.$$

This is deliberate. In the HAFF programme, the physically relevant subalgebras are selected by accessibility and stability rather than by a canonical prior factorization [5, 6]. In the Type III setting, treating the subsystem split as primitive would import exactly the kind of structure the Genesis programme is designed to derive rather than presuppose.

The single-source outputs from Foundation I are now imported only as downstream conditional objects.

Definition 2.5 (Individual manifest outputs)**.** Let S_{multi} be a multi-source datum. For each source index i, an *individual manifest output* means a pair

$$(\mathcal{A}_c^{(i)}, \omega_i)$$

obtained from the Foundation I chain for the i-th source, where $\mathcal{A}_c^{(i)} \subseteq \mathcal{A}_U$ is a von Neumann subalgebra and

$$\omega_i := \varpi_i|_{\mathcal{A}_c^{(i)}}$$

is its faithful normal manifest state.

Remark 2.6 (Independent states are load-bearing). The states ω_i are not all restrictions of the reference vector state $\langle \Omega, \cdot\, \Omega \rangle$. The vector Ω fixes a standard-form representation and the reference modular flow used below; the ϖ_i carry source-specific state data. If instead one defined every ω_i by restricting a single ambient state, then the states would agree automatically on every intersection and the overlap discrepancy of Section 4 would vanish identically.

Remark 2.7. Nothing in Definition 2.3 claims that every source in a multi-source datum automatically produces an individual manifest output. That is already the conditional content of Foundation I [1]. Foundation II takes those single-source outputs as antecedent material when the two-source main theorem is stated.

Definition 2.8 (Candidate shared sector)**.** Given individual manifest outputs $(\mathcal{A}_c^{(i)}, \omega_i)_{i=1}^N$, a *candidate shared sector* is a von Neumann subalgebra

$$\mathcal{A}_{\text{sh}}^{\text{cand}} \subseteq \bigcap_{i=1}^{N} \mathcal{A}_c^{(i)}.$$

The new source-side content begins with the joint-stability package. We state it in three layers so that later failures can be localized cleanly.

Definition 2.9 (JS-A: shared-sector invariance and semigroup compatibility)**.** Let S_{multi} be a multi-source datum and let $\mathcal{A}_{\text{sh}}^{\text{cand}} \subseteq \mathcal{A}_U$ be a candidate shared sector. We say that $(\mathcal{A}_{\text{sh}}^{\text{cand}}, \{\mathcal{E}_t^{(i)}\})$ satisfies *JS-A* if:

1. $\mathcal{E}_t^{(i)}(A) \in \mathcal{A}_{\text{sh}}^{\text{cand}}$ for every $A \in \mathcal{A}_{\text{sh}}^{\text{cand}}$, every i, and every $t \geq 0$;
2. the ambient modular flow $\sigma_s^{\omega_0,\text{tot}}$ of the reference state $\omega_0(\cdot) = \langle \Omega, \cdot\, \Omega \rangle$ satisfies
$$\sigma_s^{\omega_0,\text{tot}}(\mathcal{A}_{\text{sh}}^{\text{cand}}) = \mathcal{A}_{\text{sh}}^{\text{cand}} \qquad \forall s \in \mathbb{R};$$
3. the restricted semigroups commute on the candidate shared sector:
$$\mathcal{E}_t^{(i)} \mathcal{E}_u^{(j)}(A) = \mathcal{E}_u^{(j)} \mathcal{E}_t^{(i)}(A) \qquad \forall A \in \mathcal{A}_{\text{sh}}^{\text{cand}},\ \forall t, u \geq 0.$$

Remark 2.10 (Status of the JS-A commutativity clause). The exact commutativity requirement in Definition 2.9 is a strong regime assumption. The present paper does not claim that it follows model-independently from the bare multi-source datum. A weaker variant would replace strict commutativity by asymptotic commutativity in the long-time limit, or by commutativity only on the shared fixed-point algebra. The stronger form is used here because it makes the shared stable-carrier construction transparent; the weaker variants are left open.

Definition 2.11 (JS-B: shared-state existence and faithfulness)**.** Let $\mathcal{A}_{\text{sh}}^{\text{cand}} \subseteq \mathcal{A}_U$ be a candidate shared sector. We say that $\mathcal{A}_{\text{sh}}^{\text{cand}}$ satisfies *JS-B* if there exists a faithful normal state

$$\omega_{\text{sh}} \in (\mathcal{A}_{\text{sh}}^{\text{cand}})_*$$

such that:

1. $\omega_{\text{sh}} \circ \mathcal{E}_t^{(i)} = \omega_{\text{sh}}$ on $\mathcal{A}_{\text{sh}}^{\text{cand}}$ for every i and every $t \geq 0$;
2. ω_{sh} is compatible with the individual manifest states in the sense that
$$\omega_i|_{\mathcal{A}_{\text{sh}}^{\text{cand}}} = \omega_{\text{sh}} \qquad \forall i \in \{1, \ldots, N\}.$$

Remark 2.12 (JS-B is an agreement assumption, not a discovery). Faithfulness of each restriction follows from faithfulness of the independent ambient state ϖ_i, but equality of two restrictions does not. Clause (2) of JS-B is therefore the explicit substantive agreement input. If all ω_i were restrictions of the single reference state ω_0, clause (2) would be automatic and the discrepancy analysis would have no nonzero cases. The present datum deliberately excludes that shortcut.

Remark 2.13 (Strength of the JS-B stationarity clause). The stationarity requirement in Definition 2.11 is strong. It amounts to asking that the shared state lie in the common fixed-state sector of all source semigroups when restricted to $\mathcal{A}_{\text{sh}}^{\text{cand}}$. The present paper does not claim that such a common stationary state is generic. A weaker variant would replace exact stationarity by long-time convergence or approximate stationarity on the shared sector; those relaxations are left open.

Definition 2.14 (JS-C: compatible persistent projections)**.** Let S_{multi} be a multi-source datum, and fix a candidate shared sector $\mathcal{A}_{\text{sh}}^{\text{cand}}$. A pair of Nakajima–Zwanzig projections

$$\mathcal{P}^{(1)}, \mathcal{P}^{(2)} : \mathcal{A}_U \to \mathcal{A}_U$$

is called $\mathcal{A}_{\text{sh}}^{\text{cand}}$-*compatible* if:

1. each $\mathcal{P}^{(i)}$ is a normal idempotent projection used in the Stage A output of the i-th source chain;
2. they commute,
$$\mathcal{P}^{(1)}\mathcal{P}^{(2)} = \mathcal{P}^{(2)}\mathcal{P}^{(1)} =: \mathcal{P}_{\text{sh}};$$
3. $\mathcal{P}_{\text{sh}}(\mathcal{A}_U)$ is a von Neumann subalgebra containing $\mathcal{A}_{\text{sh}}^{\text{cand}}$, and $\mathcal{P}_{\text{sh}}|_{\mathcal{A}_{\text{sh}}^{\text{cand}}} = \text{id}$;
4. the projected predual dynamics induced by the pair admit a common dual memory kernel
$$\mathcal{K}_{\text{sh},*}(t, s)$$
on the shared projected sector, meaning that there exists a single kernel governing the shared projected dynamics on that sector; the definition does not require literal equality of the two individual source kernels before projection to the shared sector.

Remark 2.15 (Status of JS-C). Definition 2.14 is not imported from a standard open-systems textbook. It is a new named compatibility condition introduced for the present paper. The point is not to pretend that a model-independent notion of compatible Nakajima–Zwanzig projections is already standard. The point is to isolate exactly what later proofs need from such a notion. The background projection formalism is still the standard Nakajima–Zwanzig one [11, 12]; what is new here is only the multi-source compatibility layer. A weaker variant would require only that the long-time asymptotic behaviour of the two projected kernels agree on the shared sector, rather than demanding a common kernel at all times. The present paper uses the stronger form because it simplifies the Stage A construction; the weaker variant is left as an open relaxation.

Proposition 2.16 (Commuting expectations supply the algebraic part of JS-C)**.** *Let $E_i : \mathcal{A}_U \to \mathcal{M}_i$, $i = 1, 2$, be normal conditional expectations. If $E_1E_2 = E_2E_1$, then*

$$E_{12} := E_1E_2$$

is a normal conditional expectation onto $\mathcal{M}_1 \cap \mathcal{M}_2$. Consequently, if $\mathcal{A}_{\text{sh}}^{\text{cand}} \subseteq \mathcal{M}_1 \cap \mathcal{M}_2$, clauses (1)–(3) of JS-C hold with $\mathcal{P}^{(i)} = E_i$. Clause (4), existence of a common projected memory kernel, remains an additional dynamical input.

Proof. Commutation and idempotence give $E_{12}^2 = E_1^2 E_2^2 = E_{12}$. Its range is contained in both $\mathcal{M}_1$ and $\mathcal{M}_2$, while every element of the intersection is fixed by both expectations. Thus $\operatorname{Ran} E_{12} = \mathcal{M}_1 \cap \mathcal{M}_2$. Normality, complete positivity, and unitality pass to the composition; the resulting norm-one positive projection is a conditional expectation. This is the commuting-expectation/commuting-square mechanism, not an automatic property of arbitrary Nakajima–Zwanzig projections [14, 15]. □

Proposition 2.17 (Finite-dimensional two-source witness)**.** *There is a finite Type I model with independent source states satisfying JS-A, JS-B, and JS-C and with $k_{\mathrm{sh}} = 1$. In the same model, the two manifest states agree on the shared sector but have nonzero discrepancy elsewhere in their overlap.*

Proof. Let

$$\mathcal{A}_U = M_2(S) \otimes M_2(P_1) \otimes M_2(P_2)$$

in its tracial standard form, and let Z_X be Pauli Z on factor X. Write

$$D_X(A) := \tfrac{1}{2}(A + Z_X A Z_X)$$

for the dephasing expectation on factor X, tensored with the identity on the other factors. Set

$$\mathcal{P}^{(1)} = D_S D_{P_1}, \qquad \mathcal{P}^{(2)} = D_S D_{P_2}.$$

These normal UCP conditional expectations commute, and $\mathcal{P}_{\mathrm{sh}} = D_S D_{P_1} D_{P_2}$ has the full diagonal algebra as its range. For $\lambda_i > 0$, define

$$\mathcal{E}_t^{(i)} = e^{-\lambda_i t}\,\mathrm{id} + (1 - e^{-\lambda_i t})\mathcal{P}^{(i)}.$$

Idempotence of $\mathcal{P}^{(i)}$ gives the semigroup law; each map is normal, UCP, and the two semigroups commute.

Take

$$\mathcal{A}_{\mathrm{sh}}^{\mathrm{cand}} = W^*(Z_S) \otimes \mathbf{1} \otimes \mathbf{1}, \qquad \mathcal{A}_c^{(1)} = \mathcal{A}_c^{(2)} = W^*(Z_S, Z_{P_1}, Z_{P_2}).$$

The reference tracial modular flow is trivial, and both semigroups fix the candidate shared sector pointwise, proving JS-A. Let $\rho(x) := \frac{1}{2}(\mathbf{1} + xZ)$, with $|x| < 1$, and choose the independent faithful ambient states

$$\varpi_1 = \rho(1/3) \otimes \rho(1/4) \otimes \rho(1/5), \qquad \varpi_2 = \rho(1/3) \otimes \rho(-1/4) \otimes \rho(-1/5).$$

Their manifest restrictions agree faithfully on the shared sector, with common state induced by $\rho(1/3)$, and that state is stationary there. This proves JS-B. Proposition 2.16 proves the algebraic clauses of JS-C. On the pointwise-fixed shared sector the projected generator vanishes, so the common projected kernel may be taken as $\mathcal{K}_{\mathrm{sh},*} = 0$, proving clause (4).

Finally choose foreground images

$$V_{\mathrm{fg}}^{(1)} = \operatorname{span}\{Z_S, Z_{P_1}\}, \qquad V_{\mathrm{fg}}^{(2)} = \operatorname{span}\{Z_S, Z_{P_2}\}.$$

Their intersection is $\operatorname{span}\{Z_S\}$, hence $k_{\mathrm{sh}} = 1$. Yet $\omega_1(Z_{P_1}) = 1/4$ and $\omega_2(Z_{P_1}) = -1/4$, so the discrepancy is nonzero on the larger overlap algebra. This witness verifies compatibility in FD/I; its zero shared kernel and tracial modular flow are not a Type III realization of the full Foundation programme. □

3 The Two-Source Shared Manifest Sector

We now specialize to $N = 2$. The paper's main theorem is entirely two-source.

3.1 Joint source interface

Let

$$S^{(2)}_{\text{multi}} = (\mathcal{H}_U, \mathcal{A}_U, \Omega, \{\varpi_1, \varpi_2\}, \{\mathcal{E}^{(1)}_t, \mathcal{E}^{(2)}_t\})$$

be a two-source datum. Assume that each source separately satisfies the named Foundation I hypotheses required to produce a Stage A output, Stage B output, and manifest pair

$$(\mathcal{A}^{(i)}_c, \omega_i), \qquad i = 1, 2.$$

Lemma 3.1 (Joint Stage A output from compatible persistent projections)**.** *Let $S^{(2)}_{\text{multi}}$ be a two-source datum. Assume there exists a candidate shared sector $\mathcal{A}^{\text{cand}}_{\text{sh}} \subseteq \mathcal{A}^{(1)}_c \cap \mathcal{A}^{(2)}_c$ satisfying JS-C. Then the commuting product*

$$\mathcal{P}_{\text{sh}} := \mathcal{P}^{(1)}\mathcal{P}^{(2)}$$

defines a shared projected record object

$$\mathcal{O}^{\text{sh}}_A := (\mathcal{P}_{\text{sh}}, \omega_{\text{sh},t}, \mathcal{K}_{\text{sh},*}(t, s)),$$

where $\omega_{\text{sh},t}$ is the projected normal state on the shared sector and $\mathcal{K}_{\text{sh},*}(t, s)$ is the common dual memory kernel from Definition 2.14.

Proof. Commutativity of $\mathcal{P}^{(1)}$ and $\mathcal{P}^{(2)}$ implies that $\mathcal{P}_{\text{sh}}$ is again an idempotent projection. Because both $\mathcal{P}^{(1)}$ and $\mathcal{P}^{(2)}$ are normal and they commute, their composition is again normal. By JS-C, $\mathcal{P}_{\text{sh}}(\mathcal{A}_U)$ is a von Neumann subalgebra containing $\mathcal{A}^{\text{cand}}_{\text{sh}}$, and the common dual kernel $\mathcal{K}_{\text{sh},*}(t, s)$ is part of the named input. This is exactly the shared Stage A object required for later stages. □

3.2 Joint foreground overlap

For each source $i = 1, 2$, let $\mathfrak{F}^{(i)}$ denote the foreground frame produced by the conditional Foundation I finite-coordinate compression interface, with foreground realization

$$\iota_i : V_{\text{fg}}(\mathfrak{F}^{(i)}) \hookrightarrow \mathcal{A}^{\text{sa}}_U.$$

Definition 3.2 (Shared foreground overlap)**.** The *shared foreground overlap* is the operator-side intersection

$$V_{\text{fg}}(\mathfrak{F}_{\text{sh}}) := \iota_1\big(V_{\text{fg}}(\mathfrak{F}^{(1)})\big) \cap \iota_2\big(V_{\text{fg}}(\mathfrak{F}^{(2)})\big) \subseteq \mathcal{A}^{\text{sa}}_U.$$

Whenever this intersection is non-zero and finite-dimensional, we choose a basis

$$O^{\text{sh}}_1, \ldots, O^{\text{sh}}_{k_{\text{sh}}} \in V_{\text{fg}}(\mathfrak{F}_{\text{sh}})$$

and write

$$\mathcal{A}_{\mathfrak{F}_{\text{sh}}} := W^*(1, O^{\text{sh}}_1, \ldots, O^{\text{sh}}_{k_{\text{sh}}}).$$

Proposition 3.3 (Dimension bound for the shared foreground)**.** *Assume $\mathfrak{F}^{(1)}$ and $\mathfrak{F}^{(2)}$ have effective dimensions k_1^* and k_2^*, respectively, and that the shared foreground overlap of Definition 3.2 is finite dimensional. Then*

$$k_{\text{sh}} \leq \min\{k_1^*, k_2^*\}.$$

Proof. The shared foreground overlap is a subspace of each individual foreground image. Therefore its dimension cannot exceed the dimension of either image, hence cannot exceed the minimum of the two. □

Remark 3.4. Proposition 3.3 is only a bookkeeping result. It does not prove that the overlap is non-trivial. Non-triviality of the shared foreground remains a separate conditional input.

3.3 Shared stable hull

Write

$$\omega_0(\cdot) = \langle \Omega, \cdot\, \Omega \rangle$$

on the ambient algebra $\mathcal{A}_U$, and let $\sigma_s^{\omega_0,\text{tot}}$ be its ambient modular flow.

Definition 3.5 (Shared stable-carrier class)**.** Let $\mathfrak{F}_{\text{sh}}$ be a non-trivial shared foreground overlap. Define

$$\mathfrak{L}_{\text{sh}}^{\text{st}} := \left\{ \begin{array}{l} \mathcal{M} \subseteq \mathcal{A}_U : \mathcal{M} \text{ is a von Neumann subalgebra,} \\ \qquad \mathcal{A}_{\mathfrak{F}_{\text{sh}}} \subseteq \mathcal{M}, \\ \qquad \sigma_s^{\omega_0,\text{tot}}(\mathcal{M}) = \mathcal{M} \quad \forall s \in \mathbb{R}, \\ \qquad \mathcal{E}_t^{(i)}(\mathcal{M}) \subseteq \mathcal{M} \quad \forall t \geq 0,\ i = 1, 2 \end{array} \right\}.$$

Definition 3.6 (Shared stable hull)**.** Assume $\mathfrak{L}_{\text{sh}}^{\text{st}} \neq \varnothing$. The *shared stable hull* of the shared foreground is

$$\Phi_{\text{sh}}^{\text{st}}(\mathfrak{F}_{\text{sh}}) := \bigcap_{\mathcal{M} \in \mathfrak{L}_{\text{sh}}^{\text{st}}} \mathcal{M}.$$

Proposition 3.7 (Minimal shared stable hull)**.** *Assume $\mathfrak{L}_{\text{sh}}^{\text{st}} \neq \varnothing$. Then*

$$\Phi_{\text{sh}}^{\text{st}}(\mathfrak{F}_{\text{sh}})$$

is a well-defined von Neumann subalgebra of $\mathcal{A}_U$ satisfying:

1. $\mathcal{A}_{\mathfrak{F}_{\text{sh}}} \subseteq \Phi_{\text{sh}}^{\text{st}}(\mathfrak{F}_{\text{sh}})$;
2. $$\sigma_s^{\omega_0,\text{tot}}\big(\Phi_{\text{sh}}^{\text{st}}(\mathfrak{F}_{\text{sh}})\big) = \Phi_{\text{sh}}^{\text{st}}(\mathfrak{F}_{\text{sh}}) \qquad \forall s \in \mathbb{R};$$
3. $$\mathcal{E}_t^{(i)}\big(\Phi_{\text{sh}}^{\text{st}}(\mathfrak{F}_{\text{sh}})\big) \subseteq \Phi_{\text{sh}}^{\text{st}}(\mathfrak{F}_{\text{sh}}) \qquad \forall t \geq 0,\ i = 1, 2;$$
4. *it is minimal with these properties: if $\mathcal{N} \subseteq \mathcal{A}_U$ is any von Neumann subalgebra containing $\mathcal{A}_{\mathfrak{F}_{\text{sh}}}$ and stable under the ambient modular flow and both semigroups, then*

$$\Phi_{\text{sh}}^{\text{st}}(\mathfrak{F}_{\text{sh}}) \subseteq \mathcal{N}.$$

Proof. The argument is identical to the stable-hull construction in Foundation I [1], with the only change that one now imposes stability under both source semigroups. An arbitrary intersection of von Neumann subalgebras is again a von Neumann subalgebra: intersections of σ-weakly closed $*$-subalgebras remain σ-weakly closed $*$-subalgebras. Because every member of $\mathfrak{L}^{\mathrm{st}}_{\mathrm{sh}}$ contains $\mathcal{A}_{\mathfrak{F}_{\mathrm{sh}}}$, so does the intersection. Stability under the ambient modular flow and both semigroups is preserved by intersection: if $A \in \bigcap_\alpha \mathcal{M}_\alpha$, then $\mathcal{E}_t^{(i)}(A) \in \mathcal{M}_\alpha$ for every α, t, and i, and similarly for the modular flow, hence the image again lies in $\bigcap_\alpha \mathcal{M}_\alpha$. Minimality is immediate from the definition as the intersection of all subalgebras with the required properties. □

3.4 Main theorem

Theorem 3.8 (Two-source shared admissible manifest sector)**.** *Let $S^{(2)}_{\mathrm{multi}}$ be a two-source datum. Assume:*

1. ***Individual source closure.*** *Each source separately satisfies the named Foundation I hypotheses needed to produce*

$$(\mathcal{A}_c^{(1)}, \omega_1), \qquad (\mathcal{A}_c^{(2)}, \omega_2),$$

   together with foreground frames $\mathfrak{F}^{(1)}$ and $\mathfrak{F}^{(2)}$.

2. ***Joint-stability package.*** *There exists a candidate shared sector*

$$\mathcal{A}^{\mathrm{cand}}_{\mathrm{sh}} \subseteq \mathcal{A}_c^{(1)} \cap \mathcal{A}_c^{(2)}$$

   satisfying JS-A, JS-B, and JS-C.

3. ***Non-trivial shared foreground.*** *The shared foreground overlap $\mathfrak{F}_{\mathrm{sh}}$ is non-zero and admits a realization by exact shared pointer-aligned observables $O^{\mathrm{sh}}_1, \dots, O^{\mathrm{sh}}_{k_{\mathrm{sh}}}$ lying in $\mathcal{A}^{\mathrm{cand}}_{\mathrm{sh}}$.*

4. ***Admissibility input for Paper I.*** *The pair*

$$\left(\Phi^{\mathrm{st}}_{\mathrm{sh}}(\mathfrak{F}_{\mathrm{sh}}), \omega_{\mathrm{sh}}|_{\Phi^{\mathrm{st}}_{\mathrm{sh}}(\mathfrak{F}_{\mathrm{sh}})}\right)$$

   satisfies the admissibility conditions required by Paper I [2].

Then:

1. *the shared stable hull is well-defined and satisfies*

$$\Phi^{\mathrm{st}}_{\mathrm{sh}}(\mathfrak{F}_{\mathrm{sh}}) \subseteq \mathcal{A}^{\mathrm{cand}}_{\mathrm{sh}} \subseteq \mathcal{A}_c^{(1)} \cap \mathcal{A}_c^{(2)};$$

2. *the restricted state*

$$\omega^{\mathrm{st}}_{\mathrm{sh}} := \omega_{\mathrm{sh}}|_{\Phi^{\mathrm{st}}_{\mathrm{sh}}(\mathfrak{F}_{\mathrm{sh}})}$$

   is faithful on $\Phi^{\mathrm{st}}_{\mathrm{sh}}(\mathfrak{F}_{\mathrm{sh}})$;

3. *the pair*

$$\left(\Phi^{\mathrm{st}}_{\mathrm{sh}}(\mathfrak{F}_{\mathrm{sh}}), \omega^{\mathrm{st}}_{\mathrm{sh}}\right)$$

   is a shared admissible manifest sector;

4. therefore the Paper I inseparability theorem applies to that pair.

Proof. By assumption (3), the exact shared pointer-aligned observables $O_1^{\mathrm{sh}}, \dots, O_{k_{\mathrm{sh}}}^{\mathrm{sh}}$ lie in $\mathcal{A}_{\mathrm{sh}}^{\mathrm{cand}}$. Since $\mathcal{A}_{\mathrm{sh}}^{\mathrm{cand}}$ is a von Neumann subalgebra by Definition 2.4, it contains the generated shared foreground algebra:

$$\mathcal{A}_{\mathfrak{F}_{\mathrm{sh}}} \subseteq \mathcal{A}_{\mathrm{sh}}^{\mathrm{cand}}.$$

By assumption (2), the candidate shared sector satisfies JS-A, hence it is invariant under the ambient modular flow and under both source semigroups. Therefore

$$\mathcal{A}_{\mathrm{sh}}^{\mathrm{cand}} \in \mathfrak{L}_{\mathrm{sh}}^{\mathrm{st}}.$$

In particular, the class $\mathfrak{L}_{\mathrm{sh}}^{\mathrm{st}}$ is nonempty. Hence Definition 3.6 and Proposition 3.7 apply, so the shared stable hull is a well-defined von Neumann subalgebra of $\mathcal{A}_U$.

Because $\mathcal{A}_{\mathrm{sh}}^{\mathrm{cand}}$ is a member of the shared stable-carrier class, minimality gives

$$\Phi_{\mathrm{sh}}^{\mathrm{st}}(\mathfrak{F}_{\mathrm{sh}}) \subseteq \mathcal{A}_{\mathrm{sh}}^{\mathrm{cand}}.$$

By assumption (2), the candidate shared sector is already contained in $\mathcal{A}_c^{(1)} \cap \mathcal{A}_c^{(2)}$, so

$$\Phi_{\mathrm{sh}}^{\mathrm{st}}(\mathfrak{F}_{\mathrm{sh}}) \subseteq \mathcal{A}_c^{(1)} \cap \mathcal{A}_c^{(2)}.$$

This proves part (1).

By JS-B, ω_{sh} is faithful on $\mathcal{A}_{\mathrm{sh}}^{\mathrm{cand}}$. The restriction of a faithful normal state to a von Neumann subalgebra is again faithful. Therefore

$$\omega_{\mathrm{sh}}^{\mathrm{st}} = \omega_{\mathrm{sh}}|_{\Phi_{\mathrm{sh}}^{\mathrm{st}}(\mathfrak{F}_{\mathrm{sh}})}$$

is faithful on the shared stable hull. This proves part (2).

Parts (1) and (2), together with the stability properties from Proposition 3.7, show that the pair

$$\left(\Phi_{\mathrm{sh}}^{\mathrm{st}}(\mathfrak{F}_{\mathrm{sh}}), \omega_{\mathrm{sh}}^{\mathrm{st}}\right)$$

is a shared manifest sector in the sense intended by the present paper. Assumption (4) is the external admissibility input that upgrades it to a shared *admissible* manifest sector, so Paper I applies. This proves parts (3) and (4). □

Corollary 3.9 (Shared inseparability)**.** *Under the hypotheses of Theorem 3.8, geometry and modular flow on the shared sector*

$$\left(\Phi_{\mathrm{sh}}^{\mathrm{st}}(\mathfrak{F}_{\mathrm{sh}}), \omega_{\mathrm{sh}}^{\mathrm{st}}\right)$$

are not independently deformable in the sense of Paper I. In particular, any geometric reading $\mathrm{Geom}(\Phi_{\mathrm{sh}}^{\mathrm{st}}(\mathfrak{F}_{\mathrm{sh}}), \omega_{\mathrm{sh}}^{\mathrm{st}})$ *defined within the Paper I theorem domain is a* shared *geometric sector.*

Proof. This is exactly the conclusion of the Paper I theorem applied to the pair isolated by Theorem 3.8. □

Remark 3.10 (What the main theorem does *not* prove). Theorem 3.8 does not prove a general shared-world theorem. It proves something narrower: if a two-source datum already admits a candidate shared sector satisfying the named compatibility and admissibility inputs, then the overlap of the two source chains carries an honest theorem-level manifest sector to which the Paper I machinery may be applied. It also does not derive the candidate shared sector from multi-source entanglement or source dynamics alone. This is a foothold, not the full Bridge 2 closure.

Remark 3.11 (Where the theorem's new content lies). The genuine new content of Theorem 3.8 is the conditional inclusion chain

$$\text{shared foreground inclusion} \;+\; \text{JS-A} \Longrightarrow \mathcal{A}_{\text{sh}}^{\text{cand}} \in \mathfrak{L}_{\text{sh}}^{\text{st}} \Longrightarrow \Phi_{\text{sh}}^{\text{st}}(\mathfrak{F}_{\text{sh}}) \subseteq \mathcal{A}_c^{(1)} \cap \mathcal{A}_c^{(2)}.$$

The Paper I admissibility package is not derived here. It is a named external input specifying when the isolated shared stable hull lies in the theorem domain of Paper I.

4 Common/Private Decomposition

The main theorem isolates a shared admissible manifest sector. We now ask how each observer's full manifest algebra relates to that shared sector. In the Type III_1 setting, one should not treat this as a canonical factorization problem. The correct formulation is regime-dependent and uses a conditional expectation language.

Definition 4.1 (Shared/private regime assumption)**.** Let

$$\mathcal{A}_{\text{sh}} \subseteq \mathcal{A}_c^{(i)} \subseteq \mathcal{A}_U, \qquad i = 1, 2.$$

We say that the pair $(\mathcal{A}_c^{(i)}, \mathcal{A}_{\text{sh}})$ satisfies the *shared/private regime assumption* if for each i there exist:

1. a von Neumann subalgebra $\mathcal{A}_{\text{priv}}^{(i)} \subseteq \mathcal{A}_c^{(i)}$;
2. a faithful normal conditional expectation

   $$E_{\text{sh}}^{(i)} : \mathcal{A}_c^{(i)} \to \mathcal{A}_{\text{sh}}$$

   preserving the individual manifest state in the sense that

   $$\omega_i \circ E_{\text{sh}}^{(i)} = \omega_i;$$

3. a generating relation

   $$\mathcal{A}_c^{(i)} = W^*\big(\mathcal{A}_{\text{sh}} \cup \mathcal{A}_{\text{priv}}^{(i)}\big).$$

Remark 4.2. Definition 4.1 does not assert uniqueness of the private complement, and it does not assert a tensor-product or free-product factorization. It only says that, under a regime assumption, the full manifest algebra can be generated from the shared sector together with a private complement, with a faithful conditional expectation projecting back to the shared sector.

Remark 4.3 (Minimal package for common/private language). Definition 4.1 is the minimal package needed before common/private language can even be used honestly in the present paper. Without a private complement, a faithful conditional expectation onto the shared sector, and a generating relation for the full manifest algebra, the phrase "private part" remains interpretive rather than mathematical.

Proposition 4.4 (Unpacking JS-B and localizing overlap discrepancy)**.** *Assume the hypotheses of Theorem 3.8, and suppose the shared/private regime assumption holds with*

$$\mathcal{A}_{\text{sh}} := \Phi_{\text{sh}}^{\text{st}}(\mathfrak{F}_{\text{sh}}).$$

Let

$$\mathcal{A}_{\text{ov}} := \mathcal{A}_c^{(1)} \cap \mathcal{A}_c^{(2)}.$$

Define the overlap discrepancy functional

$$\delta_{12} := \omega_1|_{\mathcal{A}_{\text{ov}}} - \omega_2|_{\mathcal{A}_{\text{ov}}} \in (\mathcal{A}_{\text{ov}})_*.$$

Then:

1. *the two restricted manifest states agree on the shared sector:*

$$\omega_1|_{\mathcal{A}_{\text{sh}}} = \omega_2|_{\mathcal{A}_{\text{sh}}} = \omega_{\text{sh}}^{\text{st}}|_{\mathcal{A}_{\text{sh}}};$$

2. *equivalently,*

$$\delta_{12}|_{\mathcal{A}_{\text{sh}}} = 0;$$

3. *if $A \in \mathcal{A}_{\text{ov}}$ satisfies $\delta_{12}(A) \neq 0$, then $A \notin \mathcal{A}_{\text{sh}}$;*

Proof. For $A \in \mathcal{A}_{\text{sh}}$, JS-B clause (2), restricted from the candidate shared sector to the shared stable hull, gives

$$\omega_1(A) = \omega_{\text{sh}}^{\text{st}}(A) = \omega_2(A).$$

Therefore

$$\delta_{12}(A) = 0 \qquad \forall A \in \mathcal{A}_{\text{sh}}.$$

This proves the first two claims. The third is only the contrapositive of the second; it is not a derived theorem that independent states must agree on the whole overlap. □

Remark 4.5 (Logical status of Proposition 4.4). The agreement statement is an unpacking of the explicit JS-B assumption. The proposition's useful content is bookkeeping: once independent states are part of the datum, it identifies where a nonzero overlap discrepancy is compatible with that assumption. Proposition 2.17 shows that this nonzero case is not vacuous.

Corollary 4.6 (Shared manifest content)**.** *Under the hypotheses of Proposition 4.4, every observable in the shared sector carries common manifest content. Observer-dependent discrepancy, insofar as it is jointly accessible on $\mathcal{A}_{\text{ov}}$, can only appear outside the shared sector.*

Proof. Proposition 4.4 shows that the two restricted manifest states agree on the shared sector and that any jointly accessible disagreement must be detected off that sector. That is the formal meaning of "shared manifest content" used in the present paper. □

Remark 4.7 (Shared/private reading of Proposition 4.4). Under the shared/private regime assumption, Proposition 4.4 may be read as saying that any overlap-level disagreement is detected only outside the common manifest sector, i.e. on the observer-dependent side of the decomposition language introduced in this section.

Remark 4.8 (What Section 4 does *not* prove). Section 4 does not prove existence of a canonical private factor, nor does it derive the shared/private regime assumption from the bare two-source datum. Its role is narrower: once a shared manifest sector has been isolated and a regime assumption is granted, it makes precise where observer agreement must hold and where observer-dependent discrepancy may still appear.

Remark 4.9 (No canonical factorization claim). Corollary 4.6 should not be read as a theorem that

$$\mathcal{A}_c^{(i)} \cong \mathcal{A}_{\text{sh}} \bar{\otimes} \mathcal{A}_{\text{priv}}^{(i)}$$

or any analogous canonical factorization holds. The paper makes no such claim. It uses only the weaker and safer statement that, under a regime assumption, a shared sector and a private complement jointly generate the full manifest algebra.

5 Conditional N-Source Extension

The two-source theorem is the paper's main result. The natural next step is a conditional N-source extension. This section records the generalization in the same honest style as Foundation I: whatever can be internalized immediately is internalized, and the genuinely new input is kept explicit.

Definition 5.1 (Joint N-source Stage A compatibility)**.** Let

$$S_{\text{multi}} = (\mathcal{H}_U, \mathcal{A}_U, \Omega, \{\varpi_i\}_{i=1}^N, \{\mathcal{E}_t^{(i)}\}_{i=1}^N)$$

be a multi-source datum with individual manifest outputs $(\mathcal{A}_c^{(i)}, \omega_i)_{i=1}^N$. A family of Stage A projections $\{\mathcal{P}^{(i)}\}_{i=1}^N$ is jointly compatible with $\mathcal{A}_{\text{sh}}^{\text{cand}}$ if:

1. the normal idempotents $\mathcal{P}^{(i)}$ commute pairwise;
2. the joint product
$$\mathcal{P}_{\text{sh}}^{(N)} := \prod_{i=1}^N \mathcal{P}^{(i)}$$
has von Neumann-algebra range containing $\mathcal{A}_{\text{sh}}^{\text{cand}}$ and fixes that candidate sector pointwise;
3. the jointly projected predual dynamics admits one common kernel $\mathcal{K}_{\text{sh},*,N}(t,s)$ on the joint range.

Lemma 5.2 (Joint N-source Stage A object)**.** *Under Definition 5.1, $\mathcal{P}_{\text{sh}}^{(N)}$ is a normal idempotent and, together with the stipulated common kernel, defines a joint Stage A object.*

Proof. Pairwise commutation makes the finite product independent of order and gives

$$(\mathcal{P}_{\text{sh}}^{(N)})^2 = \prod_{i=1}^N (\mathcal{P}^{(i)})^2 = \mathcal{P}_{\text{sh}}^{(N)}.$$

Finite composition preserves normality. The range and common-kernel requirements are explicit clauses of Definition 5.1, so the resulting tuple has the required Stage A data. Pairwise two-source theorems alone would not supply this one joint kernel or guarantee that all pairwise overlap choices define the same range. □

Theorem 5.3 (Conditional N-source shared manifest sector)**.** *Let S_{multi} be an N-source datum with individual manifest outputs $(\mathcal{A}_c^{(i)}, \omega_i)_{i=1}^N$. Assume:*

1. *there exists a candidate shared sector*
$$\mathcal{A}_{\text{sh}}^{\text{cand}} \subseteq \bigcap_{i=1}^N \mathcal{A}_c^{(i)}$$
satisfying JS-A and JS-B for all sources;
2. *the joint Stage A compatibility condition of Definition 5.1 holds, and a compatible family of shared foreground overlaps generates one common foreground algebra $\mathcal{A}_{\mathfrak{F}_{\text{sh}}}$;*

3. *the N-source stable-carrier class*

$$\mathfrak{L}^{\mathrm{st}}_{\mathrm{sh},N} := \left\{ \mathcal{M} \subseteq \mathcal{A}_U : \begin{array}{l} \mathcal{M} \text{ is a von Neumann subalgebra}, \\ \mathcal{A}_{\mathfrak{F}_{\mathrm{sh}}} \subseteq \mathcal{M}, \\ \sigma_s^{\omega_0,\mathrm{tot}}(\mathcal{M}) = \mathcal{M},\ \forall s, \\ \mathcal{E}_t^{(i)}(\mathcal{M}) \subseteq \mathcal{M},\ \forall t, \forall i \end{array} \right\}$$

is nonempty and contains $\mathcal{A}^{\mathrm{cand}}_{\mathrm{sh}}$;

4. *the resulting N-source shared stable hull satisfies the admissibility hypotheses needed by Paper I.*

Then the N-source shared stable hull is a shared admissible manifest sector contained in $\bigcap_{i=1}^{N} \mathcal{A}_c^{(i)}$.

Proof. Lemma 5.2 supplies the joint Stage A object that pairwise compatibility alone does not provide. The remaining proof is the same minimality argument as in the two-source case, with the only change that one now intersects over all source labels rather than over two. The candidate shared sector is, by assumption, a member of the N-source stable-carrier class. Therefore the intersection defining the N-source shared stable hull is well-defined and lies inside the candidate sector. The remaining steps repeat the proof of Theorem 3.8. □

Remark 5.4 (What Theorem 5.3 does not contain). Theorem 5.3 does not provide:

1. a uniqueness or maximality theorem for the N-source shared sector;
2. a quantitative relationship between the number/strength of sources and geometric stability;
3. a proof that the joint Stage A and stable-carrier conditions of Definition 5.1 follow from the bare multi-source datum.

All three belong to later work.

6 Open Problems

The conditional inputs of Foundation II are not cosmetic. They isolate the genuinely unfinished parts of the programme.

Heuristically, the present two-source theorem is closer to a mean-field foothold than to a full correlated treatment: it isolates a shared effective manifest sector in a restricted regime, while genuinely cross-source structure remains the central open problem for any future multi-source closure.

1. **Model-independent joint stability.** The present paper names JS-A, JS-B, and JS-C. It does not derive them model-independently from a bare multi-source datum. The analogue of Foundation I's observer-emergence bottleneck therefore persists in multi-source form.
2. **Compatible Nakajima–Zwanzig projections.** The present notion of compatible persistent projections is a new definition, not a standard theorem. A general existence theory for such compatible projections remains open.

3. **Shared-foreground non-triviality.** The paper proves that if a non-trivial shared foreground overlap exists, then its dimension is bounded by the individual foreground dimensions. Proposition 2.17 gives one FD/I example with nonzero overlap, but no Type III or model-independent theorem forces the overlap to be nonzero.

4. **Quantitative consensus versus stability.** The paper deliberately avoids a bound of the form

$$\text{consensus strength} \Longrightarrow \text{geometric fluctuation bound}.$$

That is the separate Bridge 3 problem.

5. N**-source non-conditional closure.** The present N-source theorem is conditional. A truly non-conditional closure would require deriving a joint N-fold Stage A projection, common kernel, and common foreground from source-side dynamics.

6. **Uniqueness and maximality of the shared sector.** Even in the two-source case, the present theorem proves only existence of a minimal shared stable hull inside a candidate shared carrier. It does not prove that this sector is unique among all plausible shared manifest sectors, or maximal in any global sense.

7. **Relation to Paper II.** Paper II constrains higher-order geometry by multi-directional consistency [3]. The present paper does not yet prove how the shared manifest sector of Foundation II interacts with the Paper II small-ball remainder structure. That interface remains open.

8. **Type III and infinite-dimensional obstacles.** The common/private regime assumption is not automatic in the Type III setting. Conditional expectations, complements, and generating decompositions all become more delicate in infinite dimensions. In particular, existence of a faithful normal conditional expectation onto a candidate shared sector typically interacts with modular invariance in the sense of Takesaki and may fail outside suitable invariant regimes. Even when such expectations exist, complements and generating relations need not be unique, and non-factorial or center-bearing sectors may force a central decomposition before any shared/private language becomes well behaved.

7 Conditional Input Ledger

This section records the named inputs that remain conditional in the present paper.

1. **I0: Independent source states.** The faithful normal states ϖ_i are independent datum, and $\omega_i = \varpi_i|_{\mathcal{A}_c^{(i)}}$. They are not automatic restrictions of the reference vector state ω_0.

2. **Independent-state rebuild after retraction.** The former single-ambient-state reading is withdrawn because it made discrepancy vanish on the entire overlap. The current conditional theorem uses independent ϖ_i, and shared agreement enters explicitly as JS-B rather than as a derived fact.

3. **I1: Individual source closure.** Each source separately satisfies the named Foundation I hypotheses needed to produce an individual manifest pair $(\mathcal{A}_c^{(i)}, \omega_i)$.

4. **JS-A: Shared-sector invariance and semigroup compatibility.** The candidate shared sector is stable under all relevant semigroups and under the ambient modular flow, and the restricted semigroups commute there.

5. **JS-B: Shared-state existence and faithfulness.** A common faithful normal state exists on the candidate shared sector and is compatible with the source-specific manifest states.

6. **JS-C: Compatible persistent projections.** The pair of source-side Nakajima–Zwanzig projections commute and admit a common projected kernel on the shared sector. Commuting conditional expectations supply its algebraic clauses, but not the kernel clause.

7. **Finite compatibility witness.** Proposition 2.17 verifies JS-A/B/C and $k_{\text{sh}} = 1$ in a finite dephasing model with independent states. It is a model check, not a Type III emergence theorem.

8. **F1: Non-trivial shared foreground with exact pointer realization.** The overlap of the two source-side foreground realizations is non-zero, finite dimensional, and admits an exact pointer-aligned realization inside the candidate shared sector.

9. **A1: Admissibility input for Paper I.** The shared stable hull satisfies the admissibility package needed for the geometry–modular-flow inseparability theorem.

10. **SP1: Shared/private regime assumption.** Each individual manifest algebra admits a faithful conditional expectation onto the shared sector together with a private complement that generates the full algebra jointly with the shared sector.

11. **N1: joint N-source Stage A condition.** One jointly commuting product projection, a common range, and one common projected kernel are required. Pairwise two-source compatibility is insufficient by itself. No unused global consensus functional is retained.

A Imported Interfaces

This appendix records, in theorem-ready form, the external interfaces imported from Genesis, Foundation I, and the Consciousness Trilogy. They are not reproved here, but they are restated explicitly so that the present paper can be read without reconstructing its dependencies from the earlier series papers. What is *not* imported should also be kept clear: the multi-source notions of candidate shared sector, JS-A, JS-B, JS-C, shared stable hull, and shared/private regime are defined internally in Foundation II itself.

Proposition A.1 (Foundation I single-source closure package)**.** *Foundation I [1] isolates a conditional chain*

$$S_{\text{source}} \to \mathcal{O}_A \to \mathcal{O}_B \to (\mathcal{A}_c, \omega),$$

where:

1. *Stage A supplies a projected record object*

$$\mathcal{O}_A = (\mathcal{P}, \omega_t, \mathcal{K}_*(t,s)),$$

where $\mathcal{P} : \mathcal{A}_U \to \mathcal{A}_U$ is a normal idempotent projection and $\mathcal{K}_*(t,s)$ is a dual non-Markovian memory kernel;

2. *Stage B supplies a foreground frame $\mathfrak{F}$ with a finite-dimensional foreground space $V_{\text{fg}}(\mathfrak{F})$ and an operator realization*

$$\iota : V_{\text{fg}}(\mathfrak{F}) \hookrightarrow \mathcal{A}_U^{\text{sa}};$$

3. *Stage C supplies a manifest pair*

$$(\mathcal{A}_c, \omega) = \big(\Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F}), \omega\big),$$

where $\Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F}) \subseteq \mathcal{A}_U$ is a minimal modular-decoherence stable hull, hence a von Neumann subalgebra carrying a faithful normal restricted state;

4. *in the exact fixed-point comparison regime imported from Foundation I, one has the inclusion*

$$\Phi^{\mathrm{st}}_{\mathfrak{F}}(\mathfrak{F}) \subseteq \mathcal{A}_c^{\mathrm{HAFF}}.$$

(not used in the present paper; recorded for later reference).

Proposition A.2 (Imported single-source interfaces used in Foundation II)**.** *Whenever Foundation II invokes the i-th source chain, the only external single-source objects it uses are:*

1. *a Stage A projection $\mathcal{P}^{(i)}$, projected normal states $\omega_t^{(i)}$, and dual memory kernel $\mathcal{K}_{*,i}(t,s)$;*
2. *a Stage B foreground frame $\mathfrak{F}^{(i)}$ with realized finite-dimensional foreground image*

$$\iota_i\big(V_{\mathrm{fg}}(\mathfrak{F}^{(i)})\big) \subseteq \mathcal{A}_U^{\mathrm{sa}};$$

3. *a Stage C manifest pair*

$$(\mathcal{A}_c^{(i)}, \omega_i) = \big(\Phi^{\mathrm{st}}_{\mathfrak{F}^{(i)}}(\mathfrak{F}^{(i)}), \omega_i\big).$$

No multi-source compatibility theorem is imported at this stage. Foundation II introduces the multi-source compatibility layer itself.

Proposition A.3 (Paper I manifest inseparability theorem)**.** *Paper I [2] proves that for an admissible manifest pair $(\mathcal{A}_c, \omega)$, geometry and modular flow are not independently deformable. In particular, once a pair $(\mathcal{A}_c, \omega)$ in the theorem domain is given, the geometry and modular sectors should be treated as two constrained readings of one underlying algebraic datum. Foundation II uses this result only through its admissibility interface: whenever a shared stable hull is shown, by named external input, to belong to the Paper I theorem domain, the inseparability conclusion becomes available.*

Proposition A.4 (Paper II as a consistency constraint)**.** *Paper II [3] supplies a conditional CHM small-ball stress-tensor extraction. Its stronger Einstein-tensor identification requires additional two-derivative and transversality hypotheses and is not imported here. Foundation II uses the paper only as boundary calibration, not as a formal input to the shared-sector existence proof.*

Proposition A.5 (T-DOME source-side ingredients)**.** *The T-DOME papers motivate three single-source interfaces used upstream of the present paper:*

1. *T-DOME I motivates the Stage A projected record process with kernel $\mathcal{K}_*(t,s)$, but does not prove that non-Markovian memory is necessary for persistence [7, 10];*
2. *T-DOME II motivates finite-coordinate foreground compression. Rank saturation and symmetry breaking require the explicit model assumptions stated in Foundation I; they are not universal necessity theorems [8];*

3. *T-DOME III: self-calibration enters through a Fisher-information prior carried by the foreground structure as a specified model input, but no direct theorem from T-DOME III is used in the operator-algebraic proofs below beyond this source-side calibration role [9].*

Proposition A.6 (HAFF ambient setting)**.** *The HAFF programme supplies the ambient structural setting used here:*

1. *geometry is read from accessible algebraic structure rather than supplied as a background container [5];*

2. *accessible algebras are selected by stability under the relevant dynamics rather than by subjective choice [6];*

3. *in the exact fixed-point regime used for Foundation I and imported here, the HAFF accessible algebra*

$$\mathcal{A}_c^{\mathrm{HAFF}} \subseteq \mathcal{A}_U$$

   is a von Neumann subalgebra stable under the relevant semigroup and the ambient modular flow;

4. *fixed-point and stability ideas for completely positive semigroups motivate the use of stable carrier classes and exact pointer sectors in restricted regimes [16, 17].*

Remark A.7 (What earlier papers do and do not supply). The earlier Genesis and Consciousness papers supply only single-source objects and single-source comparison statements of the kind listed above. They do *not* supply a multi-source theorem asserting that two projections commute, that two foreground frames overlap nontrivially, or that two source chains admit a common stationary state. Those are the precise new conditional interfaces named JS-A, JS-B, and JS-C in the main text.

Remark A.8 (Conditional expectations). The common/private regime assumption uses faithful normal conditional expectations in the sense of Takesaki [13]. No claim is made here that such expectations always exist onto the shared sector in the Type III_1 setting; their existence is part of the regime assumption itself.

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Foundation III: The Perfuming Loop
A Conditional One-Step Source-Level Feedback Program

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Abstract

Foundation I [1] constructs a conditional single-source chain from a source datum S_{source} to an admissible manifest pair $(\mathcal{A}_c, \omega)$. Foundation II [2] extends this to two sources, isolating a shared manifest sector. Both chains are one-directional: source data produce manifest sectors, but the manifest sectors do not feed back to the source. The present paper, Foundation III, addresses the return arrow.

T-DOME III [6] supplies a calibration loop at the ego level under explicit residual-likelihood, convexity, drift, and hardware assumptions: the observer's prediction-error residuals drive a natural-gradient update of the reference-frame parameter σ, with tracking controlled by a local Lyapunov estimate. Paper III of the Consciousness Trilogy [5] now treats the former drift-floor theorem as withdrawn and leaves only a supplied arithmetic comparison score for finite self-models. Neither construction closes the loop at the source level: the ego adjusts its frame, but the decoherence semigroup $\{\mathcal{E}_t\}$ that generated the manifest sector in the first place is left untouched.

Foundation III defines a *source-level feedback map* that lifts the ego-level calibration signal to a controlled perturbation of the decoherence semigroup. Under named regularity and coupling conditions, we prove a conditional one-step closed-loop foothold with three components. First, the feedback-modified semigroup remains a valid source datum in the pointwise σ-weak Heisenberg-picture sense used by Foundation I v2. Second, the supplied self-description comparison functional under the closed loop satisfies a modified algebraic comparison bound that depends on the feedback coupling strength. Third, the one-directional Foundation I chain upgrades only to a conditional one-step closed loop:

$$\begin{aligned} S_{\text{source}} \longrightarrow \mathcal{O}_B \longrightarrow \text{calibration signal} \longrightarrow S_{\text{source}}[\text{updated}] \\ \longrightarrow \mathcal{O}_B[\text{updated}] \longrightarrow (\mathcal{A}_c, \omega)[\text{updated}]. \end{aligned}$$

The paper does not claim that the loop eliminates the self-description deficit. It proves only that the loop is well-posed and that the deficit transforms in a controlled way. The question of whether iterated perfuming converges to a fixed point, and what such a fixed point would mean structurally, is left as an open problem.

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1 Introduction

Foundation I [1] solved the first source-side bridge in the Consciousness Series. Starting from a source datum

$$S_{\text{source}} = (\mathcal{H}_U, \mathcal{A}_U, \Omega, \{\mathcal{E}_t\}_{t\geq 0}),$$

it isolated a conditional chain

$$S_{\text{source}} \to \mathcal{O}_A \to \mathcal{O}_B \to (\mathcal{A}_c, \omega),$$

where $\mathcal{O}_A$ is a persistent projected record process, $\mathcal{O}_B$ is a foreground frame available under the explicit conditional coding interface of Foundation I, and $(\mathcal{A}_c, \omega)$ is the manifest accessible pair. The chain is one-directional: source data produce a manifest sector, and the series then hands that sector to Paper I of the Consciousness Trilogy [3] for the geometry–modular-flow theorem and to Paper II [4] for the higher-dimensional consistency analysis.

Foundation II [2] extended that source-side analysis to the first multi-source setting. There the main result was again one-directional. Given two source chains sharing one ambient Type III_1 substrate, one obtains a conditional inclusion theorem for a shared admissible manifest sector. What remained absent in both papers was any formal return arrow from manifest calibration back to the source dynamics that generated the manifest sector in the first place. Foundation II enters the present paper only as programme boundary calibration for the later multi-source feedback problem; no theorem below imports a Foundation II result.

The missing input now comes from two already published directions. T-DOME III [6] supplied a one-step calibration signal at the ego level: prediction-residual statistics determine a natural-gradient update $\delta\sigma$, controlled by a Lyapunov tracking estimate and penalized by the ego-rigidity term $\mathcal{I}_{\text{ego}}$. Paper III of the Consciousness Trilogy [5] proposed a local self-description floor on the manifest side. That reconstruction-loss bound has since been withdrawn: a lossless code for a low-dimensional orbit defeats an n-independent positive floor. Foundation III therefore uses its displayed Fisher expression only as a supplied comparison functional, not as an imported risk or reconstruction theorem. Neither that functional nor the T-DOME calibration model, by itself, modifies the source semigroup $\{\mathcal{E}_t\}$.

The present paper addresses exactly that gap. Its aim is not to prove a global convergence theorem for iterative self-modification or a general iterated-feedback theorem. It does not claim that the self-description deficit is eliminated by feedback, and it does not claim that one can derive a feedback lift from bare multi-scale dynamics without additional structure. Its narrower goal is to identify an honest one-step theorem-level foothold for *source-level feedback from manifest calibration*.

The main design choice is to avoid a formal expression of the type

$$\mathcal{E}_t \mapsto \mathcal{E}_t + \eta\,\Gamma(\delta\sigma, \mathcal{E}_t)$$

as the *definition* of the modified dynamics. Such a first-order expression is useful heuristically, but it does not by itself preserve the semigroup law or complete positivity. Instead, the paper defines a feedback lift through an exact auxiliary semigroup $\{\mathscr{F}_r^{[\delta\sigma]}\}_{r\geq 0}$ of normal completely positive unital maps and then sets

$$\mathcal{E}_t^{[\eta]} := \mathscr{F}_{\eta t}^{[\delta\sigma]} \circ \mathcal{E}_t.$$

Once a commutation condition is imposed, the semigroup law is automatic; the first-order feedback map Γ then reappears as the $\eta \downarrow 0$ tangent of this exact construction.

This paper therefore contributes four one-step theorem-level items.

1. We define a *source-level feedback lift* from an admissible calibration signal $\delta\sigma$ to an exact Heisenberg-picture semigroup perturbation on the ambient Type III_1 algebra.

2. We prove a preservation theorem: under named regularity and commutation inputs, the modified family $\{\mathcal{E}_t^{[\eta]}\}_{t\geq 0}$ is again a pointwise σ-weakly continuous semigroup of normal completely positive unital maps on $\mathcal{A}_U$, so the ambient datum remains a valid source datum.

3. We prove a comparison theorem for a supplied local Fisher functional: if the chosen Foundation I chain is feedback-stable and its parameters remain controlled, then that functional deforms by an explicit small-coupling factor $f(\eta)$ with $f(0) = 1$. No actual deficit bound is inferred.

4. We package the result as a *one-step return datum*, showing that the chain

$$S_{\text{source}} \to \mathcal{O}_B \to \delta\sigma \to S_{\text{source}}^{[\eta]} \to \mathcal{O}_B^{[\eta]} \to (\mathcal{A}_c^{[\eta]}, \omega^{[\eta]})$$

   is mathematically well-posed under named inputs.

The scope is deliberately narrow. The paper does not solve the constructive origin of the calibration signal, which is imported from T-DOME III. It does not solve the full persistence problem for the Foundation I chain under arbitrary perturbations, which remains a conditional interface. It does not derive explicit optimal coupling windows or fixed points for iterated perfuming. Those are precisely the open problems recorded later.

The paper also keeps the same Type III discipline as Foundation II. No tensor-product factorization is assumed. All dynamics are written in the Heisenberg picture on a common ambient algebra, and continuity means pointwise σ-weak continuity rather than norm continuity. This is not cosmetic: the whole point of the paper is to modify the source semigroup without importing an external finite-dimensional controller.

The paper is organized as follows. Section 2 defines the feedback lift, the regularity package, and the modified source datum. Section 3 proves the source-datum preservation theorem. Section 4 states the extra comparison inputs needed to pass through Foundation I and then proves the comparison-functional theorem. Section 5 packages the resulting one-step return map. Section 6 records the open problems. Section 7 collects the named conditional inputs. Appendix A restates the conditional Foundation I and T-DOME III interfaces and records the withdrawal of the former Paper III floor.

2 Feedback Lift and Regularity Conditions

The first task is to isolate the additional object that Foundation III needs and the exact conditions under which it is allowed to act on the source semigroup.

Definition 2.1 (Admissible calibration signal)**.** Fix a source datum S_{source} together with a chosen Foundation I chain up to Stage B, so that one has

$$\mathcal{O}_B = (\mathcal{P}, \omega_{\mathcal{P},t}, \mathcal{K}_*(t,s), \mathfrak{F}, \pi_{\text{ego}}).$$

An *admissible calibration signal* is a vector

$$\delta\sigma \in T_\sigma \Sigma$$

in the local calibration parameter space of T-DOME III such that:

1. $\delta\sigma$ is obtained from the natural-gradient calibration step imported from T-DOME III;
2. $\sigma + \delta\sigma$ remains inside the Lyapunov neighborhood on which the T-DOME III tracking estimate is valid;
3. there exists a bound $\|\delta\sigma\| \leq R_{\text{cal}}$ for some finite calibration radius $R_{\text{cal}} > 0$.

Remark 2.2 (One-step rather than iterative calibration). Definition 2.1 is intentionally local. The paper treats a single feedback step driven by one calibration signal $\delta\sigma$. An iterated sequence $\delta\sigma_0, \delta\sigma_1, \ldots$ would require a separate fixed point or long-time theory, and nothing in the present paper should be read as supplying that theory.

Definition 2.3 (Feedback regularity package)**.** Let S_{source} be a source datum and let $\delta\sigma$ be an admissible calibration signal. A *feedback regularity package* consists of a family

$$\mathscr{F}^{[\delta\sigma]} = \{\mathscr{F}_r^{[\delta\sigma]}\}_{r\geq 0}$$

of normal maps on $\mathcal{A}_U$ satisfying the following conditions.

1. **FB-A: CP source lift.** $\{\mathscr{F}_r^{[\delta\sigma]}\}_{r\geq 0}$ is a pointwise σ-weakly continuous semigroup of normal completely positive unital maps on $\mathcal{A}_U$, with
$$\mathscr{F}_0^{[\delta\sigma]} = \mathrm{id}_{\mathcal{A}_U}, \qquad \mathscr{F}_r^{[0]} = \mathrm{id}_{\mathcal{A}_U} \quad \forall r \geq 0.$$
2. **FB-B: small-signal differentiability.** For every $A \in \mathcal{A}_U$, the map $r \mapsto \mathscr{F}_r^{[\delta\sigma]}(A)$ is σ-weakly differentiable at $r = 0$. We denote the derivative by
$$\mathfrak{G}_{\delta\sigma}(A) := \frac{d}{dr}\mathscr{F}_r^{[\delta\sigma]}(A)\bigg|_{r=0},$$
and record the normalization $\mathfrak{G}_{\delta\sigma}(\mathbf{1}) = 0$, which follows from the unitality in FB-A because $\mathscr{F}_r^{[\delta\sigma]}(\mathbf{1}) = \mathbf{1}$ for all $r \geq 0$.
3. **FB-C: chain compatibility.** The feedback lift commutes with the ambient source semigroup:
$$\mathscr{F}_r^{[\delta\sigma]}\mathcal{E}_t = \mathcal{E}_t\mathscr{F}_r^{[\delta\sigma]} \qquad \forall r, t \geq 0.$$
In addition, for the chosen Foundation I chain, the maps $\mathscr{F}_r^{[\delta\sigma]}$ preserve the named structural inputs that are required later to reconstruct Stage A, Stage B, and the manifest pair.

Remark 2.4 (Why the feedback acts through an exact semigroup). The exact feedback semigroup in Definition 2.3 is not an ornament. A first-order perturbation $\mathcal{E}_t + \eta\,\Gamma(\delta\sigma, \mathcal{E}_t)$ does not automatically preserve the semigroup property or complete positivity. The exact semigroup lift is the cleanest way to keep those properties in the Heisenberg-picture Type III setting. The first-order map Γ reappears only after the exact object has been defined.

Remark 2.5 (Meaning of the strict commutation input). The commutation requirement in FB-C is strong and should be read carefully. It does *not* say that feedback is dynamically trivial. For $\eta > 0$,
$$\mathcal{E}_t^{[\eta]} = \mathscr{F}_{\eta t}^{[\delta\sigma]} \circ \mathcal{E}_t$$
generally differs from $\mathcal{E}_t$ unless the calibration signal is zero or the coupling is turned off. The role of strict commutation is only to keep the algebra-level time ordering unchanged, so that the modified dynamics remains an exact semigroup without entering a more delicate generator-level perturbation problem. Relaxing FB-C to an approximate-commutation condition is left open.

Proposition 2.6 (A nontrivial commuting-dephasing feedback lift)**.** *The package FB-A–FB-C has a finite-dimensional normal UCP realization that depends nontrivially on $\delta\sigma$ and changes the source dynamics beyond a relabeling.*

Proof. Take $\mathcal{A}_U = M_2 \otimes M_2$, let Z_1, Z_2 be Pauli Z on the two factors, and write $\mathrm{Ad}_{Z_j}(A) = Z_j A Z_j$. Fix $\gamma, \kappa > 0$ and a continuous scalar response

$$q(\delta\sigma) := \|\delta\sigma\|^2.$$

Define bounded generators

$$\mathcal{L}_0 = \gamma(\mathrm{Ad}_{Z_1} - \mathrm{id}), \qquad \mathfrak{G}_{\delta\sigma} = \kappa q(\delta\sigma)(\mathrm{Ad}_{Z_2} - \mathrm{id}),$$

and semigroups

$$\mathcal{E}_t = e^{t\mathcal{L}_0}, \qquad \mathscr{F}_r^{[\delta\sigma]} = e^{r\mathfrak{G}_{\delta\sigma}}.$$

More explicitly,

$$\mathscr{F}_r^{[\delta\sigma]}(A) = a_r A + (1 - a_r) Z_2 A Z_2, \qquad a_r = \frac{1 + e^{-2\kappa q(\delta\sigma) r}}{2}.$$

Thus $\mathscr{F}_r^{[\delta\sigma]}$ is a normal UCP unital semigroup, equals the identity when $\delta\sigma = 0$, and has the stated generator at $r = 0$. The same formula with $(Z_2, \kappa q, r)$ replaced by (Z_1, γ, t) verifies these properties for $\mathcal{E}_t$.

Because Ad_{Z_1} and Ad_{Z_2} commute, so do the two semigroups, which is the algebraic clause of FB-C. A Foundation I input package supported on the joint diagonal algebra is fixed by both dephasings, so the named structural inputs can be chosen feedback-stable in this model. Yet for $A = \mathbf{1} \otimes X$,

$$\mathcal{E}_t(A) = A, \qquad \mathcal{E}_t^{[\eta]}(A) = e^{-2\eta\kappa q(\delta\sigma) t} A.$$

For $\eta > 0$ and $\delta\sigma \neq 0$, the lift therefore adds a real decay channel that the source semigroup did not have; it is not an inert renaming. This is an FD/I witness of compatibility, not a construction of a Type III feedback lift. □

Remark 2.7 (What strict commutation can and cannot add)*.* Proposition 2.6 shows that FB-C does not force the lift to be the identity. It also displays the limitation: the commuting generators are simultaneously diagonal on joint Pauli sectors, so feedback only reweights pre-existing joint modes and does not mix them. Creating new couplings between source sectors generally requires a near-commuting or time-ordered theory, which remains open.

Definition 2.8 (Small-coupling regime and modified semigroup)**.** Fix a feedback regularity package $\mathscr{F}^{[\delta\sigma]}$. A coupling strength $\eta \geq 0$ is in the *small-coupling regime* if $0 \leq \eta < \eta_*$ for some window $\eta_* > 0$ on which the later comparison inputs remain valid. For a first-order interpretation on a return horizon $[0, T]$, we additionally require $\eta T \ll 1$. The exact composed semigroup below exists without this perturbative estimate, but its tangent need not be uniformly small at long times. For such η, define the *feedback-modified semigroup* by

$$\mathcal{E}_t^{[\eta]} := \mathscr{F}_{\eta t}^{[\delta\sigma]} \circ \mathcal{E}_t \qquad (t \geq 0).$$

The corresponding *modified source datum* is

$$S_{\text{source}}^{[\eta]} := (\mathcal{H}_U, \mathcal{A}_U, \Omega, \{\mathcal{E}_t^{[\eta]}\}_{t \geq 0}).$$

Remark 2.9 (Heisenberg-picture convention). As in Foundation II, all semigroups in Definition 2.8 act on observables. Pointwise continuity therefore means σ-weak continuity on $\mathcal{A}_U$, not norm continuity. Nothing in the present paper uses or claims a norm-continuous generator-level perturbation theorem.

Definition 2.10 (First-order feedback map)**.** Under FB-B, define the *first-order feedback map* by

$$\Gamma(\delta\sigma, \mathcal{E}_t)(A) := t\, \mathfrak{G}_{\delta\sigma}(\mathcal{E}_t(A)), \qquad A \in \mathcal{A}_U.$$

Equivalently, for each fixed $t \geq 0$,

$$\mathcal{E}_t^{[\eta]}(A) = \mathcal{E}_t(A) + \eta\, \Gamma(\delta\sigma, \mathcal{E}_t)(A) + o(\eta)$$

in the σ-weak topology as $\eta \downarrow 0$.

Verification of the expansion. By Definition 2.8,

$$\mathcal{E}_t^{[\eta]}(A) = \mathscr{F}_{\eta t}^{[\delta\sigma]}(\mathcal{E}_t(A)).$$

Applying the $r = 0$ derivative from FB-B at $r = \eta t$ gives

$$\mathscr{F}_{\eta t}^{[\delta\sigma]}(\mathcal{E}_t(A)) = \mathcal{E}_t(A) + \eta t\, \mathfrak{G}_{\delta\sigma}(\mathcal{E}_t(A)) + o(\eta),$$

which is exactly the stated formula. □

Remark 2.11 (What Definition 2.10 does not prove). The first-order map Γ is merely the tangent of the exact feedback lift. Definition 2.10 does not claim that an arbitrary bounded normal map Γ integrates to a completely positive semigroup, nor does it characterize the most general generator-level perturbations compatible with Type III dynamics. Those questions are left open.

Remark 2.12 (The tangent is not uniform at long times). The identity

$$\Gamma(\delta\sigma, \mathcal{E}_t)(A) = t\, \mathfrak{G}_{\delta\sigma}(\mathcal{E}_t(A))$$

contains an explicit factor of t. Thus the first-order perturbation is controlled for fixed t as $\eta \downarrow 0$, or on windows with $\eta t \ll 1$ under a uniform generator bound. Small η alone does not give a perturbation uniformly small for arbitrarily long times. This is one reason the present result does not iterate to a long-time perfuming theorem.

3 Source-Datum Preservation

We now prove the first of the paper's three main results.

Lemma 3.1 (Semigroup law)**.** *Assume FB-A–FB-C. Then the family* $\{\mathcal{E}_t^{[\eta]}\}_{t\geq 0}$ *from Definition 2.8 satisfies*

$$\mathcal{E}_t^{[\eta]}\mathcal{E}_s^{[\eta]} = \mathcal{E}_{t+s}^{[\eta]} \qquad \forall s, t \geq 0.$$

Proof. For $A \in \mathcal{A}_U$,

$$\begin{aligned}\mathcal{E}_t^{[\eta]}\mathcal{E}_s^{[\eta]}(A) &= \mathscr{F}_{\eta t}^{[\delta\sigma]}\, \mathcal{E}_t\Big(\mathscr{F}_{\eta s}^{[\delta\sigma]}\, \mathcal{E}_s(A)\Big) \\ &= \mathscr{F}_{\eta t}^{[\delta\sigma]}\mathscr{F}_{\eta s}^{[\delta\sigma]}\mathcal{E}_t\mathcal{E}_s(A),\end{aligned}$$

where FB-C is used to commute $\mathcal{E}_t$ past $\mathscr{F}_{\eta s}^{[\delta\sigma]}$. Since $\{\mathscr{F}_r^{[\delta\sigma]}\}$ and $\{\mathcal{E}_t\}$ are both semigroups,

$$\mathscr{F}_{\eta t}^{[\delta\sigma]}\mathscr{F}_{\eta s}^{[\delta\sigma]}\mathcal{E}_t\mathcal{E}_s = \mathscr{F}_{\eta(t+s)}^{[\delta\sigma]}\mathcal{E}_{t+s} = \mathcal{E}_{t+s}^{[\eta]}.$$

□

Lemma 3.2 (Pointwise σ-weak continuity and positivity)**.** *Assume FB-A–FB-C. Then for every $A \in \mathcal{A}_U$,*

$$t \longmapsto \mathcal{E}_t^{[\eta]}(A)$$

is pointwise σ-weakly continuous. Moreover, each $\mathcal{E}_t^{[\eta]}$ is normal, completely positive, and unital.

Proof. Fix $A \in \mathcal{A}_U$ and a normal functional $\varphi \in (\mathcal{A}_U)_*$. By Definition 2.8,

$$\varphi(\mathcal{E}_t^{[\eta]}(A)) = \varphi\Big(\mathscr{F}_{\eta t}^{[\delta\sigma]}(\mathcal{E}_t(A))\Big).$$

The map $t \mapsto \mathcal{E}_t(A)$ is σ-weakly continuous by the source-datum assumption, and $r \mapsto \mathscr{F}_r^{[\delta\sigma]}(B)$ is σ-weakly continuous for every fixed $B \in \mathcal{A}_U$ by FB-A. To prove continuity of the composition, fix $t_0 \geq 0$ and write

$$\begin{aligned}
&\Big|\varphi\big(\mathscr{F}_{\eta t}^{[\delta\sigma]}(\mathcal{E}_t(A))\big) - \varphi\big(\mathscr{F}_{\eta t_0}^{[\delta\sigma]}(\mathcal{E}_{t_0}(A))\big)\Big| \\
&\quad\leq \Big|\varphi\big(\mathscr{F}_{\eta t}^{[\delta\sigma]}(\mathcal{E}_t(A))\big) - \varphi\big(\mathscr{F}_{\eta t}^{[\delta\sigma]}(\mathcal{E}_{t_0}(A))\big)\Big| \\
&\qquad+ \Big|\varphi\big(\mathscr{F}_{\eta t}^{[\delta\sigma]}(\mathcal{E}_{t_0}(A))\big) - \varphi\big(\mathscr{F}_{\eta t_0}^{[\delta\sigma]}(\mathcal{E}_{t_0}(A))\big)\Big|.
\end{aligned}$$

For the second term, $\mathcal{E}_{t_0}(A)$ is fixed, so pointwise σ-weak continuity of $\{\mathscr{F}_r^{[\delta\sigma]}\}_{r\geq 0}$ gives convergence to zero as $t \to t_0$.

For the first term, use the predual semigroup $\{\mathscr{F}_{r,*}^{[\delta\sigma]}\}_{r\geq 0}$. Since $\{\mathscr{F}_r^{[\delta\sigma]}\}$ is a pointwise σ-weakly continuous semigroup of normal maps, the predual family is norm-continuous on $(\mathcal{A}_U)_*$. Thus

$$\begin{aligned}
&\Big|\varphi\big(\mathscr{F}_{\eta t}^{[\delta\sigma]}(\mathcal{E}_t(A))\big) - \varphi\big(\mathscr{F}_{\eta t}^{[\delta\sigma]}(\mathcal{E}_{t_0}(A))\big)\Big| \\
&\quad= \Big|\mathscr{F}_{\eta t,*}^{[\delta\sigma]}(\varphi)\big(\mathcal{E}_t(A) - \mathcal{E}_{t_0}(A)\big)\Big| \\
&\quad\leq \Big|\big(\mathscr{F}_{\eta t,*}^{[\delta\sigma]}(\varphi) - \mathscr{F}_{\eta t_0,*}^{[\delta\sigma]}(\varphi)\big)\big(\mathcal{E}_t(A) - \mathcal{E}_{t_0}(A)\big)\Big| \\
&\qquad+ \Big|\mathscr{F}_{\eta t_0,*}^{[\delta\sigma]}(\varphi)\big(\mathcal{E}_t(A) - \mathcal{E}_{t_0}(A)\big)\Big|.
\end{aligned}$$

The first summand tends to zero because $\mathscr{F}_{\eta t,*}^{[\delta\sigma]}(\varphi) \to \mathscr{F}_{\eta t_0,*}^{[\delta\sigma]}(\varphi)$ in predual norm, while $\mathcal{E}_t(A) - \mathcal{E}_{t_0}(A)$ is norm-bounded by $2\|A\|$. The second summand tends to zero because $\mathcal{E}_t(A) \to \mathcal{E}_{t_0}(A)$ σ-weakly and $\mathscr{F}_{\eta t_0,*}^{[\delta\sigma]}(\varphi)$ is a fixed normal functional. Hence the first term also tends to zero.

Hence $\{\mathcal{E}_t^{[\eta]}\}$ is pointwise σ-weakly continuous.

Normality and complete positivity are preserved under composition of normal completely positive maps, and unitality is preserved under composition of unital maps. Since both $\mathscr{F}_{\eta t}^{[\delta\sigma]}$ and $\mathcal{E}_t$ have those properties, so does $\mathcal{E}_t^{[\eta]}$. □

Theorem 3.3 (Source-datum preservation)**.** *Let*

$$S_{\text{source}} = (\mathcal{H}_U, \mathcal{A}_U, \Omega, \{\mathcal{E}_t\}_{t\geq 0})$$

be a source datum in the sense imported from Foundation I, let $\delta\sigma$ be an admissible calibration signal, and assume that a feedback regularity package FB-A–FB-C is given. Then for every $\eta \in [0, \eta_*)$, the modified datum

$$S_{\text{source}}^{[\eta]} = (\mathcal{H}_U, \mathcal{A}_U, \Omega, \{\mathcal{E}_t^{[\eta]}\}_{t\geq 0})$$

is again a valid source datum: the family $\{\mathcal{E}_t^{[\eta]}\}_{t\geq 0}$ is a pointwise σ-weakly continuous semigroup of normal completely positive unital maps on $\mathcal{A}_U$, and Ω remains cyclic and separating for $\mathcal{A}_U$.

Proof. Lemma 3.1 gives the semigroup law and Lemma 3.2 gives pointwise σ-weak continuity, normality, complete positivity, and unitality.

The ambient standard-form triple $(\mathcal{H}_U, \mathcal{A}_U, \Omega)$ is not changed by the feedback step. In particular, Ω remains cyclic and separating for the same von Neumann algebra $\mathcal{A}_U$. Therefore $S_{\text{source}}^{[\eta]}$ satisfies the same source-datum format as S_{source}. □

Remark 3.4 (What Theorem 3.3 does not prove). The theorem proves only that the feedback-modified dynamics remains in the theorem domain of the Foundation I source-datum definition. It does not prove that the modified datum automatically reproduces the same persistent projection, the same foreground frame, or the same manifest algebra. Those are downstream interface questions and require additional inputs.

Corollary 3.5 (Exact CP lift with first-order tangent)**.** *Under the hypotheses of Theorem 3.3, the map $\Gamma(\delta\sigma, \mathcal{E}_t)$ from Definition 2.10 is the $\eta \downarrow 0$ tangent of an exact Heisenberg-picture completely positive perturbation.*

Proof. This is immediate from Theorem 3.3 together with the first-order expansion of Definition 2.10. □

4 Modified Local Comparison Functional

The preservation theorem keeps the modified dynamics inside the source-datum domain, but it does not yet produce a manifest pair. For that we need the chosen Foundation I chain to survive the perturbation and the supplied local comparison parameters to remain controlled.

Definition 4.1 (Deficit-comparison package)**.** Fix a source datum S_{source}, a chosen Foundation I chain from S_{source} to $(\mathcal{A}_c, \omega)$, and a feedback-modified datum $S_{\text{source}}^{[\eta]}$. We say that *DF-A* holds on the coupling window $[0, \eta_*)$ if:

1. **DF-A(1): feedback-stable Foundation I chain.** For every $\eta \in [0, \eta_*)$, the named conditional inputs used in the chosen Foundation I chain remain available for $S_{\text{source}}^{[\eta]}$. Hence one obtains a modified manifest pair
$$(\mathcal{A}_c^{[\eta]}, \omega^{[\eta]}).$$

2. **DF-A(2): Fisher control.** There exist continuous functions $\alpha_F, \beta_{\text{ego}} : [0, \eta_*) \to (0, \infty)$ with
$$\alpha_F(0) = \beta_{\text{ego}}(0) = 1$$
such that
$$\mathcal{I}_F^{[\eta]} \geq \alpha_F(\eta)\, \mathcal{I}_F, \qquad \mathcal{I}_{\text{ego}}^{[\eta]} \leq \beta_{\text{ego}}(\eta)\, \mathcal{I}_{\text{ego}},$$
where $\mathcal{I}_F$ and $\mathcal{I}_{\text{ego}}$ are the baseline manifest-side Fisher sensitivity and ego-rigidity constants, and $\mathcal{I}_F^{[\eta]}$, $\mathcal{I}_{\text{ego}}^{[\eta]}$ are the corresponding modified quantities. Assume also
$$\mathcal{I}_F + \mathcal{I}_{\text{ego}} > 0, \qquad \mathcal{I}_F^{[\eta]} + \mathcal{I}_{\text{ego}}^{[\eta]} > 0,$$
so the comparison functionals are defined.

3. **DF-A(3): supplied local comparison parameters.** A controlled neighborhood is equipped with a stipulated parameter $\rho(\eta) \in [0, \frac{1}{2})$, continuous in η, with baseline value $\rho(0) =: \rho_0$. This parameter and the Fisher quantities define the comparison functional below; they do not import a valid Paper III reconstruction-loss theorem.

Proposition 4.2 (Passage from the modified source datum to the modified manifest pair)**.** *Assume the hypotheses of Theorem 3.3 and DF-A(1). Then for every $\eta \in [0, \eta_*)$, the modified source datum $S_{\text{source}}^{[\eta]}$ admits a conditional Foundation I chain*

$$S_{\text{source}}^{[\eta]} \to \mathcal{O}_A^{[\eta]} \to \mathcal{O}_B^{[\eta]} \to (\mathcal{A}_c^{[\eta]}, \omega^{[\eta]}).$$

Proof. Theorem 3.3 puts $S_{\text{source}}^{[\eta]}$ inside the theorem domain of the Foundation I source-datum definition. DF-A(1) states exactly that the named conditional inputs used by the chosen Foundation I chain remain available after the feedback perturbation. The Foundation I conditional theorem can therefore be re-run with the modified datum, yielding the stated outputs. □

Definition 4.3 (Baseline and modified local comparison functionals)**.** Under DF-A, define

$$\mathfrak{b}_{\text{loc}} := \frac{(1 - 2\rho_0)\, \mathcal{I}_F}{2\big(\mathcal{I}_F + \mathcal{I}_{\text{ego}}\big)},$$

and

$$\mathfrak{b}_{\text{loc}}^{[\eta]} := \frac{(1 - 2\rho(\eta))\, \mathcal{I}_F^{[\eta]}}{2\big(\mathcal{I}_F^{[\eta]} + \mathcal{I}_{\text{ego}}^{[\eta]}\big)}.$$

These are algebraic comparison scores. The formula is retained from the former Paper III interface, but neither score is asserted here to lower bound reconstruction loss, estimation risk, or a minimax deficit.

Theorem 4.4 (Comparison of supplied local functionals)**.** *Assume the hypotheses of Proposition 4.2 and the full package DF-A(1)–DF-A(3). Then for every $\eta \in [0, \eta_*)$,*

$$\mathfrak{b}_{\text{loc}}^{[\eta]} \geq f(\eta)\, \mathfrak{b}_{\text{loc}},$$

where

$$f(\eta) := \frac{1 - 2\rho(\eta)}{1 - 2\rho_0} \, \frac{\alpha_F(\eta)\big(\mathcal{I}_F + \mathcal{I}_{\text{ego}}\big)}{\alpha_F(\eta)\mathcal{I}_F + \beta_{\text{ego}}(\eta)\mathcal{I}_{\text{ego}}}.$$

In particular, $f(0) = 1$, and by continuity $f(\eta) > 0$ for all sufficiently small η.

Proof. By Proposition 4.2, the modified datum yields a manifest pair $(\mathcal{A}_c^{[\eta]}, \omega^{[\eta]})$. For fixed $c > 0$ and $d \geq 0$, the function $x \mapsto cx/(x + d)$ is increasing where the denominator is nonzero. Therefore substituting the lower bound for $\mathcal{I}_F^{[\eta]}$ lowers the whole fraction, rather than creating a denominator sign reversal. Likewise the fraction is decreasing in its nonnegative second argument. Using DF-A(2),

$$\mathcal{I}_F^{[\eta]} \geq \alpha_F(\eta)\mathcal{I}_F, \qquad \mathcal{I}_{\text{ego}}^{[\eta]} \leq \beta_{\text{ego}}(\eta)\mathcal{I}_{\text{ego}},$$

so

$$\mathfrak{b}_{\text{loc}}^{[\eta]} \geq \frac{(1 - 2\rho(\eta))\, \alpha_F(\eta)\mathcal{I}_F}{2\big(\alpha_F(\eta)\mathcal{I}_F + \beta_{\text{ego}}(\eta)\mathcal{I}_{\text{ego}}\big)}.$$

Now factor out the baseline score from Definition 4.3:

$$\begin{aligned}\mathfrak{b}_{\text{loc}}^{[\eta]} &\geq \frac{1-2\rho(\eta)}{1-2\rho_0}\,\frac{\alpha_F(\eta)\big(\mathcal{I}_F+\mathcal{I}_{\text{ego}}\big)}{\alpha_F(\eta)\mathcal{I}_F+\beta_{\text{ego}}(\eta)\mathcal{I}_{\text{ego}}}\cdot\frac{(1-2\rho_0)\mathcal{I}_F}{2(\mathcal{I}_F+\mathcal{I}_{\text{ego}})}\\ &= f(\eta)\,\mathfrak{b}_{\text{loc}}.\end{aligned}$$

The formula for $f(0)$ follows from $\rho(0)=\rho_0$ and $\alpha_F(0)=\beta_{\text{ego}}(0)=1$. Continuity then gives positivity for sufficiently small η. □

Remark 4.5 (No monotonicity claim for $f(\eta)$). The theorem asserts only that $f(0)=1$ and that $f(\eta)$ remains positive for sufficiently small coupling. No monotonicity statement is claimed. Depending on the detailed response of the manifest Fisher sensitivity and the ego-rigidity term, one may have $f(\eta)<1$ or $f(\eta)>1$. In particular, a feedback step that increases $\mathcal{I}_F^{[\eta]}$ or decreases $\mathcal{I}_{\text{ego}}^{[\eta]}$ can raise the score rather than lower it.

Remark 4.6 (The former actual-deficit corollary is withdrawn). No saturation assumption can repair the missing premise that $\mathfrak{b}_{\text{loc}}$ is a valid positive lower bound on the actual reconstruction loss. Establishing such a bridge requires either a specified sampling/measurement model and an estimation-risk theorem, or a genuine rate-distortion bottleneck. The present paper supplies neither, so it makes no inequality claim for an actual minimax deficit.

Remark 4.7 (What Theorem 4.4 does not prove). The theorem gives a controlled deformation of a supplied algebraic score. It does not prove that the true minimax deficit is monotone in η, it does not prove that feedback reduces the deficit, and it does not prove that the deficit can be driven to zero. Those stronger claims would require a valid risk/distortion model and a transport theory for optimal self-models.

Remark 4.8 (Why the comparison factor is the right theorem target). The factor $f(\eta)$ isolates exactly what feedback changes at theorem level:

1. the supplied local coefficient $\rho(\eta)$;
2. the manifest Fisher sensitivity $\mathcal{I}_F^{[\eta]}$;
3. the ego-rigidity contribution $\mathcal{I}_{\text{ego}}^{[\eta]}$.

No stronger conclusion is claimed. In particular, the theorem does not say that a better calibration signal necessarily increases $\mathcal{I}_F^{[\eta]}$ or decreases $\mathcal{I}_{\text{ego}}^{[\eta]}$; it only says that if those quantities remain controlled, then the comparison score remains controlled as well.

5 Conditionally Well-Posed One-Step Return Map

We can now state the paper's final theorem.

Definition 5.1 (One-step return datum)**.** Under the hypotheses of Theorem 4.4, the *one-step return datum at coupling* η is the tuple

$$\mathfrak{C}_\eta := \big(S_{\text{source}}, \mathcal{O}_B, (\mathcal{A}_c,\omega), \delta\sigma, S_{\text{source}}^{[\eta]}, \mathcal{O}_B^{[\eta]}, (\mathcal{A}_c^{[\eta]},\omega^{[\eta]})\big),$$

where $\mathcal{O}_B$ and $(\mathcal{A}_c,\omega)$ are the baseline Stage B and Stage C outputs of the chosen Foundation I chain, $\delta\sigma$ is the admissible calibration signal extracted from the retained Stage B data, and $\mathcal{O}_B^{[\eta]}$ together with $(\mathcal{A}_c^{[\eta]},\omega^{[\eta]})$ are the corresponding modified outputs after the feedback step.

Theorem 5.2 (Conditional one-step return-map well-posedness)**.** *Let S_{source} be a source datum, let $\delta\sigma$ be an admissible calibration signal, and assume the packages FB-A–FB-C and DF-A(1)–DF-A(3). Then for every sufficiently small $\eta \in [0, \eta_*)$, and on any fixed return horizon $[0, T]$ with $\eta T \ll 1$, the one-step return datum $\mathfrak{C}_\eta$ of Definition 5.1 is well-defined. Equivalently, the diagram*

$$S_{\mathrm{source}} \longrightarrow \mathcal{O}_B \longrightarrow \delta\sigma \longrightarrow S_{\mathrm{source}}^{[\eta]} \longrightarrow \mathcal{O}_B^{[\eta]} \longrightarrow (\mathcal{A}_c^{[\eta]}, \omega^{[\eta]})$$

is a conditionally well-posed one-step return into the source-datum category.

Proof. The baseline arrow

$$S_{\mathrm{source}} \to \mathcal{O}_B \to (\mathcal{A}_c, \omega)$$

is the chosen Foundation I chain. The arrow

$$\mathcal{O}_B \to \delta\sigma$$

is the imported T-DOME III calibration interface. By Theorem 3.3, the feedback step

$$\delta\sigma \to S_{\mathrm{source}}^{[\eta]}$$

is well-defined at the source-datum level. By Proposition 4.2, the same conditional Foundation I chain can then be run again on $S_{\mathrm{source}}^{[\eta]}$, yielding

$$S_{\mathrm{source}}^{[\eta]} \to \mathcal{O}_B^{[\eta]} \to (\mathcal{A}_c^{[\eta]}, \omega^{[\eta]}).$$

Thus every component of $\mathfrak{C}_\eta$ is defined. The supplied comparison score is controlled by Theorem 4.4. This does not turn the one-step return into a self-consistent loop or an actual deficit theorem. □

Corollary 5.3 (Packaged one-step return map)**.** *Under the hypotheses of Theorem 5.2, there is a well-defined map*

$$\mathscr{L}_\eta : \big(S_{\mathrm{source}}, \delta\sigma\big) \longmapsto \big(S_{\mathrm{source}}^{[\eta]}, \mathcal{O}_B^{[\eta]}, (\mathcal{A}_c^{[\eta]}, \omega^{[\eta]}), \mathfrak{b}_{\mathrm{loc}}^{[\eta]}\big).$$

Proof. This is just a repackaging of Theorem 5.2 and Theorem 4.4. □

Remark 5.4 (What Theorem 5.2 does not prove). The theorem is a one-step well-posedness statement only. It does not claim that repeated application of $\mathscr{L}_\eta$ converges, that a fixed point exists, that a fixed point would be unique, or that the return map is thermodynamically optimal. It also does not address the multi-source feedback problem in which several manifest sectors update a shared ambient semigroup simultaneously.

6 Open Problems

The present note defines only a conditionally well-posed one-step return. The actual programme still leaves several substantive gaps.

1. **Constructive origin of a Type III feedback lift.** Proposition 2.6 supplies a nontrivial FD/I witness, but the general theorem still takes the source-level semigroup lift $\mathscr{F}^{[\delta\sigma]}$ as named input. A future result must derive a normal Type III lift from the calibration signal itself.

2. **Generator-level perturbation theory.** The present paper works with an exact auxiliary semigroup because it preserves complete positivity and the semigroup law cleanly. A sharper generator-level perturbation theorem on W^*-algebras, closer to the existing structural theory of dynamical semigroups on W^*-algebras [10], would have to control densely defined predual generators, domain stability, closability, and resolvent bounds in a regime where pointwise σ-weak continuity is the natural topology and norm continuity is usually unavailable. Naive first-order generator perturbations may fail to preserve complete positivity, unitality, or even semigroup well-posedness.

3. **Weakening the commutation input.** FB-C requires strict algebra-level commutation between the feedback lift and the source semigroup. A weaker theorem based on approximate commutation, asymptotic commutation, or commutation only on the relevant stable subalgebras would be more flexible and physically more natural, but it would need a replacement for the exact time-ordering argument used in Section 3. The finite witness shows explicitly that strict commutation can change decay rates while still only reweighting joint spectral sectors.

4. **Quantitative coupling thresholds.** The small-coupling window $[0, \eta_*)$ is left abstract. The programme still lacks an explicit estimate of η_* in terms of Lyapunov constants, semigroup norms, and the local BKM/Fisher geometry.

5. **Persistence of the Foundation I chain.** DF-A(1) is a real interface burden. One needs a theorem stating when persistent projections, foreground frames, and stable-hull inputs are robust under source-side feedback.

6. **A valid risk or distortion interface.** Theorem 4.4 controls only the supplied score $\mathfrak{b}_{\mathrm{loc}}$. A sampling model with an estimation-risk theorem, or a rate-distortion model with a genuine bottleneck, is required before any actual minimax deficit can be transported.

7. **Iterated perfuming and fixed points.** Once $\mathscr{L}_\eta$ is applied repeatedly, one faces an entirely new dynamical question: do the semigroups converge, cycle, or wander? The present paper does not address that long-time behavior.

8. **Thermodynamic cost of the source-level return.** T-DOME III gives model-dependent reset costs for the calibration procedure at the manifest/ego level. Foundation III has not integrated a specified irreversible implementation with the source-side semigroup perturbation.

9. **Multi-source feedback.** Foundation II handled the two-source shared-sector problem at inclusion level, but it did not address simultaneous feedback from multiple manifest sectors onto one common ambient source dynamics. Cross-source feedback remains open.

10. **Type III modular obstacles.** Even if a feedback lift is available, one still faces the usual Type III issues: modular invariance of the relevant subalgebras, possible non-uniqueness of conditional expectations, central-decomposition effects in non-factorial regimes, and the absence of a canonical bounded algebra-side generator picture that could be perturbed by finite-rank feedback data.

7 Conditional Input Ledger

The theorem chain depends on the following named inputs.

1. **FI-A: Foundation I source-to-manifest interface.** The single-source conditional chain from a source datum to a manifest pair is available in the sense restated in Appendix A.

2. **TD3-A: admissible calibration signal.** A one-step natural-gradient update signal $\delta\sigma$ exists and lies inside the T-DOME III Lyapunov neighborhood.

3. **FB-A: CP source lift.** The calibration signal admits a pointwise σ-weakly continuous semigroup lift by normal completely positive unital maps on $\mathcal{A}_U$.

4. **FB-B: small-signal differentiability.** The lifted feedback semigroup is differentiable at zero coupling so that the first-order map Γ is defined; its normalization $\mathfrak{G}_{\delta\sigma}(\mathbf{1}) = 0$ is inherited from unitality.

5. **FB-C: chain compatibility.** The feedback lift commutes with the ambient source semigroup and preserves the named structural inputs needed to rerun the chosen Foundation I chain.

6. **FB-FD: finite commuting-dephasing witness.** Proposition 2.6 verifies nontrivial $\delta\sigma$-dependence, normal UCP dynamics, strict commutation, and a source-dynamical change. It is a model check, not a Type III construction.

7. **DF-A(1): feedback-stable Foundation I chain.** The chosen Foundation I conditional inputs remain valid for $S^{[\eta]}_{\text{source}}$ on a nontrivial coupling window.

8. **DF-A(2): manifest Fisher control.** The modified manifest-side Fisher sensitivity and ego-rigidity terms satisfy the comparison inequalities controlled by $\alpha_F(\eta)$ and $\beta_{\text{ego}}(\eta)$.

9. **DF-A(3): supplied local comparison parameters.** The coefficient $\rho(\eta) < \frac{1}{2}$ and the Fisher quantities define $\mathfrak{b}^{[\eta]}_{\text{loc}}$; no Paper III reconstruction-loss theorem is inherited.

10. **Downgraded one-step return status.** The theorem-level output is a conditionally well-posed one-step return map. Its tangent is controlled only at fixed t as $\eta \downarrow 0$, or on windows $\eta t \ll 1$. Iteration, convergence, and fixed points remain open.

A Imported Interfaces

This appendix records the external results used by the body in a self-contained theorem-ready form. The statements are imported interfaces, not re-proved results.

Remark A.1 (What is and is not imported). Foundation III imports a conditional source-to-manifest interface from Foundation I and a model-level calibration interface from T-DOME III. It does not import the withdrawn Paper III reconstruction-loss theorem; only the algebraic comparison formula is retained as supplied data. Foundation II is cited in the Introduction solely to locate the later multi-source feedback problem and does not enter any proof or named input in the present paper.

Proposition A.2 (Imported Foundation I interface)**.** *Let*

$$S_{\text{source}} = (\mathcal{H}_U, \mathcal{A}_U, \Omega, \{\mathcal{E}_t\}_{t\geq 0})$$

be a source datum, where $(\mathcal{H}_U, \mathcal{A}_U, \Omega)$ is a Type III$_1$ standard-form von Neumann algebra and $\{\mathcal{E}_t\}_{t\geq 0}$ is a pointwise σ-weakly continuous semigroup of normal completely positive unital maps on $\mathcal{A}_U$. Suppose:

1. *there exists a persistent Nakajima–Zwanzig projection $\mathcal{P}$ whose projected predual dynamics admits a memory kernel $\mathcal{K}_*(t,s)$ [8, 9, 7];*

2. *a specified finite-coordinate coding model satisfies the Foundation I rank-cost, strict-improvement, and group-action assumptions and supplies a compressed foreground frame $\mathfrak{F}$;*

3. *the foreground transfer/stable-hull construction closes so as to produce a von Neumann algebra $\mathcal{A}_c \subseteq \mathcal{A}_U$ with faithful restricted state $\omega = \langle \Omega, \cdot\, \Omega\rangle|_{\mathcal{A}_c}$;*

4. *the admissibility input needed to feed Paper I [3] is available.*

Then the conditional chain

$$S_{\text{source}} \to \mathcal{O}_A \to \mathcal{O}_B \to (\mathcal{A}_c, \omega)$$

is defined.

Remark A.3. Foundation III uses only the existence of the manifest pair produced by Proposition A.2. It does not revisit the internal stable-hull construction of Foundation I. The source-datum continuity imported here is pointwise σ-weak continuity, matching Foundation I v2. The published Foundation I v1 uses the stronger phrase "strongly continuous"; the downstream Foundation I chain used here does not depend on that stronger condition.

Proposition A.4 (Conditional T-DOME III calibration interface)**.** *Let $\mathcal{O}_B$ be a Stage B output in the sense of Foundation I. In the local calibration regime of T-DOME III there exist:*

1. *a natural-gradient update signal $\delta\sigma \in T_\sigma\Sigma$;*

2. *a Lyapunov neighborhood $\mathcal{U}_{\text{cal}}$ on which the one-step update stays controlled;*

3. *a regular prior Fisher term $\mathcal{I}_{\text{ego}} \geq 0$, interpreted as ego rigidity, for the specified local estimation model.*

These data determine the one-step manifest-side calibration step used in the body of the present paper.

Remark A.5. Proposition A.4 is a one-step interface only. It does not assert global convergence of repeated calibration updates.

Definition A.6 (Supplied local Fisher comparison score)**.** For stipulated nonnegative quantities $\mathcal{I}_F$ and $\mathcal{I}_{\text{ego}}$, and $\rho \in [0, \frac{1}{2})$, define

$$\mathfrak{b}_{\text{loc}} := \frac{(1-2\rho)\,\mathcal{I}_F}{2(\mathcal{I}_F + \mathcal{I}_{\text{ego}})}.$$

The definition makes no assertion that this score lower-bounds an actual reconstruction loss or estimation risk.

Remark A.7 (Status of the former Paper III import). The original positive drift-floor theorem is withdrawn because its admissible reconstruction class includes lossless low-dimensional orbit codes. Definition A.6 retains only the algebraic expression needed for Theorem 4.4. A future sampling model may support an estimation-risk bound, and a future bottleneck may support a rate-distortion bound, but neither result is assumed here.

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Part II

Consciousness Trilogy: Geometry, Gravity, and the Structural Horizon of Self-Description

The Inseparability of Geometry and Modular Flow: A Structural Inseparability Theorem and Its Implications for the Hard Problem

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Abstract

The hard problem of consciousness presupposes that physical structure and experiential structure can be specified independently. We show that this presupposition is not available in a precise algebraic setting.

Working with Type III_1 von Neumann algebras, we identify emergent geometry as the relevant mathematical proxy for physical structure and Tomita-Takesaki modular flow as the relevant proxy for first-person temporal structure. The resulting question is whether these two sectors can be independently deformed within an admissible algebraic class. We show first that, working within the admissible class selected by $(U3)$, from the proof-relevant axiom package

$$(U1) + (C) + (Amb) + (CN_{\exists}),$$

one obtains a standard half-sided modular inclusion and hence, via the Guido-Longo-Wiesbrock construction, a canonical conformal skeleton. This yields a conformal-level Weak Inseparability theorem. We then show that the correct non-gauge local modulus is the double-interval cross-ratio family, which determines a canonical modular scale datum

$$\{K_b\}_{b\in(0,1)}.$$

Under the additional conditions $(LImp^1)$, $(CR\text{-}nd)$, and (SFa), this datum induces a Type III-compatible QFIM scale shadow

$$h_b = \mathbf{Q}_{\omega,b}(\partial_b K_b, \partial_b K_b)\, db^2,$$

At the abstract level, this yields a full geometric datum consisting of the conformal skeleton together with the scale family. In the worked examples, where the cross-ratio parameter naturally parametrizes spatial position, this datum determines the Lorentzian metric. In this setting, the full geometric datum and the extended modular data are bidirectionally inseparable.

We verify the additional Phase 2 conditions in two structurally distinct models, the Rindler/free-field chiral half-line model and the Virasoro net. Both yield the universal profile

$$h_b = C(1-b)^{-2}db^2,$$

with model dependence entering only through the overall constant. The philosophical consequence is not that the hard problem has been solved in the ordinary production sense, but that its standard formulation rests on a presupposition—independent describability—that is not available in the relevant algebraic setting. The remaining explanatory gap is thereby relocated from a gap between independently specified domains to a structural limit on self-description by observers embedded in $(\mathcal{A}, \omega)$.

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1 The Hidden Presupposition of the Hard Problem

The hard problem of consciousness is usually presented as a question about production: how do physical processes give rise to subjective experience? In that form, the problem appears deeper than the familiar "easy problems" because those concern discrimination, report, integration, control, and other functional capacities, whereas the hard problem concerns the existence of a first-person point of view and the felt character of experience itself. This contrast, introduced most influentially by Chalmers [4], has shaped almost every major debate in contemporary philosophy of mind.

Yet the canonical formulation contains a strong presupposition that is rarely isolated and almost never challenged at the level of formal structure. The presupposition is that physical structure and experiential structure can be specified independently. One first gives a complete account of the physical facts, in third-person terms, and then asks why those facts are accompanied by experience, or by this experience rather than another. Without this prior separability, the production question cannot even be posed in its usual form. The hard problem therefore depends not only on a gap between two domains, but on the prior availability of those domains as independently describable.

This presupposition is shared by positions that otherwise disagree sharply. Functionalism and mainstream physicalism accept it and then attempt to dissolve the residual gap by appeal to organization, role, or supervenience. Panpsychism accepts it as well, but seeks to avoid brute emergence by distributing proto-experiential or experiential properties more fundamentally. Illusionism also inherits the same starting point: it grants the independent specification of physical structure and then argues that what seemed left over on the experiential side is a cognitive misdescription. Integrated Information Theory takes a different route, but still treats consciousness as something to be attached to or read off from an already specified structure by means of an additional quantity. In all of these cases, the debate concerns how the bridge should be built. The deeper assumption that there are two independently specifiable shores is left intact.

The present paper takes aim at that assumption. It does not begin by proposing a new psychophysical law, a new emergent mechanism, or a new primitive consciousness quantity. It begins by asking whether the independent-describability presupposition survives translation into a precise mathematical framework. If "physical structure" is represented not by an unconstrained intuitive picture of matter but by an emergent geometric structure, and if "first-person time" is represented by modular flow in the sense of Tomita-Takesaki theory, then the hard problem can be reformulated as a structural question: can these two sectors be independently varied inside a single admissible algebraic setting?

The central claim of this paper is that, under a precise Type III_1 algebraic framework, they cannot. At the conformal level, emergent causal geometry and conformal modular data are already mutually rigid. With three additional Phase 2 inputs, the same becomes true at the full Lorentzian level: complete geometry and extended modular data cannot be independently deformed. This does not amount to the claim that the hard problem has simply been "solved." It amounts to a more exact and, in one sense, more radical thesis: the standard production form of the hard problem depends on a presupposition that is not available in the relevant mathematical setting.

This is why the main theorem of the paper is an inseparability theorem rather than a reduction theorem. The point is not that one side reduces to the other, nor that one side causally generates the other, but that both emerge as two readings of one underlying algebraic organization. In philosophical terms, the paper argues that the usual mind-body contrast is not fun-

damental enough to underwrite the standard hard-problem question. In mathematical terms, it shows that the admissible deformations of geometry and modular structure do not factor into two independent sectors.

The argument unfolds in three stages. First, we translate the relevant philosophical notions into precise mathematical objects: emergent geometry for the object-side physical structure, and modular flow for first-person temporal structure. Second, we prove the conditional Phase 1 result: working with the carrier specified by $(U3)$, from the proof-relevant axiom package

$$(U1) + (C) + (Amb) + (CN_{\exists})$$

one canonically obtains a conformal skeleton, together with a conformal-level inseparability theorem. Third, we explain how the full metric scale is added by the cross-ratio modular datum and its Type III QFIM shadow, yielding a conditional full inseparability theorem under the additional Phase 2 package

$$(\text{LImp}^1) + (\text{CR-nd}) + (\text{SFa}).$$

Two worked examples, one based on the Rindler/free-field chiral half-line model and one on the Virasoro net, show that the Phase 2 conditions are not merely formal placeholders but are realized in structurally distinct settings.

The resulting philosophical payoff is twofold. First, the hard problem is reframed: the primary error lies not in failing to build a bridge, but in assuming that the bridge must connect independently specifiable domains. Second, the residual explanatory gap is relocated rather than denied. Once independent describability fails, the remaining question is no longer how matter produces experience, but why a finite observer embedded in a single algebraic structure cannot completely internalize the relation between structural invariants and lived articulation. That second issue belongs to the limits of self-description, not to a production gap between two ontologically detached realms.

The rest of the paper is organized as follows. Section 2 identifies the mathematical counterparts of physical structure and first-person time, and formulates the independent-describability issue as a question of independent deformability. Section 3 states the algebraic framework and axioms. Section 4 proves the conditional Phase 1 result: from Paper I's shell interface and dynamical package to a canonical conformal skeleton and Weak Inseparability. Section 5 develops the Phase 2 completion via the cross-ratio modular scale datum and the full inseparability theorems. Section 6 identifies abstract sufficient conditions for the chart axiom $(LImp^1)$: for Diff-covariant nets it is fully discharged, and for non-Diff-covariant nets the remaining burden is isolated to transportability and regularity assumptions. Section 7 verifies the remaining Phase 2 conditions in two worked examples. Section 8 then returns to the hard problem directly and argues that the theorem shows its standard formulation rests on a presupposition not available in the stated algebraic setting, while relocating the remaining explanatory gap to the self-referential horizon of finite observers. Section 9 discusses the broader mathematical and physical significance, and Section 10 states the explicit non-claims and open problems.

1.1 Logical Roadmap

The argument has two phases, each with a clear input-output structure.

Phase 1 (Sections 3–4). Input: the axiom package $(U1) + (C) + (Amb) + (CN_{\exists})$, working with the carrier specified by $(U3)$. Output: a canonical conformal skeleton via the Guido-Longo-Wiesbrock reconstruction, and a conformal-level Weak Inseparability theorem (Theorem 4.8). The key new step is that half-sided modular inclusion is *derived* from the modular-decoherence scaling axiom (C), not assumed.

Phase 2 (Section 5). Input: the additional conditions $(LImp^1)+(CR\text{-}nd)+(SFa)$. Output: a full geometric datum (conformal skeleton + scale shadow) and bidirectional Full Inseparability (Theorem 5.7). The key new object is the cross-ratio modular scale datum $\{K_b\}$ and its Araki-Hessian QFIM shadow h_b.

Section 6 provides abstract sufficient conditions for $(LImp^1)$. Section 7 verifies all Phase 2 conditions in two worked examples. Sections 8–10 draw philosophical consequences, discuss broader significance, and state open problems.

2 Mathematical Identification: Geometry as Physics, Modular Flow as First-Person Time

The philosophical argument of this paper rests on a mathematical theorem. For the theorem to bear on the hard problem, the translation between philosophical concepts and mathematical objects must be made explicit and defended. This section fixes the three identifications that underwrite the rest of the paper: emergent geometry as the mathematical content of physical structure, Tomita-Takesaki modular flow as the mathematical content of first-person time, and independent deformability as the mathematical translation of independent describability.

2.1 Emergent Geometry as Physical Structure

In the standard formulation of the hard problem, "physical structure" is left deliberately broad. It may refer to microphysical states, functional organization, or spacetime geometry, depending on the author. For the present purposes, the relevant notion of physical structure is not an arbitrary state description but the emergent spacetime geometry associated with a quantum-algebraic datum.

The reason for this choice is not merely technical convenience. In the algebraic approach to quantum field theory, spacetime is not a fixed background container. It is recovered from the algebraic relations among observables. The programme initiated by Wiesbrock and developed by Guido, Longo, and others has shown that, under appropriate conditions, the inclusion structure of von Neumann algebras determines a conformal geometry [17, 5], and in favorable cases a full Lorentzian metric. In this setting, "physical structure" is not something given prior to the algebra. It is something read off from the algebra.

This is the correct level of description for the hard problem, because the hard problem is not about the relation between consciousness and some particular physical substrate such as neurons or silicon. It is about the relation between the subjective and the objective as structural categories. If the objective-physical side is itself an emergent reading of algebraic data, then the question of whether it can be independently specified becomes a question about the algebraic structure, not about a particular material realization.

We therefore identify:

$$\text{physical structure} \longleftrightarrow \text{Geom}(\mathcal{N}, \omega),$$

where Geom denotes the emergent geometry extracted from a family $\mathcal{N}$ of von Neumann algebras in a state ω. At the Phase 1 level, this is a conformal skeleton. At the full level, it includes the metric scale.

2.2 Modular Flow as First-Person Time

The second identification is less familiar in the philosophy of mind but is equally well motivated from the algebraic side.

Given a von Neumann algebra $\mathcal{A}$ and a faithful normal state ω, the Tomita-Takesaki theorem [14] produces a canonical one-parameter automorphism group σ_t^ω of $\mathcal{A}$, the modular automorphism group. This group has three properties that make it the natural mathematical counterpart of first-person temporal structure.

First, uniqueness. The modular flow is the unique automorphism group satisfying the KMS condition at inverse temperature $\beta = 1$ with respect to ω. No external clock or background time parameter is needed to define it. It is intrinsic to the pair $(\mathcal{A}, \omega)$.

Second, algebraic relativity. Different choices of algebra $\mathcal{A}$ within the same ambient structure yield different modular flows. This is not a defect. It is the mathematical expression of the fact that different observational perspectives give rise to different temporal experiences. The Unruh effect is the most familiar physical instance [15]: an accelerated observer's modular flow differs from that of an inertial observer, and the difference is experienced as a thermal bath.

Third, structural inseparability at the operator level. The Tomita-Takesaki polar decomposition

$$S_\Omega = J_\Omega \Delta_\Omega^{1/2}$$

is unique. The modular operator Δ and the modular conjugation J are not independently choosable components. One cannot have Δ without J, nor J without Δ. Since Δ^{it} generates the modular flow and J implements the exchange between the algebra and its commutant, this already hints at the inseparability that the main theorem will establish at the geometric level: the temporal structure (Δ) and the spatial/duality structure (J) are two factors of a single operator.

We therefore identify:

$$\text{first-person time} \longleftrightarrow \sigma_t^\omega = \Delta_\omega^{it}(\cdot)\Delta_\omega^{-it}.$$

More precisely, the full modular sector includes not only the one-parameter flow but also the modular Hamiltonian $K_\omega = -\ln \Delta_\omega$, the modular conjugation J_ω, and, at the extended level, the local modular scale data $\{K_b\}$ introduced in Phase 2.

2.3 From Independent Describability to Independent Deformability

With the two identifications in place, the philosophical question can be stated mathematically.

The hard problem, in its standard form, presupposes that one can give a complete specification of the physical facts and then ask, as a further question, why those facts are accompanied by experience. Translated into the present framework, this becomes: can one specify $\mathrm{Geom}(\mathcal{N}, \omega)$ without thereby fixing $\mathrm{Mod}(\mathcal{N}, \omega)$, or conversely?

The natural mathematical formalization of "independent specification" is independent deformability. Two structures are independently describable, in the relevant sense, if and only if there exist admissible deformations of the underlying algebraic data that change one while leaving the other invariant. If no such deformation exists, then specifying one structure automatically constrains the other, and the two cannot be treated as independently given.

This is a precise, falsifiable mathematical question. It does not depend on intuitions about zombies, inverted qualia, or conceivability arguments. It depends on the rigidity properties of the map

$$(\mathcal{N}, \omega) \longmapsto \big(\mathrm{Geom}(\mathcal{N}, \omega),\ \mathrm{Mod}(\mathcal{N}, \omega)\big)$$

within the admissible class defined by the axiom package.
The hard problem, so translated, becomes:

> Does there exist an admissible deformation of $(\mathcal{N}, \omega)$ that changes Geom while preserving Mod, or vice versa?

The inseparability theorems of this paper answer: no. At the conformal level (Phase 1), the answer is unconditional within the axiom class. At the full scale-sensitive level (Phase 2), the answer is conditional on three regularity inputs that are verified in two worked examples.

2.4 What This Translation Does and Does Not Claim

Two caveats are essential.

First, the identification of modular flow with first-person time is a structural correspondence, not an identity claim. We do not assert that modular flow "is" consciousness in any reductive sense. We assert that modular flow is the correct mathematical object that plays the structural role assigned to first-person temporal experience in the hard-problem debate. The correspondence is justified by the three properties listed above: intrinsicness, perspectival relativity, and operator-level inseparability from spatial structure.

Second, the identification of emergent geometry with physical structure is restricted to the level of spacetime geometry. It does not claim to capture all of physics. In particular, it does not address the relation between consciousness and specific material substrates such as neural tissue. The claim is that at the level of structural categories relevant to the hard problem, emergent geometry is the right proxy for the objective-physical side. Whether this extends to more fine-grained physical descriptions is an open question that lies beyond the scope of the present paper.

3 Algebraic Framework and Axioms

We now state the minimal algebraic framework in which the failure of independent deformability can be proved. The point of this section is not to give a general introduction to Type III_1 factors or to algebraic quantum field theory. It is to isolate the exact package of assumptions needed for the Phase 1 theorem and to indicate, in each case, why that assumption belongs in the present hard-problem-led setting.

3.1 Ambient Datum and Admissible Object

We work in the standard form of an ambient von Neumann algebra

$$(\mathcal{H}_U, \mathcal{A}_U, \Omega),$$

where Ω is cyclic and separating for $\mathcal{A}_U$. This is the minimal regularity needed for Tomita-Takesaki theory and is satisfied in all standard AQFT models [6]. Inside $\mathcal{A}_U$ we select an accessible von Neumann subalgebra

$$\mathcal{A}_c \subset \mathcal{A}_U,$$

and define the reference state

$$\omega(X) := \langle \Omega, X\Omega \rangle.$$

The pair

$$(\mathcal{A}_c, \omega)$$

is the basic object of the theory. In the intended applications, $\mathcal{A}_c$ is of Type III_1. This is not an arbitrary technical preference. Type III_1 is the natural habitat of local quantum field theoretic algebras [6], and it is precisely the setting in which modular theory is rich enough to support the present route from algebraic accessibility data to emergent geometry.

The ambient algebra $\mathcal{A}_U$ should be thought of as the unreduced field of possible observables, while $\mathcal{A}_c$ is the algebra actually stabilized or selected by the accessibility criterion. The question of the paper is then asked not about the entire ambient structure at once, but about what can be extracted from the selected pair $(\mathcal{A}_c, \omega)$.

3.2 Inherited Accessibility Conditions

The first two carrier assumptions are motivated by the HAFF setting [8], but are stated here as independent Paper I inputs.

(U1) Faithfulness. The restriction of the reference state to the accessible algebra is faithful:

$$\omega|_{\mathcal{A}_c} \text{ is faithful.}$$

Equivalently, Ω is cyclic and separating for $\mathcal{A}_c$.

This is the minimal condition that allows modular theory to be applied to $(\mathcal{A}_c, \omega)$. In the present philosophical translation, $(U1)$ says that the allegedly "first-person" sector is not attached to a degenerate algebraic viewpoint. The accessible algebra must support a genuine intrinsic modular structure.

(U3) Stable carrier. The accessible algebra is supplied as a von Neumann subalgebra that is faithful and forward-stable under the decoherence semigroup:

$$\mathcal{E}_t(\mathcal{A}_c) \subset \mathcal{A}_c, \qquad \forall t \geq 0,$$

Here $\{\mathcal{E}_t\}_{t\geq 0}$ is the decoherence semigroup acting on the ambient algebra: a strongly continuous one-parameter semigroup of normal completely positive unital maps on $\mathcal{A}_U$ satisfying the Markov property $\mathcal{E}_0 = \mathrm{id}$ and $\mathcal{E}_{s+t} = \mathcal{E}_s \circ \mathcal{E}_t$. Its physical role is to model the progressive loss of coherence between the accessible algebra and its environment. No identity $\mathcal{A}_c = \mathrm{Fix}(\mathcal{E})$, and no uniqueness or maximality property, is assumed here. (For the motivating construction, see [8].)

The role of $(U3)$ is only to specify the carrier on which the modular construction is performed. The Phase 1 proofs use forward stability, not a selection or uniqueness theorem. Canonical selection of $\mathcal{A}_c$ from a larger registry of accessible effects, probe channels, records, or exact fixed points is a separate interface problem.

Throughout this paper, we work with a carrier satisfying $(U3)$. No claim below promotes that supplied carrier to the unique accessible algebra.

3.3 The New Dynamical Axiom

The crucial new input is the scaling relation between decoherence and modular evolution.

(C) Modular-decoherence scaling covariance. For all $s \in \mathbb{R}$ and $t \geq 0$,

$$\mathcal{E}_t \circ \mathrm{Ad}\,\Delta_{\mathcal{A}_c}^{is} = \mathrm{Ad}\,\Delta_{\mathcal{A}_c}^{is} \circ \mathcal{E}_{e^{2\pi s}t}.$$

Here $\mathrm{Ad}\,\Delta_{\mathcal{A}_c}^{is}$ denotes the outer conjugation action of the modular unitary on the ambient operator algebra.

This is the only genuinely new dynamical axiom introduced in Phase 1. Its content is that modular evolution does not merely preserve the accessible algebra tautologically. It rescales the effective decoherence depth. In the language of the hard problem, (C) is the first place where the object-side and act-side structures are explicitly tied together. It is therefore the real replacement for the earlier non-scrambling intuition.

Remark 3.1 (Physical plausibility of (C)). Axiom (C) holds whenever the modular flow of the accessible algebra acts as a dilation on the decoherence time parameter. This is the case in the Bisognano-Wichmann setting, where modular flow is a Lorentz boost and decoherence depth scales with the boost parameter, and more generally in any Rindler-type wedge algebra. The axiom is a nontrivial dynamical input: it ties the intrinsic modular time of the accessible algebra to the extrinsic decoherence structure of the ambient algebra. Its domain of validity beyond the HAFF setting [8] is an open question.

Remark 3.2 (Finite-dimensional bounded-generator obstruction). There is no nontrivial finite-dimensional realization of (C) with a bounded generator. If $\mathcal{E}_t = e^{t\mathcal{L}}$, differentiating $\mathcal{E}_t\sigma_s = \sigma_s\mathcal{E}_{e^{2\pi s}t}$ at $t = 0$ gives

$$\sigma_s^{-1}\mathcal{L}\sigma_s = e^{2\pi s}\mathcal{L}.$$

Because the modular action is isometric, similarity by σ_s preserves the operator norm. Hence $\|\mathcal{L}\| = e^{2\pi s}\|\mathcal{L}\|$ for every s, forcing $\mathcal{L} = 0$. Thus a nontrivial model must leave the finite-dimensional bounded-generator regime; this remark is an obstruction, not an existence claim for the Type III case.

3.4 Ambient Shell Realization

To turn (C) into a geometric statement, one needs an external family on which the rescaling can act.

(Amb) Operational shell interface. Fix $\varepsilon > 0$. For O in the norm unit ball $\mathrm{Ball}(\mathcal{A}_U)$, define

$$d_\tau(O) := \sup_{0\le t\le\tau} \|\mathcal{E}_t(O) - O\|, \qquad \mathcal{S}_\varepsilon(\tau) := \{O \in \mathrm{Ball}(\mathcal{A}_U) : d_\tau(O) \le \varepsilon\},$$

and supply the ambient shell family by

$$\mathcal{A}(\tau) := W^*\big(\mathcal{A}_c \cup \mathcal{S}_\varepsilon(\tau)\big), \qquad \tau > 0.$$

The ambient modular action $\sigma_s = \mathrm{Ad}\,\Delta_{\mathcal{A}_c}^{is}$ is assumed to act on $\mathcal{A}_U$ by isometric $*$-automorphisms. The following closure and standardness properties are interface inputs:

1. for every $\tau > 0$,
$$\mathcal{A}_c \subset \mathcal{A}(\tau) \subset \mathcal{A}_U;$$
2. if $\tau_1 < \tau_2$, then
$$\mathcal{A}(\tau_1) \supset \mathcal{A}(\tau_2);$$
this nesting also follows directly from $d_{\tau_1} \le d_{\tau_2}$;
3. the accessible algebra is the core limit:
$$\mathcal{A}_c = \bigcap_{\tau>0} \mathcal{A}(\tau);$$
4. Ω is cyclic and separating for every $\mathcal{A}(\tau)$.

The point of (Amb) is to make shell membership and von Neumann closure explicit. It is a Paper I interface input: Foundation I does not derive this membership rule, its core identity, or its standardness. Crucially, modular covariance is not included in (Amb); shell scaling will be derived from the independent dynamical axiom (C).

3.5 Existential Conormality

The final input required for the Phase 1 conformal reconstruction is a second-direction existence condition.

(CN$_\exists$) Existential conormality. There exists some $\mu_0 > 0$ such that, for

$$\mathcal{C}_{\mu_0} := \mathcal{B}'_{\mu_0} \cap \mathcal{A}_c,$$

the vector Ω is cyclic.

The algebras $\mathcal{B}_\mu$ will be constructed from (C) and (Amb) in Section 4; the present point is only that one cyclicity assumption at one scale is enough to close the second direction and trigger the GLW reconstruction.

Philosophically, $(CN_\exists)$ is the minimal extra input needed to pass from a single half-sided modular direction to a full conformal skeleton. It should therefore be read neither as an arbitrary technical patch nor as a hidden second dynamics. It is an existence-type condition ensuring that the accessible algebra contains enough internal relational richness to support the second lightlike direction.

3.6 What Is Explicitly Not Assumed

One older condition is intentionally removed.

(U2) is deleted. The statement that the modular group preserves the accessible algebra,

$$\sigma_t^{\mathcal{A}_c}(\mathcal{A}_c) = \mathcal{A}_c,$$

is automatic from Tomita-Takesaki theory [14] and therefore carries no genuine selection content.

This deletion matters conceptually. The paper does not rest on a tautological "modular stability" assumption. The real content enters through (C), where modular evolution acts outwardly on the shell family rather than merely inwardly on $\mathcal{A}_c$ itself.

It is also worth noting that the stronger exhaustion condition

$$\text{OE}: \quad W^*\Big(\bigcup_{\tau>0} \mathcal{A}(\tau)\Big) = \mathcal{B}(\mathcal{H}_U)$$

is not part of the minimal Phase 1 package. It is only relevant for stronger scalar-core statements that are not needed for the present conformal-level theorem.

3.7 The Minimal Phase 1 Package

We may now summarize the standing assumptions. Throughout the paper we work within the admissible class selected by $(U3)$. The proof-relevant Phase 1 package is:

$$(U1) + (C) + (Amb) + (CN_\exists).$$

The logic of this package is straightforward.

- $(U1)$ makes modular theory available on the accessible algebra.
- (C) ties modular evolution to decoherence depth.
- (Amb) gives that depth a concrete shell realization inside the ambient algebra.
- $(CN_{\exists})$ provides the minimal extra cyclicity needed to reconstruct the second direction and hence the conformal skeleton.

The remainder of Phase 1 consists in proving that this proof-relevant package is already enough to derive a standard half-sided modular inclusion, a canonical GLW Möbius net, and therefore a conformal-level inseparability theorem. That is the task of the next section.

4 From Axioms to Conformal Skeleton (Phase 1)

4.1 From Axiom C to Standard Half-Sided Modular Inclusion

The first structural point of the paper is that the half-sided modular inclusion needed by the Wiesbrock-GLW machinery is not postulated. It is derived conditionally from Paper I's dynamical axiom (C) and shell interface (Amb). In particular, the present route differs sharply from the earlier formulation in which HSMI appeared as an external ansatz. Here it is the output of the Phase 1 assumptions.

Throughout this section we assume $(U1)$, (C), and the explicit shell interface (Amb), and we work with the carrier specified by $(U3)$. We write

$$\sigma_t := \sigma_t^{\mathcal{A}_c}$$

for the modular automorphism group of $(\mathcal{A}_c, \Omega)$ and

$$J := J_{\mathcal{A}_c}$$

for the associated modular conjugation. By (Amb), we are given a decreasing family of ambient shell algebras

$$\mathcal{A}_c \subset \mathcal{A}(\tau) \subset \mathcal{A}_U, \qquad \tau > 0,$$

with

$$\mathcal{A}_c = \bigcap_{\tau>0} \mathcal{A}(\tau),$$

and whose parameter τ is interpreted as a decoherence window.

4.2 Outer Scaling from Modular-Decoherence Covariance

The real content of axiom (C) is that the modular flow of the accessible algebra does not merely preserve $\mathcal{A}_c$ tautologically. Rather, it rescales the ambient shell parameter. This is the first point at which the older $(U2)$-style formulation must be replaced.

Proposition 4.1 (Outer scaling law)**.** *Conditional on the shell interface* (Amb) *and the independent dynamical covariance axiom* (C)*,*

$$\sigma_s(\mathcal{A}(\tau)) = \mathcal{A}(e^{-2\pi s}\tau), \qquad \forall s \in \mathbb{R},\ \tau > 0.$$

Proof. For $O \in \mathrm{Ball}(\mathcal{A}_U)$, axiom (C) and isometry of σ_s give

$$\begin{aligned} d_{e^{-2\pi s}\tau}(\sigma_s O) &= \sup_{0\le t\le e^{-2\pi s}\tau} \|\mathcal{E}_t(\sigma_s O) - \sigma_s O\| \\ &= \sup_{0\le t\le e^{-2\pi s}\tau} \|\mathcal{E}_{e^{2\pi s}t}(O) - O\| \\ &= d_\tau(O), \end{aligned}$$

where the last line uses $u = e^{2\pi s}t$. Thus $\sigma_s \mathcal{S}_\varepsilon(\tau) = \mathcal{S}_\varepsilon(e^{-2\pi s}\tau)$. Since modular flow preserves $\mathcal{A}_c$, applying σ_s to the generating sets and passing to their von Neumann closures proves the equality. □

This result is the correct algebraic replacement for the old non-scrambling intuition: modular time rescales decoherence depth. It is already an inseparability statement in weak form, because the shell parameter and the modular parameter are no longer independently adjustable.

4.3 Outer Standardness

The next observation is simple but essential: the ambient shells are standard for the same vector Ω. This is where the ambient realization is used as a regularity hypothesis rather than as extra dynamics.

Lemma 4.2 (Outer standardness)**.** *For every $\tau > 0$, the vector Ω is cyclic and separating for $\mathcal{A}(\tau)$.*

Proof. Since $\mathcal{A}_c \subset \mathcal{A}(\tau)$ and Ω is cyclic for $\mathcal{A}_c$ by $(U1)$, it is cyclic for $\mathcal{A}(\tau)$. Since $\mathcal{A}(\tau) \subset \mathcal{A}_U$ and Ω is separating for $\mathcal{A}_U$ by the ambient datum (Section 3.1), separation passes to the subalgebra $\mathcal{A}(\tau)$. □

This standardness will allow us to transport the outer shell family into an internal family inside $\mathcal{A}_c$ by Tomita-Takesaki duality.

4.4 The J-Dual Internal Filtration

The key construction is to turn the outer shell family into a family of subalgebras of the accessible algebra itself. This is the point at which the earlier interiorization axiom becomes unnecessary.

For $\mu > 0$, define

$$\mathcal{B}_\mu := J\,\mathcal{A}(1/\mu)'\,J.$$

The next proposition records the properties of this family needed for the HSMI step.

Proposition 4.3 (J-dual internal filtration)**.** *The family $\{\mathcal{B}_\mu\}_{\mu>0}$ satisfies:*

1. *$\mathcal{B}_\mu \subset \mathcal{A}_c$ for all $\mu > 0$;*
2. *if $\mu_1 > \mu_2$, then $\mathcal{B}_{\mu_1} \subset \mathcal{B}_{\mu_2}$;*
3. *Ω is cyclic and separating for each $\mathcal{B}_\mu$;*
4. *the modular group rescales the family according to*

$$\sigma_s(\mathcal{B}_\mu) = \mathcal{B}_{e^{-2\pi s}\mu};$$

5. *the family generates the accessible algebra:*

$$W^*\Big(\bigcup_{\mu>0}\mathcal{B}_\mu\Big)=\mathcal{A}_c.$$

Proof. The inclusion $\mathcal{A}_c\subset\mathcal{A}(1/\mu)$ implies

$$\mathcal{A}(1/\mu)'\subset\mathcal{A}_c',$$

hence

$$\mathcal{B}_\mu=J\,\mathcal{A}(1/\mu)'\,J\subset J\,\mathcal{A}_c'\,J=\mathcal{A}_c,$$

which proves (1).

For (2), if $\mu_1>\mu_2$, then $1/\mu_1<1/\mu_2$, so the monotonicity of the outer shell family gives

$$\mathcal{A}(1/\mu_1)\supset\mathcal{A}(1/\mu_2),$$

hence

$$\mathcal{A}(1/\mu_1)'\subset\mathcal{A}(1/\mu_2)'$$

and therefore

$$\mathcal{B}_{\mu_1}\subset\mathcal{B}_{\mu_2}.$$

For (3), Lemma 4.2 shows that Ω is cyclic and separating for $\mathcal{A}(1/\mu)$. Therefore Ω is also cyclic and separating for $\mathcal{A}(1/\mu)'$. Since $J\Omega=\Omega$ and J is antiunitary, the same holds for

$$J\,\mathcal{A}(1/\mu)'\,J=\mathcal{B}_\mu.$$

For (4), using the Tomita relation

$$\Delta^{is}J=J\Delta^{-is},$$

we compute

$$\sigma_s(\mathcal{B}_\mu)=\Delta^{is}J\,\mathcal{A}(1/\mu)'\,J\Delta^{-is}=J\Delta^{-is}\mathcal{A}(1/\mu)'\Delta^{is}J=J\,\sigma_{-s}\big(\mathcal{A}(1/\mu)\big)'\,J.$$

Now Proposition 4.1 gives

$$\sigma_{-s}\big(\mathcal{A}(1/\mu)\big)=\mathcal{A}(e^{2\pi s}/\mu),$$

so

$$\sigma_s(\mathcal{B}_\mu)=J\,\mathcal{A}(e^{2\pi s}/\mu)'\,J=\mathcal{B}_{e^{-2\pi s}\mu}.$$

Finally, (5) follows from the core limit in (Amb):

$$\mathcal{A}_c=\bigcap_{\tau>0}\mathcal{A}(\tau)\quad\Longrightarrow\quad\mathcal{A}_c'=W^*\Big(\bigcup_{\tau>0}\mathcal{A}(\tau)'\Big).$$

Applying $J(\cdot)J$ yields

$$\mathcal{A}_c=J\mathcal{A}_c'J=W^*\Big(\bigcup_{\tau>0}J\,\mathcal{A}(\tau)'\,J\Big)=W^*\Big(\bigcup_{\mu>0}\mathcal{B}_\mu\Big).$$

□

This construction is the first genuinely new structural step of the Phase 1 chain. The internal family is not imposed from outside. It is obtained functorially from the shell realization and modular data.

4.5 Half-Sided Modular Inclusion as a Theorem

We may now isolate the fixed inclusion that drives the Wiesbrock machinery.

Theorem 4.4 (HSMI from modular-decoherence covariance)**.** *For any fixed $\mu_0 > 0$, the inclusion*

$$(\mathcal{B}_{\mu_0} \subset \mathcal{A}_c, \Omega)$$

is a half-sided modular inclusion, namely

$$\sigma_t(\mathcal{B}_{\mu_0}) \subset \mathcal{B}_{\mu_0}, \qquad \forall t \leq 0.$$

Proof. By Proposition 4.3(3), Ω is cyclic and separating for both $\mathcal{B}_{\mu_0}$ and $\mathcal{A}_c$. By Proposition 4.3(4),

$$\sigma_t(\mathcal{B}_{\mu_0}) = \mathcal{B}_{e^{-2\pi t}\mu_0}.$$

If $t \leq 0$, then $e^{-2\pi t} \geq 1$, hence $e^{-2\pi t}\mu_0 \geq \mu_0$. By Proposition 4.3(2),

$$\mathcal{B}_{e^{-2\pi t}\mu_0} \subset \mathcal{B}_{\mu_0}.$$

Therefore

$$\sigma_t(\mathcal{B}_{\mu_0}) \subset \mathcal{B}_{\mu_0}, \qquad t \leq 0.$$

□

The important point is conceptual as much as technical: in the present framework, HSMI is not a geometric input imported from AQFT. It is the algebraic consequence of a scaling law that ties modular flow to decoherence depth.

4.6 The Wiesbrock-Borchers Positive Generator

Once the fixed inclusion is available, the standard Wiesbrock-Borchers conclusion follows.

Corollary 4.5 (Positive translation generator for the fixed inclusion)**.** *For each fixed $\mu_0 > 0$, there exists a unique positive self-adjoint operator*

$$P_{\mu_0} \geq 0$$

such that, with

$$U_{\mu_0}(a) := e^{-iaP_{\mu_0}}, \qquad a \geq 0,$$

one has

1. $U_{\mu_0}(a)\Omega = \Omega$ *for all* $a \geq 0$*;*
2. *the Borchers covariance relation*

$$\Delta_{\mathcal{A}_c}^{is} U_{\mu_0}(a) \Delta_{\mathcal{A}_c}^{-is} = U_{\mu_0}(e^{-2\pi s}a)$$

holds for all $s \in \mathbb{R}$ *and* $a \geq 0$.

Proof. This is the standard Wiesbrock theorem for a fixed half-sided modular inclusion [17], together with Borchers covariance [2]. □

At this stage we do not claim a pointwise identification between the entire family $\{\mathcal{B}_\mu\}_{\mu>0}$ and the Wiesbrock orbit generated by P_{μ_0}. Such an identification requires an additional normalization or uniqueness argument and is not needed for the Phase 1 main chain.

This completes the first half of Phase 1. We have shown that the accessible algebra canonically carries the half-sided modular structure needed to enter the Möbius reconstruction theorem.

4.7 GLW Conformal Skeleton and Weak Inseparability

The second half of Phase 1 upgrades the fixed half-sided modular inclusion to a full conformal skeleton. The only additional input is an existential conormality condition. Once this is imposed, the Guido-Longo-Wiesbrock reconstruction theorem produces a canonical strongly additive local Möbius net on S^1, and with it the conformal-level version of inseparability.

Throughout this section we assume, in addition to the hypotheses of Section 4.1, the conormality condition $(CN_\exists)$: there exists $\mu_0 > 0$ such that the vector Ω is cyclic for

$$\mathcal{C}_{\mu_0} := \mathcal{B}'_{\mu_0} \cap \mathcal{A}_c.$$

4.8 The Second Direction from Existential Conormality

The role of $(CN_\exists)$ is very precise. It is not used to construct the half-sided inclusion itself. That has already been achieved in Section 4.1. Rather, it upgrades the fixed inclusion to a standard one in the sense required by the GLW theorem.

For $\mu > 0$, define

$$\mathcal{C}_\mu := \mathcal{B}'_\mu \cap \mathcal{A}_c.$$

Proposition 4.6 (The second direction)**.** *Under $(U1)$, (C), (Amb), and $(CN_\exists)$, the family $\{\mathcal{C}_\mu\}_{\mu>0}$ satisfies:*

1. *$\mathcal{C}_\mu \subset \mathcal{A}_c$ and Ω is separating for $\mathcal{C}_\mu$;*
2. *the modular group rescales the family by*
$$\sigma_t(\mathcal{C}_\mu) = \mathcal{C}_{e^{-2\pi t}\mu};$$
3. *for all $t \geq 0$,*
$$\sigma_t(\mathcal{C}_\mu) \subset \mathcal{C}_\mu;$$
4. *if Ω is cyclic for one $\mathcal{C}_{\mu_0}$, then by modular propagation it is cyclic for all $\mathcal{C}_\mu$.*

In particular, the fixed inclusion

$$(\mathcal{B}_{\mu_0} \subset \mathcal{A}_c, \Omega)$$

is standard.

Proof. The inclusion $\mathcal{C}_\mu \subset \mathcal{A}_c$ is immediate from the definition. Separation of Ω follows from $(U1)$, since $\omega|_{\mathcal{A}_c}$ is faithful and faithfulness passes to subalgebras.

For modular covariance,

$$\sigma_t(\mathcal{C}_\mu) = \sigma_t(\mathcal{B}'_\mu \cap \mathcal{A}_c) = \sigma_t(\mathcal{B}_\mu)' \cap \sigma_t(\mathcal{A}_c) = \mathcal{B}'_{e^{-2\pi t}\mu} \cap \mathcal{A}_c = \mathcal{C}_{e^{-2\pi t}\mu},$$

where we used Proposition 4.3(4) and the tautological invariance of $\mathcal{A}_c$ under its own modular group.

If $t \geq 0$, then $e^{-2\pi t} \leq 1$, so the monotonicity of the $\mathcal{B}_\mu$ family implies the reverse monotonicity of the relative commutants:

$$\mathcal{C}_{e^{-2\pi t}\mu} \subset \mathcal{C}_\mu.$$

Thus

$$\sigma_t(\mathcal{C}_\mu) \subset \mathcal{C}_\mu.$$

Finally, $(CN_\exists)$ gives cyclicity for one $\mathcal{C}_{\mu_0}$, and the modular covariance formula propagates this cyclicity along the entire orbit. Since Section 4.1 already gives cyclicity and separation for $\mathcal{B}_{\mu_0}$ and $\mathcal{A}_c$, the inclusion is standard. □

This is the exact point at which the second direction closes. No stronger global conormality assumption is needed for the fixed inclusion that feeds the GLW theorem.

4.9 The GLW Reconstruction Theorem

Once standardness is available, the conformal skeleton appears by a known AQFT theorem. The novelty in the present paper is not the GLW theorem itself, but the conditional construction of its input from the explicit Paper I shell interface and dynamical package.

Theorem 4.7 (Conformal skeleton via Guido-Longo-Wiesbrock)**.** *Fix $\mu_0 > 0$ as in $(CN_\exists)$. Then the standard half-sided modular inclusion*

$$(\mathcal{B}_{\mu_0} \subset \mathcal{A}_c, \Omega)$$

determines a unique strongly additive local Möbius covariant net

$$I \longmapsto \mathcal{A}_{\mathrm{GLW}}(I)$$

on S^1 such that

$$\mathcal{A}_{\mathrm{GLW}}(0, +\infty) = \mathcal{A}_c, \qquad \mathcal{A}_{\mathrm{GLW}}(1, +\infty) = \mathcal{B}_{\mu_0},$$

and Ω is the vacuum vector for the corresponding positive-energy representation of $\mathrm{PSL}(2, \mathbb{R})$.

Proof. This is precisely the Guido-Longo-Wiesbrock reconstruction theorem for standard half-sided modular inclusions [5]. □

Two clarifications are important here.

First, at this stage no separate relative-position hypothesis is needed. Standard HSMI is enough to reconstruct the Möbius net. The previously suspected relative-position bottleneck concerns later identification questions, not the existence of the conformal skeleton itself.

Second, the net produced here is an abstract but canonical conformal object attached to the fixed inclusion. The remaining question of identifying this GLW net pointwise with the pre-given shell filtrations or later cross-ratio families belongs to the next phase of the program.

4.10 The Conformal Skeleton as the Output of Modular Data

Theorem 4.7 already yields the first inseparability statement of the paper. The data entering the standard inclusion are modular data; the output is a conformal geometry in the minimal sense appropriate to one spatial and one temporal dimension, namely a causal/conformal skeleton.

To make this precise, let

$$\mathrm{Mod}_{\mathrm{conf}}(\mathcal{A}_c, \omega)$$

denote the conformal modular sector determined in Phase 1: the modular group of $(\mathcal{A}_c, \Omega)$, the fixed-inclusion positive generator P_{μ_0}, and the corresponding Möbius representation recovered from the standard HSMI. Let

$$\mathrm{Conf}(\mathcal{A}_c, \omega)$$

denote the conformal skeleton, equivalently the GLW net together with its Möbius vacuum representation.

Theorem 4.8 (Conditional Weak Inseparability)**.** *Conditional on the supplied shell interface (Amb), the independent dynamical axiom (C), $(U1)$, $(CN_\exists)$, and the carrier condition $(U3)$, the conformal skeleton and the conformal modular sector cannot be independently varied. More precisely:*

1. *the standard HSMI data determine a unique conformal skeleton;*
2. *the conformal skeleton, once normalized by the half-line identification above, determines the conformal modular sector;*
3. *therefore any admissible deformation that preserves one of these two structures must preserve the other.*

Proof. Item (1) is exactly Theorem 4.7.

For item (2), the Möbius net comes equipped with a positive-energy representation of $\mathrm{PSL}(2,\mathbb{R})$. Once the half-line normalization

$$\mathcal{A}_{\mathrm{GLW}}(0,+\infty) = \mathcal{A}_c, \qquad \mathcal{A}_{\mathrm{GLW}}(1,+\infty) = \mathcal{B}_{\mu_0}$$

is fixed, the corresponding dilation and translation subgroups recover the modular action and the positive translation generator associated with the inclusion. In other words, the conformal skeleton rigidly determines the conformal modular sector.

Item (3) is immediate. If an admissible deformation preserves the conformal modular sector, item (1) forces the conformal skeleton to remain fixed. Conversely, if it preserves the conformal skeleton, item (2) forces the modular sector to remain fixed. Hence the two structures are not independently deformable. □

This result is weak only in the sense that it stops at the conformal level. It does not yet recover metric scale. That is the purpose of Step I and Step II. But it already establishes the central conceptual point of Phase 1:

> causal/conformal geometry and modular time are not two separately specified ingredients;
> they are two readings of the same standard half-sided modular data.

4.11 What Phase 1 Has Actually Closed

It is useful to isolate the precise mathematical output of the section before moving on to metric scale.

1. HSMI is not separately assumed. It is a conditional theorem derived from the explicit interface (Amb) and dynamical axiom (C).
2. The second direction is no longer a global mystery. It is isolated into the existence-type conormality input $(CN_{\exists})$.
3. The conformal skeleton is no longer conjectural. It is reconstructed canonically by GLW from the fixed standard inclusion.
4. The first inseparability theorem is already in hand: conformal geometry and conformal modular data are mutually rigid.

What remains open after Phase 1 is not the existence of the conformal skeleton, but the recovery of metric scale from local modular data. That is why the next phase turns to the cross-ratio family and to the canonical local modular scale datum

$$\{K_b\}_{b\in(0,1)}.$$

5 From Conformal Skeleton to Full Geometry (Phase 2)

Phase 1 established that conformal geometry and conformal modular data are inseparable. But the conformal skeleton determines only the causal structure, the light cones. It does not determine distances. A conformal class of Lorentzian metrics contains infinitely many metrics related by Weyl rescaling. To reach the full inseparability statement, we must show that the metric scale is also locked to modular data.

This is the task of Phase 2. The strategy has three steps: identify the correct local geometric modulus, extract a canonical modular scale datum from it, and show that this datum determines the metric scale via a Type III-compatible quantum Fisher information metric.

5.1 Why Single-Interval Deformations Fail

The first instinct is to deform a single interval within the GLW net and track how the modular Hamiltonian changes. This fails for a precise reason.

On S^1, the group $\mathrm{PSL}(2,\mathbb{R})$ acts transitively on the space of intervals. Any single-interval deformation, such as moving one endpoint while fixing the other, is a Möbius transformation. Since the vacuum state ω is Möbius-invariant, pulling back the restricted state along such a deformation produces a constant family. The quantum Fisher information metric of a constant family is zero.

The lesson is that single-interval endpoint motion is a gauge orbit, not a geometric deformation. To obtain a nontrivial modulus, one needs at least four boundary points.

5.2 The Cross-Ratio as the Minimal Non-Möbius Modulus

Consider a nested double-interval configuration within the GLW net:

$$\mathcal{M}_{\text{in}}(b) \subset \mathcal{M}_{\text{out}}, \qquad b \in (0,1).$$

In the standard half-line gauge, this means

$$\mathcal{M}_{\text{out}} = \mathcal{A}_{\text{GLW}}(0,\infty), \qquad \mathcal{M}_{\text{in}}(b) = \mathcal{A}_{\text{GLW}}(b,1).$$

The four boundary points are $0, b, 1, \infty$. The Möbius group has three real parameters, so after gauge-fixing, exactly one real modulus survives: the cross-ratio

$$\chi = b.$$

This is the minimal non-Möbius geometric degree of freedom available in the GLW net. Varying b genuinely changes the relative position of the inner and outer algebras in a way that no Möbius transformation can undo.

5.3 The Canonical Modular Scale Datum

For each $b \in (0,1)$, the inclusion $\mathcal{M}_{\text{in}}(b) \subset \mathcal{M}_{\text{out}}$ determines a modular operator $\Delta_{\mathcal{M}_{\text{in}}(b),\Omega}$ via the Tomita-Takesaki theorem applied to $\mathcal{M}_{\text{in}}(b)$ with respect to the vacuum vector Ω.

Definition 5.1 (Canonical modular scale datum)**.**

$$K_b := -\ln \Delta_{\mathcal{M}_{\text{in}}(b),\Omega}, \qquad b \in (0,1).$$

The family $\{K_b\}_{b\in(0,1)}$ is the canonical local modular scale datum. It is determined entirely by the GLW net and the vacuum state. No additional choices are involved.

Proposition 5.2 (Cross-ratio is the minimal non-Möbius modulus)**.** *A nested double-interval configuration in the GLW net has exactly one Möbius-invariant modulus, the cross-ratio $\chi = b$. In particular, the family $\{K_b\}$ captures the minimal local modular scale information that is not already contained in the conformal skeleton.*

This is the theorem-level output of Step I. It requires no additional axioms beyond Phase 1.

5.4 The Chart Axiom and the Scale Shadow

To convert the modular scale datum into a Riemannian metric on the cross-ratio parameter space, we need to differentiate the family $\{K_b\}$ and evaluate a quadratic form. This requires regularity that is not automatic from the abstract GLW construction.

Axiom (LImp^1). For each reference point $b_0 \in (0,1)$, there exists a neighborhood U_{b_0} and a unitary family

$$W_b \in \mathcal{U}(\mathcal{M}_{\text{out}}), \qquad b \in U_{b_0},$$

satisfying:

1. $W_{b_0} = \mathbf{1}$;
2. $W_b \mathcal{M}_* W_b^* = \mathcal{M}_{\text{in}}(b)$ for a fixed reference algebra $\mathcal{M}_* = \mathcal{M}_{\text{in}}(b_0)$;
3. $b \mapsto \omega(W_b A W_b^*)$ is C^1 for each $A \in \mathcal{M}_*$;
4. the tangent vector $\dot{\phi}_{b_0}$ at $b = b_0$ is represented by the modular Hamiltonian variation $\dot{K}_{b_0} = \frac{d}{db}|_{b=b_0} K_b$ via the standard Araki correspondence.

This axiom packages three regularity conditions into one: the existence of a local implementer (CR1), C^1 smoothness of the pulled-back state family (CR2), and tangent compatibility with the modular Hamiltonian variation (CR3).

Axiom (CR-nd) (Non-degeneracy). The Araki-Hessian quadratic form $\mathbf{Q}_{\omega,b}$ is strictly positive on the modular Hamiltonian variation:

$$\mathbf{Q}_{\omega,b}(\partial_b K_b, \partial_b K_b) > 0, \qquad \forall\, b \in (0,1).$$

In the worked examples, this follows from the positivity of the stress-tensor two-point function. It fails only if the cross-ratio deformation produces no state-level change, a physically degenerate situation.

Under these two conditions, the scale shadow is a genuine Riemannian metric on the cross-ratio parameter space.

Theorem 5.3 (Scale shadow, conditional on (LImp^1) + (CR-nd))**.** *Under the Phase 1 axioms and the additional conditions $(LImp^1)$ and $(CR\text{-}nd)$, the canonical modular scale datum induces a Riemannian scale shadow*

$$h_b = \mathbf{Q}_{\omega,b}(\partial_b K_b, \partial_b K_b)\, db^2,$$

where $\mathbf{Q}_{\omega,b}$ is the positive quadratic form induced by the Araki relative entropy Hessian [1] at the state $\omega|_{\mathcal{M}_{\text{in}}(b)}$ via the $(LImp^1)$ chart.

Remark 5.4. In the special case where the chart is of exponential type (CR-exp), this reduces to the familiar variance formula

$$h_b = \text{Var}_\omega(\partial_b K_b)\, db^2.$$

The general formula via $\mathbf{Q}_{\omega,b}$ does not require this specialization.

5.5 Geometric Datum: Conformal Skeleton + Slice Scale

The conformal skeleton from Phase 1 determines the causal structure, equivalently the conformal causal structure. The scale shadow $\{h_b\}$ determines the complementary scale information along the cross-ratio slice family. At the abstract level established in this paper, the pair $(\mathrm{Conf}, \{h_b\})$ is the full geometric datum. In the worked examples of Section 6, the cross-ratio parameter b naturally parametrizes the spatial position of the inner interval endpoint within the outer interval. The scale shadow h_b therefore directly defines a Riemannian metric on the spatial slice, and together with the conformal causal structure determines a Lorentzian metric.

Corollary 5.5 (Full geometric datum, 1+1 dimensions)**.** *In the 1+1 dimensional setting of the GLW net, the conformal skeleton* $\mathrm{Conf}(\mathcal{N}, \omega)$ *together with the scale family* $\{h_b\}_{b\in(0,1)}$ *defines the full geometric datum*

$$\mathrm{Geom}_{\mathrm{full}}(\mathcal{N}, \omega) := \big(\mathrm{Conf}(\mathcal{N}, \omega),\ \{h_b\}_{b\in(0,1)}\big).$$

In the worked examples of Section 6, the cross-ratio parameter b naturally parametrizes the spatial position of the inner interval endpoint within the outer interval. The scale shadow h_b therefore directly defines a Riemannian metric on the spatial slice, and together with the conformal causal structure determines a Lorentzian metric.

Remark 5.6. The step from scale shadow to Lorentzian metric relies on the model-level identification of the cross-ratio parameter with spatial position. At the abstract level, Corollary 5.5 provides a full geometric datum; the Lorentzian interpretation is verified in the worked examples but not proved in full generality. The extension to higher dimensions is an open problem (Section 10.2).

5.6 Forward and Bidirectional Full Inseparability

With the full geometric datum in hand, we can state the main theorems.

Definition 5.7 (Extended modular data)**.**

$$\mathrm{Mod}_{\mathrm{ext}}(\mathcal{N}, \omega) := \big(\mathrm{Mod}_{\mathrm{conf}}(\mathcal{N}, \omega),\ \{K_b\}_{b\in(0,1)}\big).$$

Two extended modular data $\mathrm{Mod}^{(1)}_{\mathrm{ext}}$ and $\mathrm{Mod}^{(2)}_{\mathrm{ext}}$ are identified if their conformal modular sectors agree and their canonical scale families $\{K_b\}$ agree up to the trivial equivalences specified in (SFa). All equalities of $\mathrm{Mod}_{\mathrm{ext}}$ below are understood in this quotient sense.

Theorem 5.8 (Forward Full Inseparability)**.** *Under the Phase 1 axioms and* $(LImp^1)$ + $(CR\text{-}nd)$*, if two admissible objects have the same extended modular data, then they have the same full geometric datum:*

$$\mathrm{Mod}^{(1)}_{\mathrm{ext}} = \mathrm{Mod}^{(2)}_{\mathrm{ext}} \quad \Longrightarrow \quad \mathrm{Geom}^{(1)}_{\mathrm{full}} = \mathrm{Geom}^{(2)}_{\mathrm{full}}.$$

Proof. By Phase 1 Weak Inseparability, the conformal modular data determine the conformal skeleton. By Theorem 5.3, the canonical modular scale family $\{K_b\}$ determines the scale shadow $\{h_b\}$. By Corollary 5.4, the conformal skeleton and scale shadow together determine the full geometric datum. □

The forward direction says: you cannot change the full geometric datum without changing the modular data. But can you change the modular data without changing the full geometric datum? This requires an additional faithfulness condition.

Axiom (SFa) (Scale faithfulness). The canonical scale map $\{K_b\} \mapsto \{h_b\}$ is faithful within the admissible class: if $h_b^{(1)} = h_b^{(2)}$ for all b, then $K_b^{(1)} = K_b^{(2)}$ for all b, up to trivial equivalences that do not affect the scale map.

Theorem 5.9 (Forward result and conditional converse)**.** *Under the Phase 1 axioms,* $(LImp^1)$*, and* $(CR\text{-}nd)$*, the derived implication is*

$$\mathrm{Mod}_{\mathrm{ext}}^{(1)} = \mathrm{Mod}_{\mathrm{ext}}^{(2)} \quad \Longrightarrow \quad \mathrm{Geom}_{\mathrm{full}}^{(1)} = \mathrm{Geom}_{\mathrm{full}}^{(2)}.$$

If the additional scale-faithfulness axiom (SFa) *is supplied, then the converse implication also holds, and hence*

$$\mathrm{Geom}_{\mathrm{full}}^{(1)} = \mathrm{Geom}_{\mathrm{full}}^{(2)} \quad \Longleftrightarrow \quad \mathrm{Mod}_{\mathrm{ext}}^{(1)} = \mathrm{Mod}_{\mathrm{ext}}^{(2)}.$$

Proof. The displayed forward implication is Theorem 5.6. For the conditional converse: by the definition of $\mathrm{Geom}_{\mathrm{full}}$, equality of full geometries determines both the conformal skeleton and the scale family $\{h_b\}$. The conformal skeleton determines the conformal modular data by Phase 1 Weak Inseparability. The scale family determines the canonical modular scale family $\{K_b\}$ by (SFa). Together, $\mathrm{Mod}_{\mathrm{ext}}^{(1)} = \mathrm{Mod}_{\mathrm{ext}}^{(2)}$. □

The converse is therefore not an independently derived theorem: its injective step is exactly the model-level axiom (SFa). Under the identifications defended in Section 2, the mathematical counterpart of physical structure (the full geometric datum $\mathrm{Geom}_{\mathrm{full}}$) and the mathematical counterpart of algebraic modular time (extended modular data) cannot be independently specified within this conditional package. Modular flow supplies an algebraic time parameter only. Identifying that parameter with phenomenal or first-person time is an additional interpretive move, not a conclusion of Theorems 4.8 or 5.9.

6 Abstract Sufficient Conditions for $(LImp^1)$

The Phase 2 theorems are conditional on the chart axiom $(LImp^1)$, the non-degeneracy condition $(CR\text{-}nd)$, and the faithfulness condition (SFa). Of these three, $(LImp^1)$ is the most structurally demanding: it requires a local unitary implementer family with C^1 regularity and tangent compatibility. This section identifies two abstract routes for discharging $(LImp^1)$: a full discharge for Diff-covariant nets, and a decomposition of the remaining non-Diff burden into transportability, inner implementability, and regularity.

The conditions $(CR\text{-}nd)$ and (SFa) are not addressed here. They appear to depend on dynamical and spectral properties of the specific model (e.g., the positivity of the stress-tensor two-point function for $(CR\text{-}nd)$, and the injectivity of the variance map for (SFa)), and are better treated as model-level verifications rather than consequences of structural regularity.

6.1 Decomposition of $(LImp^1)$

It is useful to isolate the three independent requirements packaged into $(LImp^1)$:

1. **Local transport.** For each b near a reference point b_0, there exists an isomorphism $\alpha_b \colon \mathcal{M}_* \to \mathcal{M}_{\mathrm{in}}(b)$ that extends to an automorphism of $\mathcal{M}_{\mathrm{out}}$.

2. **Unitary implementability.** The automorphism α_b is inner in $\mathcal{M}_{\text{out}}$: there exists $W_b \in \mathcal{U}(\mathcal{M}_{\text{out}})$ with $\alpha_b = \text{Ad}(W_b)$.

3. C^1 **regularity and tangent compatibility.** The map $b \mapsto \omega(W_b A W_b^*)$ is C^1 for each $A \in \mathcal{M}_*$, and the tangent at b_0 is represented by the modular Hamiltonian variation $\dot{K}_{b_0}$ via the Araki correspondence.

Different sufficient conditions address different subsets of these requirements.

6.2 Why Split Property Alone Is Not Sufficient

The split property for the inclusion $\mathcal{M}_{\text{in}}(b) \subset \mathcal{M}_{\text{out}}$ guarantees the existence of a Type I factor $\mathcal{N}$ interpolating between the two algebras, and hence a local tensor product decomposition of the Hilbert space. This is relevant to requirement (2): in the presence of a split, the "annular region" between the inner and outer algebras has enough room to localize unitary implementers.

However, the split property does not by itself produce the transport of requirement (1). It tells you that the relative position of $\mathcal{M}_{\text{in}}(b)$ inside $\mathcal{M}_{\text{out}}$ is "standard" in a measure-theoretic sense, but it does not provide a canonical family of automorphisms mapping $\mathcal{M}_*$ to $\mathcal{M}_{\text{in}}(b)$ as b varies. That family must come from an external source: either a symmetry group acting on the net, or an explicit transportability assumption.

Nor does the split property address requirement (3). The C^1 regularity of the pulled-back state family and its tangent compatibility with the modular Hamiltonian variation are analytic conditions that depend on the smoothness of whatever generates the transport, not on the split structure.

In summary: split property assists with unitary implementability but does not generate transport or regularity. It is a facilitating condition, not a sufficient one.

6.3 Diff-Covariant Nets

The cleanest sufficient condition for $(LImp^1)$ is diffeomorphism covariance of the net.

Theorem 6.1 (Diff-covariance implies $(LImp^1)$)**.** *Let $I \mapsto \mathcal{A}(I)$ be a local net on S^1 (or on the half-line) that is covariant under a strongly continuous positive-energy representation of* Diff(S^1) *(respectively, of orientation-preserving diffeomorphisms of the half-line fixing the boundary). Then $(LImp^1)$ holds for the cross-ratio family $\{\mathcal{M}_{\text{in}}(b)\}_{b\in(0,1)}$ inside $\mathcal{M}_{\text{out}}$.*

Proof. Fix a reference point $b_0 \in (0,1)$ and set $\mathcal{M}_* = \mathcal{M}_{\text{in}}(b_0)$. For each b in a neighborhood of b_0, choose a smooth family of diffeomorphisms f_b satisfying $f_{b_0} = \text{id}$, $f_b(b_0) = b$, $f_b(1) = 1$, with support contained in the outer interval.

Transport. The diffeomorphism covariance of the net gives $f_b(\mathcal{M}_*) = \mathcal{M}_{\text{in}}(b)$, and since $\text{supp}(f_b)$ is contained in the outer interval, the corresponding automorphism α_b restricts to an automorphism of $\mathcal{M}_{\text{out}}$. This establishes requirement (1).

Unitary implementability. The positive-energy representation provides unitaries $U(f_b)$ implementing α_b. Since f_b is supported in the outer interval, $U(f_b) \in \mathcal{M}_{\text{out}}$ by Haag duality (or by the localization properties of the Diff representation). Set $W_b := U(f_b)$. This establishes requirement (2).

C^1 regularity. The map $b \mapsto f_b$ is smooth by construction. The strong continuity of the Diff representation ensures that $b \mapsto U(f_b)$ is strongly continuous, and the smoothness of f_b upgrades this to C^1 regularity of the pulled-back state family $b \mapsto \omega(W_b A W_b^*)$.

Tangent compatibility. The infinitesimal generator of the family $U(f_b)$ at $b = b_0$ is a smeared stress-energy tensor supported in the outer interval. By the relation between stress-energy deformations and modular Hamiltonian variations in conformal field theory [3], this generator represents the modular Hamiltonian variation $\dot{K}_{b_0}$ via the Araki correspondence. This establishes requirement (3).

Nontriviality. The diffeomorphisms f_b are not Möbius transformations (they fix boundary points that a Möbius transformation would move). Therefore the pulled-back state family is nonconstant, and the QFIM is nonzero. □

This theorem upgrades the Virasoro worked example of Section 7.2 from a model verification to an instance of a general result: any Diff-covariant conformal net automatically satisfies $(LImp^1)$.

6.4 Transportability-Based Sufficiency

For nets that are not Diff-covariant, a second sufficient condition can be formulated by combining an explicit transportability hypothesis with the injectivity of the outer algebra.

Definition 6.2 (Local transportability)**.** The cross-ratio family $\{\mathcal{M}_{\text{in}}(b)\}$ is *locally transportable* inside $\mathcal{M}_{\text{out}}$ if, for each $b_0 \in (0,1)$, there exists a neighborhood U_{b_0} and a family of automorphisms $\alpha_b \in \text{Aut}(\mathcal{M}_{\text{out}})$, $b \in U_{b_0}$, with $\alpha_{b_0} = \text{id}$ and $\alpha_b(\mathcal{M}_*) = \mathcal{M}_{\text{in}}(b)$.

Theorem 6.3 (Transportability-based sufficiency for $(LImp^1)$)**.** *Suppose:*

1. *the cross-ratio family is locally transportable inside $\mathcal{M}_{\text{out}}$;*
2. *$\mathcal{M}_{\text{out}}$ is an injective (equivalently, hyperfinite) Type III$_1$ factor;*
3. *the transport family $b \mapsto \alpha_b$ is C^1 in the sense that $b \mapsto \omega(\alpha_b(A))$ is C^1 for each $A \in \mathcal{M}_*$, with tangent compatible with the modular Hamiltonian variation.*

Then $(LImp^1)$ holds.

Proof. Requirement (1) of $(LImp^1)$ is exactly hypothesis (1). For requirement (2): by hypothesis (2) and the Connes theorem [14] ($\text{Out}(\mathcal{M}) = \{e\}$ for injective Type III$_1$ factors), every automorphism of $\mathcal{M}_{\text{out}}$ is inner. Hence $\alpha_b = \text{Ad}(W_b)$ for some $W_b \in \mathcal{U}(\mathcal{M}_{\text{out}})$. Requirement (3) is hypothesis (3). □

Remark 6.4. If the inclusion $\mathcal{M}_{\text{in}}(b) \subset \mathcal{M}_{\text{out}}$ additionally satisfies the split property, then the Type I interpolation allows W_b to be chosen with support localized in the annular region between the two algebras. This localization can improve the analytic control needed for the C^1 regularity of hypothesis (3), but it is not required for the theorem as stated.

It is important to be clear about what this theorem does and does not achieve. Hypotheses (1) and (3) —transportability and C^1 tangent compatibility —are already the two main substantive components of $(LImp^1)$ itself. The theorem's contribution is to identify the precise algebraic condition (injectivity of the outer factor) that bridges the gap between "transport exists as an automorphism" and "transport is unitarily implemented inside the outer algebra." For nets whose local algebras are known to be injective Type III$_1$ factors —which includes all local algebras in standard AQFT models —this bridge is automatic.

6.5 Status of the Three Phase 2 Conditions

The results of this section change the status of $(LImp^1)$ as follows:

- For Diff-covariant nets, $(LImp^1)$ is fully discharged by Theorem 6.1. No additional assumptions are needed.
- For non-Diff-covariant nets with injective Type III_1 local algebras, $(LImp^1)$ reduces to local transportability plus C^1 regularity (Theorem 6.3). The remaining burden is to verify that the cross-ratio family admits a smooth transport, which is a geometric property of the net rather than an abstract axiom.
- $(CR\text{-}nd)$ and (SFa) remain model-level verifications. They depend on dynamical and spectral properties (positivity of the stress-tensor two-point function, injectivity of the scale map) that are not captured by structural regularity conditions.

The net effect is that, for Diff-covariant nets, the Phase 2 conditionality is reduced from three independent axioms to two model-level verifications. For non-Diff-covariant nets, the situation is more modest: $(LImp^1)$ is decomposed and partially discharged, but the transportability and regularity inputs remain substantive.

7 Verification: Two Worked Examples

The abstract Phase 2 result is deliberately conditional. Of the three conditions, $(LImp^1)$ now has abstract sufficient conditions (Section 6): it is fully discharged for Diff-covariant nets, and for other nets the remaining burden is isolated to transportability and regularity. The remaining conditions $(CR\text{-}nd)$ and (SFa) are verified in concrete models. This section provides that verification in two structurally distinct examples.

The examples are chosen to do different kinds of work. The Rindler/free-field chiral half-line model shows that the Phase 2 package is realized in the same physical family that motivated much of the original HAFF intuition. The Virasoro-net example then shows that the construction is not a free-field accident and, in particular, that the chart axiom becomes especially natural once $\mathrm{Diff}(S^1)$ covariance is built in.

7.1 Worked Example I: Rindler / Free-Field Chiral Half-Line

The first example must be formulated carefully. The correct Phase 2 object is not the family of translated Rindler wedges. That family is useful as a sanity check for the Phase 1 half-sided modular structure, but it is the wrong object for the cross-ratio construction because global translations leave the vacuum invariant and thus produce a constant pulled-back state family. The correct model is the chiral half-line realization associated with the Rindler horizon.

We therefore take:

$$\mathcal{M}_{\text{out}} = \mathcal{A}(0,\infty), \qquad \mathcal{M}_{\text{in}}(b) = \mathcal{A}(b,1), \qquad b \in (0,1),$$

in the vacuum representation of a chiral free-field net. The canonical modular scale datum is then

$$K_b := -\ln \Delta_{\mathcal{M}_{\text{in}}(b),\Omega}.$$

7.1.1 Verification of $(LImp^1)$

In this model, the required local chart is supplied by the local diffeomorphism covariance of the free-field chiral net. Specifically, we assume covariance under orientation-preserving diffeomorphisms of the half-line that fix the boundary, implemented by unitaries in the corresponding local algebras. Fix a reference point $b_0 \in (0,1)$ and set

$$\mathcal{M}_* := \mathcal{A}(b_0, 1).$$

Choose a smooth local family of diffeomorphisms

$$f_b : (0,\infty) \to (0,\infty), \qquad b \in U_{b_0},$$

such that

1. $f_{b_0} = \mathrm{id}$;
2. $f_b(1) = 1$;
3. $f_b(b_0) = b$;
4. the support of f_b is contained in the outer half-line.

This covariance gives unitaries

$$W_b := U(f_b) \in \mathcal{U}(\mathcal{M}_{\text{out}})$$

with

$$W_b \mathcal{M}_* W_b^* = \mathcal{M}_{\text{in}}(b).$$

Because the deformation is genuinely boundary-fixed rather than Möbius, the family is nontrivial; because $b \mapsto f_b$ is smooth, the pulled-back state family is C^1; and because the infinitesimal generator is represented by the stress tensor, the tangent of the state family is compatible with the modular Hamiltonian variation. Thus $(LImp^1)$ holds in the model.

7.1.2 Verification of $(CR\text{-}nd)$

In a vacuum chiral conformal theory, the modular Hamiltonian of an interval (a,b) has the local CHM form [3]

$$K_{(a,b)} = 2\pi \int_a^b \frac{(x-a)(b-x)}{b-a}\, T_{00}(x)\, dx.$$

Specializing to the present family gives

$$K_b = 2\pi \int_b^1 \frac{(x-b)(1-x)}{1-b}\, T_{00}(x)\, dx,$$

and differentiation yields

$$\partial_b K_b = -2\pi \int_b^1 \frac{(1-x)^2}{(1-b)^2}\, T_{00}(x)\, dx.$$

The corresponding Araki-Hessian quadratic form is positive and nonzero, so the scale shadow is non-degenerate. More explicitly, one obtains

$$h_b = C_R (1-b)^{-2} db^2, \qquad C_R > 0.$$

Hence $(CR\text{-}nd)$ is satisfied.

7.1.3 Verification of (SFa)

In this model, faithfulness reduces to a one-dimensional injectivity statement. Since

$$h_b = C_R(1-b)^{-2}db^2$$

is strictly monotone in b, the metric profile determines the parameter b, and hence determines the corresponding modular datum K_b up to the trivial equivalences already allowed in the definition of (SFa). Therefore the scale map is faithful in the required sense.

The upshot is that the entire Phase 2 package is realized in the Rindler/free-field chiral half-line model:

$$(\text{LImp}^1), \quad (\text{CR-nd}), \quad (\text{SFa}).$$

7.2 Worked Example II: The Virasoro Net

The second example shows that the preceding mechanism is not tied to free fields. Let

$$I \longmapsto \mathcal{A}_{\text{Vir},c}(I)$$

be the Virasoro net at central charge c, with vacuum vector Ω. As in the first example, we work in the standard half-line gauge:

$$\mathcal{M}_{\text{out}} = \mathcal{A}_{\text{Vir},c}(0,\infty), \qquad \mathcal{M}_{\text{in}}(b) = \mathcal{A}_{\text{Vir},c}(b,1).$$

7.2.1 Verification of $(LImp^1)$

What was model-dependent in the first example becomes built in here. The Virasoro net is Diff(S^1)-covariant, so $(LImp^1)$ follows directly from Theorem 6.1. Concretely, one can choose a smooth boundary-fixed family

$$f_b \in \text{Diff}(S^1), \qquad b \in U_{b_0},$$

with

$$f_{b_0} = \text{id}, \qquad f_b(b_0) = b, \qquad f_b(1) = 1,$$

and support contained in the outer half-line. The positive-energy Diff(S^1) representation yields unitaries

$$W_b := U(f_b) \in \mathcal{U}(\mathcal{M}_{\text{out}})$$

such that

$$W_b \mathcal{A}(b_0,1) W_b^* = \mathcal{A}(b,1).$$

The nontriviality, smoothness, and tangent compatibility required by $(LImp^1)$ follow immediately from the non-Möbius character of f_b, the smoothness of the Diff(S^1) action, and the fact that the infinitesimal action is generated by the Virasoro stress tensor. Thus $(LImp^1)$ holds here in a cleaner and more canonical way than in the free-field model.

7.2.2 Verification of $(CR\text{-}nd)$

The interval modular Hamiltonian again has the CHM form

$$K_b = 2\pi \int_b^1 \frac{(x-b)(1-x)}{1-b}\, T(x)\, dx,$$

where T is now the Virasoro stress tensor. Differentiating gives the same structural formula as before:

$$\partial_b K_b = -2\pi \int_b^1 \frac{(1-x)^2}{(1-b)^2}\, T(x)\, dx.$$

Using the Virasoro two-point function

$$\langle T(x)T(y)\rangle \sim \frac{c/2}{(x-y)^4},$$

one obtains, after standard regularization, a positive finite coefficient

$$\mathbf{Q}_{\omega,b}(\partial_b K_b, \partial_b K_b) = \frac{C_{\text{Vir}}(c)}{(1-b)^2}, \qquad C_{\text{Vir}}(c) > 0.$$

Hence

$$h_b = C_{\text{Vir}}(c)(1-b)^{-2} db^2,$$

and $(CR\text{-}nd)$ holds.

7.2.3 Verification of (SFa)

Exactly as in the first example, the scale profile is strictly injective in the cross-ratio parameter. Therefore the full scale family determines b, and hence determines the local modular datum up to the same trivial equivalences. Thus (SFa) is satisfied in the Virasoro model as well.

7.3 Comparison and Universal Scaling

The significance of the two examples lies less in their individual computations than in what they share.

1. Both models realize the same abstract Phase 2 package:
$$(\text{LImp}^1) + (\text{CR-nd}) + (\text{SFa}).$$
2. Both produce the same scale profile:
$$h_b = C(1-b)^{-2} db^2.$$
3. The model dependence enters only through the coefficient:
   - in the free-field model through the normalization of the stress-tensor sector;
   - in the Virasoro model through the central charge c.

This is the strongest available evidence, within the present paper, that the scale shadow is not a model-specific artifact. What appears to be universal is the cross-ratio scaling law itself, which is fixed by the CHM structure of interval modular Hamiltonians. What remains model-sensitive is only the overall constant.

From the point of view of the paper's philosophical thesis, this matters because it shows that the full inseparability mechanism is not tied to one special solvable example. The rigidity claim is abstract; the examples show that the additional Phase 2 inputs are concretely realizable in more than one structural setting, and that the theorem's conditions arise naturally where one would most expect them to arise.

8 Implications for the Hard Problem

For the purposes of this section, we take the mathematical results of the previous sections as given. At the conformal level, Theorem 4.8 already shows that conformal geometry and conformal modular data are not independently deformable. The later Phase 2 results strengthen this to the full scale-sensitive level by tying the metric scale to the canonical modular scale datum. The philosophical question is therefore no longer whether one can attach a speculative interpretation to a formal analogy. It is whether the standard formulation of the hard problem survives once its key presupposition is written in mathematically precise terms.

8.1 The Failure of the Presupposition

Section 1 identified the hidden presupposition of the hard problem: independent describability. Section 2 translated it into independent deformability. The inseparability theorems of Sections 4 and 5 now supply the answer.

At the Phase 1 level, Theorem 4.8 shows that conformal geometry and conformal modular data are already locked together. At the full level, Theorem 5.7 shows that, under $(LImp^1)$, $(CR\text{-}nd)$, and (SFa), the full geometric datum and the extended modular data cannot be independently varied. Conditional on the identifications defended in Section 2, the presupposition is not available at either level.

The strongest claim justified by the theorem is therefore not that the hard problem has been "solved" in the ordinary sense. It is that its canonical production form is ill-posed. The question "how does matter produce mind?" asks for a derivation across a divide that, in the present framework, is not mathematically available as an independent divide. One cannot first specify the geometric-physical side and only afterwards ask how the first-person side appears. They are two structurally inseparable readings of the same algebraic datum $(\mathcal{A}, \omega)$.

8.2 Relocating the Explanatory Gap

Once the production picture is rejected, a residual question remains. Even if geometry and modular flow are inseparable, why should a given modular invariant correspond to a given phenomenal quality? This is legitimate, but it is not the original hard problem. It concerns the interpretive reach of an observer embedded in the very structure being described.

The proper location of the remaining gap is not between physical and experiential structure as two externally juxtaposed domains. A possible research direction places it within the self-description limits of a finite agent modeled by $(\mathcal{A}, \omega)$. Paper G's discussion of self-reference [9] motivates this direction, but it does not prove a generic fixed-point no-go or an exhaustive regress-versus-collapse dichotomy. No theorem of the present paper therefore infers a phenomenal limitation from Paper G.

On that interpretation, the explanatory gap may be treated as structural rather than merely technical [7]. Establishing such a horizon would require a separate formal bridge from algebraic self-reference to phenomenal description; the analogy alone does not supply it.

This changes what counts as philosophical success. The goal is no longer to show how an independently specified physical world outputs experience. It is to understand why an embedded agent necessarily encounters a limit in its attempt to completely internalize the relation between structural invariants and lived articulation. The residual gap is relocated and reclassified, not removed.

8.3 Comparison with Existing Positions

The closest philosophical relatives of the present framework are positions that deny a brute separation between mind and world. Even so, the relation is exact only up to a point. The following comparison is necessarily compressed; each position listed has significant internal diversity, and the characterizations below refer to the canonical or most common formulation.

Position	Relation to the present paper
Physicalism / functionalism	Shares the desire for structural rigor, but typically preserves an explanatory asymmetry: physical facts are taken as independently specifiable, and consciousness must be shown to follow. The present theorem denies that asymmetry in its standard form.
Panpsychism	Avoids emergence by treating experience as basic, but inherits the combination problem. The present framework does not posit basic micro-experiences; it treats modular structure and geometry as co-emergent readings of one algebraic organization.
Dual-aspect monism	The nearest metaphysical relative. The decisive difference is that here the dual-aspect claim is not left at the level of metaphysical slogan; it is tied to a concrete inseparability theorem.
IIT	Also seeks a mathematically articulated bridge, but introduces an additional quantity Φ as a privileged measure. The present route instead treats inseparability as internal to modular and geometric structure, not as the effect of an extra consciousness functional.
Illusionism	Rejects the hard problem by denying that phenomenal consciousness needs the explanatory role usually assigned to it. The present framework does not deny experience. It denies the independent specification presupposition that makes experience look like a separately generated add-on.
Russellian monism	Very close in spirit insofar as it challenges the exhaustiveness of third-person structure. But Russellian monism often speaks of an intrinsic categorical basis behind structural physics. The present framework is more austere: both geometry and first-person time are readings of one algebraic structure, without appeal to a hidden intrinsic substrate beyond that structure.

The upshot is that the present work should not be advertised as a simple algebraic version of any single existing position. It is closest to a theorem-backed dual-aspect view, but even that label is only approximate. The real novelty lies in replacing a metaphysical balance claim with a rigidity statement.

A structural correspondence with the Yogacara fourfold analysis is recorded in Appendix A.

9 Broader Significance

The philosophical argument of the preceding sections is the primary contribution of this paper. The mathematical and physical content, however, has independent interest that is worth recording briefly.

9.1 Mathematical Significance

The Wiesbrock-GLW programme [17, 5] established that half-sided modular inclusions reconstruct conformal symmetry. What remained open was the passage from conformal skeleton to metric scale using only modular data. The present paper isolates a precise conditional route to a full geometric datum in 1+1 dimensions: the cross-ratio modular scale datum $\{K_b\}$ serves as the canonical non-gauge geometric modulus, and the Araki relative entropy Hessian [1] serves as the Type III-compatible QFIM that converts it into a Riemannian scale shadow. The route is conditional on $(LImp^1)$, $(CR\text{-}nd)$, and (SFa), all three of which are verified in two structurally distinct models. For Diff-covariant nets, Section 6 further shows that $(LImp^1)$ is automatic, leaving only $(CR\text{-}nd)$ and (SFa) as model-level verifications. In the worked examples, where the cross-ratio parameter naturally parametrizes the spatial slice, this datum assembles into the Lorentzian metric.

The universal scaling law $h_b \sim (1-b)^{-2}db^2$, shared by both worked examples, is a structural consequence of the Casini-Huerta-Myers form [3] of interval modular Hamiltonians. It is not a free-field accident. The model-dependent information enters only through the overall constant, which encodes data such as the central charge.

9.2 Physical Significance

The slogan "it from qubit" and the ER=EPR programme assert that entanglement structure determines spacetime geometry [11, 13, 16, 12]. These claims have so far been tied to the AdS/CFT correspondence. The present result establishes a version of the same thesis without assuming holographic duality. The axiom package selected by $(U3)$, together with the axiom package $(U1)+(C)+(Amb)+(CN_{\exists})+(LImp^1)+(CR\text{-}nd)+(SFa)$, makes no reference to anti-de Sitter space, conformal field theory on a boundary, or bulk-boundary correspondence. Inseparability is a property of Type III_1 algebraic structure under appropriate regularity [18], not a special consequence of holography.

This also resolves two open issues from the earlier HAFF programme (Paper D) [8]. The non-uniqueness of the map from algebraic deformations to state deformations is absorbed by the cross-ratio datum and the chart axiom, which together fix the relevant deformation direction. The signature mismatch between the positive-definite QFIM and the Lorentzian metric is resolved by restricting the QFIM to spatial slices and letting the conformal skeleton supply the causal structure.

10 Explicit Non-Claims and Open Problems

10.1 What This Paper Does Not Claim

1. It does not claim to have solved the hard problem of consciousness. It claims to have shown that the standard production formulation rests on a presupposition that is not available in the relevant mathematical framework.

2. It does not claim that modular flow "is" consciousness. The identification in Section 2 is a structural correspondence: modular flow plays the mathematical role that first-person temporal structure plays in the philosophical debate.

3. It does not claim that the inseparability theorem holds unconditionally for all Type III_1 factors. Phase 1 is unconditional within the stated axiom class. Phase 2 is conditional

on $(LImp^1)$, $(CR\text{-}nd)$, and (SFa). For Diff-covariant nets, $(LImp^1)$ is discharged by Theorem 6.1; the remaining two conditions are verified in two models but not proved in full generality.

4. It does not claim that the Buddhist correspondence in Section 7.4 participates in any proof. It is included as a structural comparison, not as evidence.

5. It does not claim that the results extend to dimensions higher than 1+1. The GLW construction produces a Möbius net on S^1. Higher-dimensional generalizations require additional HSMI directions and remain open.

6. It does not claim that consciousness is a quantum effect. The argument concerns the structural relationship between geometry and modular flow at the level of algebraic quantum theory, not a physical mechanism involving quantum coherence in biological systems.

10.2 Open Problems

1. Abstract sufficient conditions for $(CR\text{-}nd)$ and (SFa). Section 6 provides sufficient conditions for $(LImp^1)$ (Diff-covariance or split-assisted transportability), but the remaining two conditions are still verified only in specific models. A general criterion for $(CR\text{-}nd)$, perhaps in terms of spectral properties of the stress-energy tensor, would further reduce the conditionality.

2. Foliation compatibility beyond 1+1 dimensions. In the present 1+1 dimensional setting, the cross-ratio family is a single-parameter family and the assembly is straightforward. In higher dimensions, the scale data from different cross-ratio slices must satisfy nontrivial mutual consistency conditions. Formulating and verifying these conditions is an open problem tied to the higher-dimensional extension.

3. Higher-dimensional extension. In $d > 1 + 1$, the conformal group is larger and the cross-ratio construction must be replaced by a higher-dimensional analogue. Whether the inseparability mechanism survives this extension is the most important structural open question.

4. A formal self-description bridge. Paper G currently supplies philosophical motivation, not a fixed-point no-go theorem. A separate model with a typed syntax/observable interface and an explicit capacity functional would be required before modular invariants could yield a quantitative self-description bound.

5. The relationship between the present framework and Integrated Information Theory. Both seek mathematically articulated accounts of consciousness. Whether the inseparability theorem implies constraints on Φ or vice versa is unexplored.

A Yogacara Fourfold Correspondence

The Yogacara fourfold analysis provides a structural correspondence for the architecture that emerges in this paper. The correspondence does not enter any proof; it is included as a disciplined interpretive comparison that becomes newly intelligible once the mathematical theorem is in place. Each position listed below has significant internal diversity within the Buddhist philosophical tradition; the characterizations refer to the canonical formulation.

Layer	Phenomenological role	Mathematical counterpart
相分	the presented object-side	emergent geometry Geom
见分	the act-side of cognition	modular flow σ_t^{ω}
自证分	reflexive awareness	the self-calibrating informational loop of T-DOME Paper III [10]
证自证分	the boundary of reflexivity	the incompleteness limit of Paper G [9]

The structural fit is that object-side and act-side are not heterogeneous substances but internally linked aspects of a single cognitive event; reflexivity does not continue indefinitely into an infinite hierarchy, but closes through a higher-order boundary condition. This is exactly the pattern suggested by the inseparability framework: geometry and modular flow are inseparable at the base level, and reflexive closure is marked by incompleteness rather than by defect.

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Higher-Dimensional Consistency of Geometry and Modular Flow: Small-Ball Modular Data and Linearized Gravity

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Abstract

Paper I established that, in 1+1 dimensions, emergent geometry and modular flow are not independently deformable. The higher-dimensional sequel cannot proceed by simply replacing the one-parameter cross-ratio family with a higher-dimensional analogue. For ball-shaped regions in $d+1$ dimensions, the relevant non-gauge datum is not an individual nested configuration but the full family of modular Hamiltonians attached to all balls and all local rest frames. The new question is therefore one of consistency: when can this multi-directional family of modular data be explained by a single geometry?

We study this question in the CHM ball framework for conformal nets with finite-dimensional conformal covariance, in the linearized regime. First, we show that the small-ball limit of the modular energy uniquely recovers the local stress tensor from the all-directions family of modular data. Second, for a local geometric functional satisfying linearity, locality, covariance, gauge invariance, and the correct scaling behavior, we additionally assume that its leading coefficient is a local symmetric rank-2 tensor of exactly two-derivative order and is divergence-free. Under those independently supplied premises, the linearized Lovelock classification identifies that coefficient with the linearized Einstein tensor up to normalization. Matching the modular and geometric small-ball expansions then yields the linearized Einstein equation within the stated domain. Thus, in higher dimensions, inseparability does not merely lock geometry and modular flow together; the consistency of that locking across directions takes the form of linearized gravitational constraint equations.

The resulting theorem is conditional but precise. It does not attempt full generality, nonlinear dynamics, or arbitrary Type III_1 nets. Its proof domain is restricted to CHM ball data in conformally covariant theories, in the linearized regime. Within that domain, however, the conclusion is sharp: the linearized Einstein equation appears as the consistency condition required for a single geometry to remain compatible with the full modular data family.

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1 Introduction

Paper I[1] showed that, in $1+1$ dimensions, the standard hard-problem presupposition of independent describability fails because geometry and modular flow cannot be independently deformed. That result depended on a one-parameter datum: the cross-ratio family of double intervals. In higher dimensions, the mathematical problem changes in a decisive way. There is no longer a single distinguished scalar modulus whose variation carries the entire scale information. The higher-dimensional sequel must therefore identify a different object.

The central claim of the present paper is that the correct object is the full family of ball modular Hamiltonians, indexed not only by center and radius but also by local timelike direction. Once this is recognized, the problem becomes a consistency problem rather than a parametrization problem. A single ball gives only one local modular datum. A family of balls in many directions gives many such data. The question is whether they can all be explained by one and the same geometry.

Our conditional answer is that, in the CHM ball framework and at linear order, consistency yields an Einstein coefficient equation once a local, two-derivative, divergence-free geometric coefficient is supplied. The CHM data alone recover the local stress tensor; they do not select the geometric operator.

The proof strategy is intentionally narrow. We do not attempt a nonlinear theorem, a fully general Type III_1 result, or an "if and only if" in complete abstractity. Instead we isolate a tractable domain in which the statement can be made cleanly: conformal nets with finite-dimensional conformal covariance, CHM modular Hamiltonians for balls, the vacuum or near-vacuum linearized regime, and a local geometric functional satisfying a small list of natural structural axioms. Within that domain, the theorem is short. The real work lies in identifying the correct small-ball expansions on both the modular and geometric sides.

The paper is organized in the reverse order of discovery. Section 2 defines the higher-dimensional modular datum and explains why the natural $1+1$-dimensional cross-ratio strategy has no direct higher-dimensional analogue. Section 3 formulates the small-ball consistency condition that replaces assembly in higher dimensions. Section 4 states and proves the main theorem. Section 5 records the free-field worked example that motivated the final theorem statements. Section 6 discusses the converse direction and the limits of the argument. Section 7 places the result in the broader Consciousness Trilogy, and Section 8 lists the natural open problems.

Remark 1.1 (Relation to Paper I). Paper II is mathematically self-contained. It uses Paper I only for motivation and context: the conceptual shift from "independent describability" to "independent deformability," and the observation that in $1+1$ dimensions the cross-ratio family carries the full scale information. No theorem from Paper I is invoked as a premise in any proof of Paper II. The CHM ball framework, the geometric axioms (G1)–(G5), and the small-ball consistency condition (MC) are all defined and used within the present paper.

2 Multi-Directional Modular Data in Higher Dimensions

2.1 From Cross-Ratio to Ball Families

Paper I[1] relied on the fact that, in $1+1$ dimensions, the correct non-gauge local modulus is the cross-ratio of a double-interval configuration. That datum is one-dimensional, and the resulting assembly problem is trivial. There is only one parameter direction, so there is no compatibility problem between different directional slices.

In higher dimensions, the naive analogue fails. A single ball is a conformal gauge orbit. Even a nested pair of balls does not produce a nontrivial scalar modulus in the way that a double interval does in $1+1$ dimensions. The correct higher-dimensional object therefore cannot be "a better cross-ratio." The no-go is structural: the relevant conformal degrees of freedom are not carried by one isolated nested configuration.

What replaces the cross-ratio is the family of all ball modular data. The local region is now a ball

$$B(\mathbf{x}_0, R) \subset \mathbb{R}^d,$$

or, more invariantly, a small causal diamond with center x_0, radius R, and rest-frame normal u^μ. Each such region gives one modular Hamiltonian. The higher-dimensional datum is not one scalar parameter but a family

$$\{K_{B,u}\}_{(B,u)\in\mathcal{B}},$$

where $\mathcal{B}$ denotes the set of all sufficiently small balls in all local rest frames.

2.2 CHM Ball Hamiltonians

In the vacuum of a conformal theory, the modular Hamiltonian of a ball is local and universal. In a rest frame adapted to the ball, the Casini–Huerta–Myers[2] formula gives

$$K_{B,u} = 2\pi \int_{B_u} d^d x \, \frac{R^2 - |\mathbf{x}-\mathbf{x}_0|^2}{2R} \, T_{\mu\nu}(x)\, u^\mu u^\nu.$$

The importance of this formula for the present paper is twofold. First, it supplies an explicit local kernel. Second, it shows that what the ball modular Hamiltonian probes is not an arbitrary operator but the rest-frame energy density $T_{uu} := T_{\mu\nu}u^\mu u^\nu$.

This already reveals the first higher-dimensional subtlety. If one restricts to a single fixed time slice, then the family of all balls only probes T_{00}. That is enough to recover the Hamiltonian constraint source, but not enough to recover the full stress tensor. To reconstruct the full tensorial source, the family must vary not only in center and radius but also in local timelike direction. This is why "multi-directional modular datum" is not rhetorical excess but a necessary mathematical strengthening.

2.3 The Canonical Datum

We therefore take the higher-dimensional canonical modular datum to be the full ball family

$$K_{B,u} := -\ln \Delta_{\mathcal{A}(B_u),\Omega},$$

understood in the CHM class where the modular Hamiltonian admits the local stress-tensor representation above. This datum is canonical in exactly the sense needed for the sequel to Paper I[1]: it is the minimal non-gauge family rich enough to encode the directional dependence that becomes nontrivial in dimensions higher than $1+1$.

The conceptual shift is worth making explicit. In Paper I[1], geometry and modular flow were locked by a one-parameter scale datum. In the present paper, they are locked by the requirement that an entire family of local modular energies be compatible with one geometry at once. What becomes nontrivial in higher dimensions is therefore not "finding the right modulus," but enforcing coherence across a family of moduli.

3 The Small-Ball Consistency Condition

3.1 Why Small Balls First

The strongest possible version of Paper II would formulate a global consistency condition on the modular data of all balls and derive the linearized Einstein equation from that global condition. We do not begin there. The small-ball regime already contains the local dynamical content, while avoiding the additional integrability problems of a global ball-space formulation. If the thesis fails even in the small-ball limit, the global version has no chance. If it succeeds there, the globalization problem becomes a genuine strengthening rather than a prerequisite.

3.2 Modular Small-Ball Expansion

Let $\delta\omega$ be a linearized perturbation of the vacuum, and assume the local stress tensor is smooth in a neighborhood of a point x_0. Then the CHM kernel gives an immediate small-ball asymptotic:

$$\delta\langle K_{B,u}\rangle = \frac{2\pi V_d}{d+2}\, R^{d+1}\, \delta\langle T_{\mu\nu}(x_0)\rangle u^\mu u^\nu + O(R^{d+3}),$$

where $V_d = \pi^{d/2}/\Gamma(d/2+1)$ is the volume of the unit d-ball. The key point is not the coefficient but the structural content: the leading small-ball term is the quadratic form of the local stress tensor evaluated on the timelike direction u^μ.

Because the family is taken over all timelike u^μ, the leading term uniquely determines the full symmetric tensor $\delta\langle T_{\mu\nu}(x_0)\rangle$. This is the first half of the main theorem. The all-directions small-ball modular datum is already strong enough to recover the complete local matter source.

3.3 Geometric Small-Ball Functional

The geometric side is intentionally abstract. We do not assume a particular holographic dictionary, RT formula[6], or bulk construction. Instead we assume that the geometry supplies, for each small ball and each direction, a linearized functional

$$\delta\mathcal{G}_{B,u}[h],$$

depending on the metric perturbation $h_{\mu\nu}$, subject to five structural requirements:

1. linearity in h;
2. locality in the small-ball limit;
3. covariance under local Lorentz transformations;
4. invariance under linearized gauge transformations;
5. the same leading scaling R^{d+1} as the modular side.

These assumptions are not arbitrary. They are exactly what is needed for a small-ball geometric quantity to be comparable to the CHM modular energy without smuggling in additional dynamics by hand.

3.4 Consistency

The higher-dimensional consistency condition can now be stated in its minimal form.

Small-ball modular consistency (MC). For all sufficiently small balls and all local timelike directions u^μ, the modular small-ball functional and the geometric small-ball functional agree:

$$\delta\mathcal{G}_{B,u}[h] = \delta\langle K_{B,u}\rangle.$$

This is the higher-dimensional replacement for the trivial assembly of Paper I[1]. In $1+1$ dimensions, there is only one parameter direction, so the scale datum needs no consistency check beyond existence. In higher dimensions, the requirement that all directional slices fit one geometry is itself the dynamical content.

4 Conditional Einstein Coefficient Matching

4.1 Modular Extraction Proposition

The modular side is summarized by the following statement.

Proposition 4.1. *In the CHM ball framework, for a linearized perturbation of the vacuum with a smooth local stress tensor, the all-directions small-ball modular datum has leading expansion*

$$\delta\langle K_{B,u}\rangle = \frac{2\pi V_d}{d+2}\, R^{d+1}\, \delta\langle T_{\mu\nu}(x_0)\rangle u^\mu u^\nu + O(R^{d+3}).$$

The proof is the small-ball Taylor expansion of the CHM kernel, together with the symmetry of the ball that kills the linear term. The resulting leading coefficient is universal and independent of the detailed model.

Corollary 4.2. *The all-directions leading small-ball term uniquely determines the full symmetric tensor $\delta\langle T_{\mu\nu}(x_0)\rangle$ at each point.*

This is immediate because a symmetric rank-2 tensor is uniquely fixed by its quadratic form on all timelike directions.

4.2 Missing Premises for Geometric Identification

The former claim that (G1)–(G5) alone select the linearized Einstein tensor is withdrawn. Those conditions do not restrict derivative order, and even at two derivatives the relevant coefficient must separately be required to be transverse. The following premises are therefore not consequences of (G1)–(G5):

(G-der2) The leading coefficient of $\delta\mathcal{G}_{B,u}[h]$ is a local symmetric rank-2 tensor, linear in h, of exactly two derivative order.

(G-div) That coefficient is divergence-free on the Minkowski background.

When (G-der2), (G-div), linearized gauge invariance, and Lorentz covariance are supplied, the standard linearized Lovelock classification identifies the coefficient with $\alpha_d G^{\mathrm{lin}}_{\mu\nu}$ up to normalization [5]. This is a conditional coefficient identification, not a uniqueness result derived from (G1)–(G5).

4.3 Conditional Coefficient Matching

With the missing geometric premises displayed, coefficient matching is immediate.

Theorem 4.3 (Conditional Einstein coefficient matching)**.** *Assume:*

1. *CHM modular Hamiltonians for all sufficiently small balls;*
2. *conformal covariance of the net (finite-dimensional conformal group* $\mathrm{SO}(d,2)$*);*
3. *linearization around a Minkowski vacuum background;*
4. *existence of a geometric functional satisfying the small-ball axioms (G1)–(G5) above;*
5. *the additional premises (G-der2) and (G-div);*
6. *small-ball modular consistency (MC).*

Then

$$G^{\mathrm{lin}}_{\mu\nu} = \kappa_d\, \delta\langle T_{\mu\nu}\rangle,$$

with $\kappa_d = \frac{2\pi V_d}{\alpha_d(d+2)}$.

The proof first uses the explicitly supplied (G-der2) and (G-div), together with gauge invariance and covariance, to invoke the linearized Lovelock classification. It then compares the resulting $G^{\mathrm{lin}}_{\mu\nu}u^\mu u^\nu$ coefficient with the CHM stress-tensor coefficient. Equality for all timelike u^μ forces equality of the tensors themselves.

Remark 4.4 (Conditionality and consistency). Assumptions (4)–(5) are the most significant: neither the geometric functional nor its two-derivative transverse leading coefficient is derived from the algebraic data. The theorem says that *if* such a functional exists and agrees with the modular side for all sufficiently small balls, *then* its leading local content is forced to be Einsteinian. The internal construction of $\delta\mathcal{G}_{B,u}$ from modular data alone remains the most significant open problem (Section 8.3).

The matching is also dynamically consistent: the linearized Einstein tensor satisfies the Bianchi identity $\partial^\mu G^{\mathrm{lin}}_{\mu\nu} = 0$, and the stress tensor satisfies the conservation law $\partial^\mu \delta\langle T_{\mu\nu}\rangle = 0$. Both sides of the equation are independently divergence-free.

4.4 Interpretation

This theorem is the higher-dimensional analogue of the inseparability theorem of Paper I[1], but with one crucial new feature. In $1+1$ dimensions, inseparability rules out independent deformation. In higher dimensions, the demand that the inseparable structures remain mutually consistent across all directions forces the geometry to obey linearized gravitational dynamics.

That is the sense in which gravity enters the trilogy. It is not introduced as an additional law attached from outside the modular-geometric relation. It appears as the consistency condition required for that relation to survive directional proliferation.

5 Worked Example: Free Conformal Scalar Field in $d+1$ Dimensions

This section verifies the main theorem in the simplest higher-dimensional model where all ingredients are explicit: the free massless conformal scalar field in $d+1$ dimensions. The verification has three parts: the stress tensor and its perturbation, the small-ball modular expansion, and the consistency check against the linearized Einstein equation.

5.1 The Model

Let ϕ be a free massless scalar field in $d+1$ Minkowski spacetime, with action

$$S = -\frac{1}{2}\int d^{d+1}x\,(\partial\phi)^2.$$

The stress-energy tensor is

$$T_{\mu\nu} = \partial_\mu\phi\,\partial_\nu\phi - \frac{1}{2}\eta_{\mu\nu}(\partial\phi)^2,$$

with the conformal improvement term understood when $d > 1$. In the vacuum state $|\Omega\rangle$, $\langle T_{\mu\nu}\rangle = 0$.

5.2 Perturbation: Classical Source

For the purpose of verifying the small-ball expansion, we do not need the full machinery of coherent states. It suffices to take any smooth, conserved classical stress tensor as the source. Let ϕ_{cl} be a smooth solution of the free wave equation $\Box\phi_{\text{cl}} = 0$, and define

$$\delta\langle T_{\mu\nu}(\mathbf{x})\rangle := \partial_\mu\phi_{\text{cl}}\,\partial_\nu\phi_{\text{cl}} - \frac{1}{2}\eta_{\mu\nu}(\partial\phi_{\text{cl}})^2,$$

with the conformal improvement understood when $d > 1$. This is the classical stress tensor of the on-shell configuration ϕ_{cl}. It is smooth, symmetric, and satisfies the conservation law

$$\partial^\mu\delta\langle T_{\mu\nu}\rangle = 0$$

as a direct consequence of $\Box\phi_{\text{cl}} = 0$.

The "linearized regime" of the theorem refers to linearization in the metric perturbation $h_{\mu\nu}$, not in the matter amplitude. The classical source $\delta\langle T_{\mu\nu}\rangle$ is a fixed, smooth tensor field that enters the right-hand side of the linearized Einstein equation. No conflict arises between the quadratic nature of the stress tensor in ϕ_{cl} and the linearized-gravity framework.

5.3 Modular Side: Small-Ball Expansion

For a ball $B(\mathbf{x}_0, R)$ in the rest frame with timelike normal $u^\mu = (1, \mathbf{0})$, the CHM formula gives

$$\delta\langle K_B\rangle = 2\pi\int_{|\mathbf{x}-\mathbf{x}_0|<R} d^dx\,\frac{R^2 - |\mathbf{x}-\mathbf{x}_0|^2}{2R}\,\delta\langle T_{00}(\mathbf{x})\rangle.$$

Expanding $\delta\langle T_{00}(\mathbf{x})\rangle$ around $\mathbf{x}_0$:

$$\delta\langle T_{00}(\mathbf{x})\rangle = \delta\langle T_{00}(\mathbf{x}_0)\rangle + \partial_i\delta\langle T_{00}\rangle(\mathbf{x}_0)(x^i - x_0^i) + O(|\mathbf{x}-\mathbf{x}_0|^2).$$

The linear term integrates to zero by the spherical symmetry of the ball. The leading contribution is therefore

$$\delta\langle K_B\rangle = 2\pi\,\delta\langle T_{00}(\mathbf{x}_0)\rangle \int_{|\mathbf{y}|<R} d^d y\,\frac{R^2-|\mathbf{y}|^2}{2R} + O(R^{d+3}).$$

The radial integral evaluates to

$$\int_{|\mathbf{y}|<R} d^d y\,\frac{R^2-|\mathbf{y}|^2}{2R} = \frac{\Omega_{d-1}}{2R}\int_0^R r^{d-1}(R^2-r^2)\,dr = \frac{\Omega_{d-1}}{2R}\left(\frac{R^{d+2}}{d}-\frac{R^{d+2}}{d+2}\right) = \frac{\Omega_{d-1}\,R^{d+1}}{d(d+2)},$$

where $\Omega_{d-1} = \frac{2\pi^{d/2}}{\Gamma(d/2)}$ is the surface area of the unit $(d-1)$-sphere. Since $V_d = \frac{\Omega_{d-1}}{d}$, this simplifies to

$$\int_{|\mathbf{y}|<R} d^d y\,\frac{R^2-|\mathbf{y}|^2}{2R} = \frac{V_d\,R^{d+1}}{d+2}.$$

Therefore

$$\delta\langle K_B\rangle = \frac{2\pi V_d}{d+2}\,R^{d+1}\,\delta\langle T_{00}(\mathbf{x}_0)\rangle + O(R^{d+3}).$$

For a boosted ball with timelike normal u^μ, the same computation in the adapted frame gives

$$\delta\langle K_{B,u}\rangle = \frac{2\pi V_d}{d+2}\,R^{d+1}\,\delta\langle T_{\mu\nu}(\mathbf{x}_0)\rangle\,u^\mu u^\nu + O(R^{d+3}),$$

confirming Proposition 4.1 in the free-field model.

5.4 Full Tensor Recovery

In the free-field classical source, $\delta\langle T_{\mu\nu}\rangle$ is a smooth symmetric tensor field. By Corollary 4.2, the all-directions small-ball data uniquely determine this tensor at each point. In the present model, this can be verified directly: the quadratic form $u^\mu u^\nu\,\delta\langle T_{\mu\nu}(\mathbf{x}_0)\rangle$ for all timelike u determines $\delta\langle T_{\mu\nu}(\mathbf{x}_0)\rangle$ because the timelike cone spans the full symmetric tensor space.

5.5 Conservation Law

The classical stress tensor satisfies the conservation law

$$\partial^\mu \delta\langle T_{\mu\nu}\rangle = 0$$

as established in Section 5.2 from the on-shell condition $\Box\phi_{\text{cl}} = 0$. This is the necessary compatibility condition for the comparison with the linearized Einstein tensor, which satisfies the Bianchi identity $\partial^\mu G^{\text{lin}}_{\mu\nu} = 0$. In the free-field model, both sides of the would-be Einstein equation are independently divergence-free, so the matching is dynamically consistent and not merely algebraic.

5.6 Consistency Check

Suppose a linearized metric perturbation $h_{\mu\nu}$ is sourced by the free-field stress tensor via the semiclassical Einstein equation

$$G^{\text{lin}}_{\mu\nu}[h] = 8\pi G_N\,\delta\langle T_{\mu\nu}\rangle.$$

If one additionally supplies a geometric small-ball functional whose leading coefficient has the (G-der2) and (G-div) form used in Theorem 4.3, then it has leading term

$$\delta\mathcal{G}_{B,u}[h] = \alpha_d\, R^{d+1}\, G^{\mathrm{lin}}_{\mu\nu}(\mathbf{x}_0)\, u^\mu u^\nu + O(R^{d+3}) = \alpha_d \cdot 8\pi G_N\, R^{d+1}\, \delta\langle T_{\mu\nu}(\mathbf{x}_0)\rangle\, u^\mu u^\nu + O(R^{d+3}).$$

Comparing with the modular side:

$$\delta\langle K_{B,u}\rangle = \frac{2\pi V_d}{d+2}\, R^{d+1}\, \delta\langle T_{\mu\nu}(\mathbf{x}_0)\rangle\, u^\mu u^\nu + O(R^{d+3}).$$

The consistency condition (MC) requires these to be equal. Matching coefficients:

$$\alpha_d \cdot 8\pi G_N = \frac{2\pi V_d}{d+2},$$

which fixes the gravitational coupling in terms of the geometric normalization constant α_d and the spatial dimension d. This is not a derivation of G_N from first principles — the constant α_d is part of the geometric functional's normalization — but it shows that the coefficient comparison is compatible with the linearized Einstein equation: if such an equation holds, the modular and geometric small-ball data match at leading order.

5.7 What the Example Does and Does Not Show

The free-field example verifies three things:

1. The small-ball modular expansion (Proposition 4.1) is not a formal artifact. It is realized with explicit, computable coefficients in a standard QFT model.

2. The conservation law on the modular side matches the Bianchi identity on the geometric side, so the comparison is dynamically consistent.

3. The coefficient comparison is compatible with a semiclassical Einstein interpretation: if a metric perturbation satisfying the linearized Einstein equation exists, then the consistency condition (MC) is satisfied at leading order.

The example does not construct the geometric functional and therefore does not establish the existence premise of the conditional theorem. It assumes the semiclassical Einstein equation and checks compatibility of that assumption with the CHM modular coefficient. It does not show that (MC) is the unique route to the Einstein equation, nor does it address the converse direction.

6 Converse, Limits, and Relation to Faulkner–Jacobson

6.1 Why the Theorem Is Not an Equivalence

Theorem 4.3 is a conditional coefficient-matching theorem. It begins with the small-ball modular family and additionally assumes a single geometric explanation with the two-derivative and transverse premises (G-der2) and (G-div). Under those supplied premises it concludes that the leading coefficients obey the linearized Einstein equation. A reverse implication would require substantially more structure. One would need a faithful reconstruction map

$$h_{\mu\nu} \longmapsto \{\delta\mathcal{G}_{B,u}[h]\}_{(B,u)}$$

defined for all sufficiently small balls, with the correct normalization and covariance properties, and with agreement not only at leading order but at the level of the full ball-dependent functional. The linearized Einstein equation by itself does not provide such a map.

This distinction is not pedantic. The coefficient comparison in Section 4 identifies the leading local tensorial term,

$$G^{\mathrm{lin}}_{\mu\nu}(x_0) \propto \delta\langle T_{\mu\nu}(x_0)\rangle,$$

because equality for all timelike u^μ forces equality of the corresponding quadratic forms. But that argument is deliberately local and asymptotic. It does not control the $O(R^{d+3})$ remainder, it does not determine the full dependence of $\delta\mathcal{G}_{B,u}[h]$ on the ball parameters, and it does not show that every solution of the linearized Einstein equation arises from a geometric functional satisfying (G1)–(G5), (G-der2), and (G-div). The converse direction therefore belongs to a stronger future theorem, not to the present one.

6.2 Precise Relation to the Faulkner–Jacobson Line

The closest existing line of work runs from the CHM ball modular Hamiltonian through the entanglement first law to the Faulkner[3]–Jacobson[4] derivations of linearized gravity (see also [8]). The overlap is genuine and should be stated without exaggeration. In both settings, ball-shaped regions are essential, the CHM kernel supplies the universal local weighting, and the comparison across sufficiently small balls in all directions forces a rank-2 tensor equation whose leading local content is Einsteinian. At the level of small-ball tensor algebra, both routes exploit the same rigid fact: on a Minkowski background, the only local, symmetric, gauge-invariant, two-derivative tensor linear in $h_{\mu\nu}$ is the linearized Einstein tensor up to normalization [5].

The difference lies in what is taken as primitive. Faulkner-type arguments begin from an entropy variation on the geometric side, typically mediated by holographic ingredients such as the RT formula[6] and, in the quantum-corrected setting, JLMS[7]. The present paper begins instead from the family of modular Hamiltonians itself and asks under which supplied geometric conditions one geometry can explain that family. The geometric side is therefore left abstract: we do not assume a bulk dual, a surface-area formula, or an entanglement-entropy interpretation. We do assume a small-ball geometric functional satisfying (G1)–(G5), (G-der2), and (G-div). The result is thus explicit about where the geometric input enters.

It is therefore best to regard the two routes as overlapping but not identical. If one instantiates $\delta\mathcal{G}_{B,u}[h]$ by an entropy or area variation of the appropriate kind, then the present framework reproduces the local coefficient comparison of the Faulkner–Jacobson argument. Without such an instantiation, Theorem 4.3 remains a conditional consistency statement rather than an internal derivation of gravity.

6.3 Limits of the Present Result

The domain of validity should also be kept explicit. First, the modular side is restricted to the CHM ball class in conformally covariant theories, so the theorem does not apply to arbitrary Type III_1 nets. Second, the argument is local and small-ball. It extracts the leading tensorial content near each point, but it does not yet provide a global all-balls integrability theorem. Third, the analysis is linearized around a Minkowski vacuum background. Extension to other maximally symmetric backgrounds (de Sitter, anti-de Sitter) would require replacing $G^{\mathrm{lin}}_{\mu\nu}$ by the appropriate background-relative linearized Einstein operator; this is a natural but nontrivial generalization. Nothing in the present paper proves the full nonlinear Einstein equation.

Fourth, the geometric functional and the premises (G-der2) and (G-div) are axiomatized rather than internally derived from the algebraic data. This is the main remaining mathematical burden. The theorem shows that once these inputs exist and agree with the modular side for all sufficiently small balls, their leading coefficients match in Einstein form. It does not explain why an admissible algebraic model should supply those inputs. Globalization, nonlinear order, and internal construction are therefore genuine next problems rather than hidden assumptions.

7 Broader Physical and Conceptual Implications

Paper I[1] gives a conditional algebraic inseparability result. The present paper asks what an explicitly restricted higher-dimensional coefficient comparison adds to that programme.

In $1+1$ dimensions, the conditional inseparability statement is a rigidity result. In higher dimensions, if a geometric functional also satisfies (G-der2) and (G-div), all-direction consistency gives a linearized Einstein coefficient equation. The dynamics is therefore identified only inside this supplied geometric regime.

This has three consequences for the trilogy as a whole.

First, it isolates the extra premises required to obtain an Einstein coefficient. The equation is not derived from modular consistency alone: the two-derivative and divergence-free geometric structure is supplied. Within that regime, all-direction consistency fixes the coefficient matching.

Second, it gives a conditional higher-dimensional extension of Paper I's programme. The gravitational coefficient equation follows only within the stated CHM, linearized, and supplied-geometric regime; it is not a general dynamical consequence of modular data.

Third, the $O(R^{d+3})$ remainder remains uncontrolled. This paper does not construct coefficient directions, a Fisher map, or a self-description budget from that remainder. Accordingly it supplies no remainder-to-self-description bridge for Paper III.

8 Open Problems

8.1 Globalization

The present theorem is small-ball and local. It extracts the leading tensorial content near each point but does not provide a global all-balls integrability theorem. A global version would require controlling the full ball-dependent functional $\delta\mathcal{G}_{B,u}[h]$, not just its small-ball asymptotic. This is a genuine strengthening that would bring the result closer to the Faulkner[3]–Jacobson[4] line, where the entanglement first law is applied to all balls simultaneously.

8.2 Nonlinear Order

The $O(R^{d+3})$ and higher terms in the small-ball expansion may encode nonlinear corrections to the Einstein equation. In the Jacobson[4] thermodynamic derivation, the passage from linearized to full nonlinear Einstein equations requires an additional equilibrium or maximal-entropy assumption. Whether a similar upgrade is available in the present modular-consistency framework is an open question. This is the natural bridge to a later gravity paper, not something to force into the present one.

8.3 Internal Construction of the Geometric Functional

At present, the geometric small-ball functional $\delta\mathcal{G}_{B,u}[h]$, (G-der2), and (G-div) are axiomatized, not constructed from purely algebraic data. The theorem identifies an Einstein coefficient only after those inputs are supplied. Constructing the functional and deriving its derivative order and transversality from modular data would be required to remove this conditionality.

8.4 No Current Coefficient-Tower Bridge

The higher-order remainder is only an asymptotic error term in the present analysis. No independent coefficient directions or Fisher map are constructed here, so it cannot be passed to Paper III as a physical control tower or quantitative self-description deficit. Establishing such a bridge would require new definitions and estimates beyond the current small-ball result.

8.5 Relation to Entanglement Equilibrium

Jacobson's[4] entanglement equilibrium approach derives the full nonlinear Einstein equation from the assumption that the vacuum maximizes entanglement entropy for small causal diamonds. The present framework does not assume entropy maximization. Whether the modular consistency condition (MC) implies, or is implied by, an entanglement equilibrium condition is an unexplored connection that could unify the two approaches.

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Finite Self-Models and the Open Explanatory Gap: A Conditional Arithmetic Analysis

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Abstract

Paper G offers a programme-level diagnosis of possible self-reference limits, while Paper II retains a leading small-ball stress-tensor extraction with an $O(R^{d+3})$ remainder. Neither result supplies a fixed-point no-go theorem, a physical Fisher spectrum, or a self-description tower. This paper tests a proposed quantitative bridge by defining a local Araki-relative-entropy reconstruction loss for a stipulated finite statistical bottleneck.

The former local minimax drift-floor theorem is withdrawn. A finite Fisher ceiling does not by itself imply a positive reconstruction loss: on a short modular orbit, an admissible two-outcome channel can preserve the orbit coordinate exactly and achieve zero loss. A future positive risk theorem would therefore need a typed observation model that excludes such lossless-orbit constructions and relates estimator risk to the reconstruction loss.

The surviving formal result is a conditional arithmetic inversion. If a stipulated positive sequence satisfies

$$\log I_k = -\alpha k \log k + O(k)$$

and the available budget grows at polynomial scale, then its maximal admissible index obeys

$$K(n) = O\left(\frac{\log n}{\log\log n}\right).$$

This lemma proves neither a physical coefficient/Fisher spectrum nor a self-model resource bound. The explanatory gap therefore remains open: the paper identifies conditions required for a future quantitative rebuild but does not establish a finite self-description horizon.

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1 Introduction

Paper G[3] motivates the possibility that self-reference limits an accessibility-based description. Its regress and collapse scenarios are philosophical diagnostics, not a fixed-point no-go theorem or an exhaustive dichotomy. Paper III uses that discussion only as motivation.

Paper II[2] moved the discussion to a more concrete level. In the higher-dimensional CHM ball framework, Paper II retains the leading stress-tensor extraction at order R^{d+1}, with a remainder $O(R^{d+3})$. It does not construct independent coefficient directions, a Fisher map, or a self-description tower from that remainder.

The present paper defines a local Araki-relative-entropy reconstruction loss for a stipulated finite statistical bottleneck, but withdraws the claimed universal drift floor. Estimation risk in a specified noisy experiment cannot be identified with reconstruction loss for a channel that may preserve the orbit coordinate exactly. The surviving formal result is instead a conditional arithmetic inversion for a stipulated positive sequence and budget. The surrounding programme draws on:

1. the BKM/QFIM structure already used in Paper I[1];
2. the self-referential Fisher and ego-rigidity formalism of T-DOME Paper III[4];
3. the leading CHM expansion and remainder order recorded in Paper II[2].

The arithmetic statement assumes, rather than derives, a positive sequence I_k with

$$\log I_k = -\alpha k \log k + O(k),$$

and a polynomial-scale budget. Under those stipulations its maximal index obeys

$$K(n) = O\left(\frac{\log n}{\log\log n}\right).$$

This proves neither a physical Fisher spectrum nor a self-model budget.

The trilogy-level interpretation also depends on Paper I's supplied shell interface (Amb), its dynamical covariance axiom (C), and the scale-faithfulness axiom (SFa). Modular flow supplies algebraic time only; a phenomenal-time reading is conditional and interpretive. T-DOME Paper III supplies an ego-rigidity prior model, not a theorem that turns it into a reconstruction floor.

The paper is organized as follows. Section 2 defines finite observers, admissible self-models, and the self-description deficit. Section 3 records why the former drift-floor theorem is withdrawn. Section 4 proves the conditional arithmetic inversion. Section 5 retains only the exact CHM moment coefficient and explains why it does not supply a Fisher spectrum. Section 6 explains the trilogy-level significance of the result, and Section 7 states the remaining limitations and open problems.

2 Finite Observers and the Self-Description Deficit

2.1 Finite Observers

Motivated by Paper G's architectural discussion, we first specify what counts as a finite statistical self-model. This definition does not inherit a no-go theorem from Paper G.

Definition 2.1 (Finite observer)**.** A finite observer of complexity n is a triple

$$(\mathcal{A}_{\text{obs}}, \omega_{\text{obs}}, \mathfrak{M}_n),$$

where:

1. $\mathcal{A}_{\text{obs}} \subset \mathcal{A}_c$ is the observer's accessible subalgebra;
2. $\omega_{\text{obs}} = \omega|_{\mathcal{A}_{\text{obs}}}$ is the restricted state;
3. $\mathfrak{M}_n$ is an n-dimensional statistical model used internally to represent the observer's own accessible state.

The complexity parameter n measures effective statistical degrees of freedom, not arbitrary description length. This is deliberate. Paper III is not about Kolmogorov complexity in full generality. It is about the finite number of statistically usable directions available to an observer trying to model itself.

2.2 Modular Fisher Sensitivity

The Tomita–Takesaki modular automorphism group σ_t^ω of a faithful normal state ω on a von Neumann algebra satisfies $\omega \circ \sigma_t^\omega = \omega$ for all t. The orbit is therefore trivial at the level of states, and the associated tangent vector vanishes. Since the modular orbit itself carries no state-level Fisher information, we replace it with the intrinsic accessible-unitary perturbation family as the operational proxy for modular sensitivity.

Definition 2.2 (Modular Fisher sensitivity)**.** Fix a self-adjoint element $h \in \mathcal{A}_{\text{obs}}$ and define the perturbed family

$$\omega_s^{(h)} := \omega_{\text{obs}} \circ \text{Ad}(e^{ish}), \qquad s \in (-\epsilon, \epsilon).$$

The modular Fisher sensitivity of ω_{obs} along h is

$$\mathcal{I}_F(\omega_{\text{obs}}, h) := \mathbf{Q}_{\omega_{\text{obs}}}\left(\left.\frac{d\omega_s^{(h)}}{ds}\right|_{s=0}, \left.\frac{d\omega_s^{(h)}}{ds}\right|_{s=0} \right).$$

The modular Fisher sensitivity of the observer is

$$\mathcal{I}_F(\omega_{\text{obs}}) := \sup\big\{ \mathcal{I}_F(\omega_{\text{obs}}, h) : \ h = h^* \in \mathcal{A}_{\text{obs}}, \ \|h\| \le 1 \big\}.$$

The supremum is taken over unit-norm self-adjoint perturbations in the observer's own accessible algebra. It is finite whenever the BKM form is bounded on the unit ball of $\mathcal{A}_{\text{obs}}$-tangent directions, which holds in the local regime used throughout this paper. The quantity $\mathcal{I}_F(\omega_{\text{obs}})$ measures the observer's maximal intrinsic sensitivity to unitary perturbations of its own accessible state. It is strictly positive whenever ω_{obs} is not a trace state on $\mathcal{A}_{\text{obs}}$.

Throughout the paper, the term *local modular drift family* refers to the one-parameter family $\{\omega_s^{(h)}\}$ for a fixed perturbation h. When the supremum in Definition 2.2 is attained, we take h to be a maximizer; when it is not, we use a standard η-optimization: for each $\eta > 0$, choose $h_\eta = h_\eta^* \in \mathcal{A}_{\text{obs}}$ with $\|h_\eta\| \le 1$ and $\mathcal{I}_F(\omega_{\text{obs}}, h_\eta) \ge \mathcal{I}_F(\omega_{\text{obs}}) - \eta$, carry out the proof for h_η, and let $\eta \downarrow 0$. Despite the name, this family is not the Tomita–Takesaki modular orbit (which is state-preserving) but the accessible-unitary perturbation family that serves as its operational proxy at the level of Fisher information. This is the family along which the observer's self-description is hardest to compress.

2.3 Admissible Self-Models

The theorem chain does not require a literal embedding of the self-model into a matrix algebra. That stronger route would force us to solve a much harder operator-algebraic problem than the present paper needs. What matters here is the existence of a finite statistical bottleneck.

Definition 2.3 (Admissible self-model)**.** An admissible n-dimensional self-model is a pair

$$(\Pi_n, \Psi_n),$$

where:

1. $\Pi_n : \mathcal{S}(\mathcal{A}_{\text{obs}}) \to \mathfrak{M}_n$ is the state-space map induced by a normal unital completely positive map from a finite-dimensional algebra into $\mathcal{A}_{\text{obs}}$, so that Petz monotonicity of relative entropy applies;
2. $\Psi_n : \mathfrak{M}_n \to \mathcal{S}(\mathcal{A}_{\text{obs}})$ is a recovery map;
3. Π_n factors through the observer's own accessible state space, so no external "God's-eye" parameter is introduced;
4. near the reference state ω_{obs}, the Fisher matrix on $\mathfrak{M}_n$ can be whitened:
$$\mathbf{I}_\omega^{(n)} = \text{diag}(\lambda_1, \ldots, \lambda_n), \qquad \lambda_a \le \mathcal{I}_F(\omega_{\text{obs}}).$$

The last condition is the substantive standing assumption of the theorem, not a mere coordinate normalization. It says that the compressed model has only n effective Fisher directions, and none of them is sharper than the observer's own modular sensitivity scale. This is the exact place where the theorem absorbs finite-dimensionality.

2.4 Self-Description Deficit

The correct loss function is not a metric but a divergence. In a Type III setting there is no preferred trace-class density-matrix picture, while Araki relative entropy is native to the operator algebraic formulation [5].

Definition 2.4 (Self-description deficit)**.** Define the BKM ball of radius δ around ω_{obs} by

$$\mathcal{B}_\delta(\omega_{\text{obs}}) := \big\{\omega \in \mathcal{S}(\mathcal{A}_{\text{obs}}) : \ \mathbf{Q}_{\omega_{\text{obs}}}(\omega - \omega_{\text{obs}},\, \omega - \omega_{\text{obs}}) < \delta^2\big\}.$$

Fix once and for all a $\delta > 0$ small enough that Araki relative entropy admits its quadratic expansion on $\mathcal{B}_\delta(\omega_{\text{obs}})$, and large enough that the ego-rigidity prior π_{ego} of T-DOME Paper III[4] can be supported on a segment of the local modular drift family inside $\mathcal{B}_\delta(\omega_{\text{obs}})$. The self-description deficit of complexity n is

$$\varepsilon(n) := \inf_{(\Pi_n, \Psi_n)} \sup_{\omega \in \mathcal{B}_\delta(\omega_{\text{obs}})} S(\omega \parallel \Psi_n \circ \Pi_n(\omega))\,,$$

where the infimum is taken over all admissible n-dimensional self-models.

Near a fixed state, Araki relative entropy has the familiar BKM/QFIM quadratic expansion

$$S(\omega \| \omega + \delta\omega) = \frac{1}{2}\mathbf{Q}_\omega(\delta\omega, \delta\omega) + o(\|\delta\omega\|^2),$$

which is the local information-geometric structure already used in Paper I[1] and in Petz's monotone-metric framework [7]. The minimax form is essential: it rules out trivial constant recovery maps that memorize the reference state while failing arbitrarily nearby. Thus $\varepsilon(n)$ measures optimal local uniform reconstruction loss in the intrinsic Type III geometry of states.

3 The Fisher Ceiling for Admissible Self-Models

3.1 Petz Monotonicity and Finite Fisher Budget

The first ingredient is standard.

Lemma 3.1 (Petz monotonicity)**.** *For any normal coarse-graining* Π_n*,*

$$S(\omega\|\sigma) \geq S(\Pi_n(\omega)\|\Pi_n(\sigma)).$$

Proof. This is the monotonicity of relative entropy under coarse-graining [6]. The present paper uses it only in the local regime, where it implies that compression cannot increase BKM/Fisher information. □

Whitening then converts finite statistical dimension directly into two distinct ceilings: one for the specific modular drift channel, and one for the total trace budget across all retained directions.

Proposition 3.2 (Directional and total Fisher ceilings)**.** *Let* (Π_n, Ψ_n) *be an admissible* n*-dimensional self-model. Then:*

1. *along the local modular drift family through* ω_{obs}*, the Fisher information retained after compression satisfies*
$$\mathcal{I}^{(n)}_{\text{drift}} \leq \mathcal{I}_F(\omega_{\text{obs}});$$
2. *the total retained data Fisher information satisfies*
$$\mathcal{I}^{(n)}_{\text{data}} := \operatorname{tr} \mathbf{I}^{(n)}_\omega \leq n\, \mathcal{I}_F(\omega_{\text{obs}}).$$

Proof. (1) Apply Lemma 3.1 to the one-parameter modular drift family. In the local quadratic regime, monotonicity of relative entropy implies monotonicity of the associated BKM/Fisher metric, so compression cannot increase Fisher information for that specific scalar parameter.

(2) In Fisher-whitened coordinates, the compressed Fisher matrix is

$$\mathbf{I}^{(n)}_\omega = \operatorname{diag}(\lambda_1, \ldots, \lambda_n), \qquad \lambda_a \leq \mathcal{I}_F(\omega_{\text{obs}}).$$

Summing over the n effective directions gives

$$\operatorname{tr} \mathbf{I}^{(n)}_\omega = \sum_{a=1}^{n} \lambda_a \leq n\, \mathcal{I}_F(\omega_{\text{obs}}).$$

□

3.2 Withdrawal of the Drift-Floor Claim

The former local minimax drift-floor theorem and its corollary are withdrawn. The argument applied a van Trees estimation-risk bound to a compression/recovery loss without specifying a noisy observation model that prevents the orbit coordinate from being retained.

The obstruction can be implemented by an internal finite-dimensional channel. Let $s \mapsto \omega_s$ be a nonconstant smooth local drift orbit. Choose an effect $0 \leq M \leq \mathbf{1}$ in $\mathcal{A}_{\text{obs}}$ with $\frac{d}{ds}\omega_s(M)|_{s=0} \neq 0$. The normal unital completely positive map

$$\Phi : \mathbb{C}^2 \longrightarrow \mathcal{A}_{\text{obs}}, \qquad \Phi(a, b) = aM + b(\mathbf{1} - M),$$

induces the binary statistical coordinate

$$\Pi_1(\omega_s) = \big(p(s), 1 - p(s)\big), \qquad p(s) := \omega_s(M).$$

By the inverse-function theorem, after restricting to a sufficiently short orbit segment, p is injective. Define the recovery map on its image, and extend it arbitrarily off that image, by

$$\Psi_1\big(p(s), 1 - p(s)\big) = \omega_s.$$

This code uses an observable in the accessible algebra rather than an external parameter, and it factors through a finite-dimensional commutative algebra as required by Definition 2.3. Then

$$S(\omega_s \| \Psi_1 \circ \Pi_1(\omega_s)) = 0$$

for every s on the encoded segment. A positive Bayes estimation risk can arise only after a likelihood or sampling channel restricts access to s; it is not a lower bound on this lossless orbit reconstruction. The previous proof therefore confused estimation risk with reconstruction loss. Petz monotonicity and the stipulated Fisher ceilings above remain bookkeeping facts within their stated model, but they imply no nonzero reconstruction floor.

4 A Conditional Arithmetic Inversion

Paper II[2] supplies a leading CHM coefficient and an $O(R^{d+3})$ remainder, not an infinite family of normalized Fisher directions. We therefore separate the arithmetic statement from any physical interpretation.

Let $\{I_k\}_{k\geq 1}$ be a *stipulated* positive sequence and let $B(n) \geq 1$ be a *stipulated* budget. Define

$$K(n) := \max\left\{ K \geq 1 : \sum_{j=1}^{K} I_j^{-1} \leq B(n) \right\},$$

with $K(n) = 0$ if the set is empty. No claim is made here that the I_k are Fisher informations of physical coefficient directions or that $B(n)$ is a self-model resource.

Lemma 4.1 (Conditional arithmetic inversion)**.** *Suppose*

$$\log I_k = -\alpha k \log k + O(k), \qquad \alpha > 0,$$

and $\log B(n) = O(\log n)$. *Then*

$$K(n) = O\left(\frac{\log n}{\log\log n}\right).$$

Proof. If $K = K(n) \geq 1$, the final nonnegative summand satisfies $I_K^{-1} \leq B(n)$. Hence

$$\alpha K \log K + O(K) = -\log I_K \leq \log B(n) = O(\log n).$$

For sufficiently large K, the $K \log K$ term dominates the linear remainder, so inversion gives $K = O(\log n / \log\log n)$. Bounded K obeys the same asymptotic bound. □

This lemma is exactly the surviving claim. It establishes neither a self-model budget nor a physical Fisher spectrum. In particular, rescaling coefficient coordinates changes any candidate I_k, so a physical application requires a separately justified normalization, likelihood, and Fisher map.

5 Exact CHM Moment Without a Fisher-Spectrum Claim

The following exact integral is retained as an algebraic check. It does not construct the coefficient directions or Fisher map absent from Paper II.

5.1 Exact d=1 Signal Factor

In $d = 1$, the CHM modular Hamiltonian is

$$K_B = \frac{\pi}{R} \int_{-R}^{R} (R^2 - x^2)\, T_{00}(x)\, dx.$$

Expanding $T_{00}(x)$ around the ball center gives

$$T_{00}(x) = \sum_{k \geq 0} \frac{x^{2k}}{(2k)!} \partial_x^{2k} T_{00}(0) + \cdots .$$

Therefore the coefficient multiplying the $2k$-th derivative is

$$c_k^{(d=1)} = \frac{\pi}{R} \cdot \frac{1}{(2k)!} \int_{-R}^{R} (R^2 - x^2) x^{2k}\, dx \cdot R^{-(2k+2)}.$$

The radial moment is exact:

$$\int_{-R}^{R} (R^2 - x^2) x^{2k}\, dx = \frac{4R^{2k+3}}{(2k+1)(2k+3)}.$$

Hence

$$c_k^{(d=1)} = \frac{4\pi}{(2k+1)(2k+3)(2k)!},$$

which shows exact factorial suppression:

$$c_k^{(d=1)} \sim \frac{\text{const}}{(2k)!\, k^2}.$$

The factorial suppression of $c_k^{(d=1)}$ is normalization-dependent signal information. It does not by itself determine a fluctuation kernel, a renormalized Fisher information I_k, or a cumulative inverse-information budget. Consequently this calculation does not instantiate Lemma 4.1; it only shows what one factor in a future, fully specified model would look like.

6 Trilogy Implications

6.1 Paper G as Motivation Only

Paper G[3] supplies philosophical motivation for studying self-description, but no formal bridge maps its syntax-level scenarios to the reconstruction loss defined here. The withdrawn drift-floor argument does not quantify Paper G, and this paper asserts no generic fixed-point obstruction.

6.2 Relocating the Gap in Paper II Terms

Paper II[2] supplies a leading CHM stress-tensor term and an $O(R^{d+3})$ remainder. It does not supply normalized coefficient directions or a Fisher map. Paper III therefore does not reinterpret the remainder as a physical control tower. Lemma 4.1 applies only after a positive sequence and budget are stipulated independently.

6.3 Trilogy-Level Interpretation

The trilogy-level reading is conditional on Paper I's shell interface (Amb), dynamical axiom (C), and scale-faithfulness assumption (SFa), as well as Paper II's restricted CHM regime. These inputs supply algebraic modular time and conditional geometric statements, not a phenomenal-time identification. The surviving arithmetic lemma does not locate or quantify an explanatory gap.

7 Limitations and Open Problems

The withdrawals identify four concrete tasks for any future rebuild.

1. **A typed observation model.** A future risk theorem must specify a likelihood or sampling channel that prevents lossless orbit coding, then distinguish estimator risk from state reconstruction loss.

2. **Normalized coefficient directions.** A physical use of Lemma 4.1 requires independently constructed coefficient perturbations, a renormalized kernel, and a normalization under which the Fisher sequence is invariantly defined.

3. **A Paper II bridge.** The remainder order alone is insufficient. One must construct a map from small-ball coefficient data to the statistical model before any resource comparison is meaningful.

4. **Ego-rigidity interface.** The T-DOME prior may enter a specified Bayesian estimation problem, but it supplies no reconstruction floor by itself. Its operational channel and boundary conditions must be given before van Trees can be used.

8 Conclusion

Paper III tests a proposed quantitative route to the explanatory gap. Paper G supplies programme-level motivation, while Paper II supplies a leading stress-tensor extraction and a controlled remainder order; it does not identify that remainder as a physical self-description carrier. The present paper defines a finite-self-model reconstruction loss and records why the former universal drift-floor inference fails.

A finite Fisher ceiling alone does not imply a nonzero local minimax reconstruction loss. The lossless-orbit construction shows that a finite channel can preserve the relevant modular-orbit coordinate exactly on a short segment. A positive lower bound may still arise under an additional typed observation model that excludes such encodings and connects estimation risk to reconstruction loss, but a precise characterization of those conditions remains open.

Separately, the conditional arithmetic lemma shows that a stipulated positive sequence with $\log I_k = -\alpha k \log k + O(k)$ has maximal admissible index $K(n) = O(\log n / \log\log n)$ under a

polynomial-scale budget. It does not establish a physical Fisher spectrum, a self-model budget, or a finite self-description horizon. The explanatory gap is therefore not eliminated in general, but it is not quantitatively located by the present results.

A Trilogy Key Premises Summary

The following list records the external inputs and their restricted roles. Paper III is not self-contained with respect to these trilogy interfaces.

1. **Paper I algebraic interface** (Paper I[1]). The Araki relative entropy of two nearby faithful normal states on a Type III von Neumann algebra admits a local quadratic expansion $S(\omega\|\omega+\delta\omega) = \frac{1}{2}\mathbf{Q}_\omega(\delta\omega,\delta\omega) + o(\|\delta\omega\|^2)$, where $\mathbf{Q}_\omega$ is the BKM/QFIM quadratic form. This is the information-geometric structure used in Definitions 2.2 and 2.4 and throughout Section 3. The broader trilogy interpretation also assumes Paper I's shell interface (Amb), dynamical covariance (C), and scale-faithfulness (SFa). These give conditional algebraic structure; modular flow is algebraic time, not a derived phenomenal-time variable.

2. **Small-ball expansion with $O(R^{d+3})$ remainder** (Paper II[2]). In the CHM ball framework for conformal nets, the modular energy of a small ball of radius R admits the expansion $\delta\langle K_{B,u}\rangle = \frac{2\pi V_d}{d+2} R^{d+1}\delta\langle T_{\mu\nu}\rangle u^\mu u^\nu + O(R^{d+3})$. The leading coefficient extracts the local stress tensor, while the higher-order remainder is uncontrolled. Paper II supplies neither an infinite coefficient tower nor a Fisher map; Section 5 uses only a separate exact $d = 1$ moment calculation.

3. **Self-reference motivation** (Paper G[3]). Paper G discusses regress and collapse scenarios but does not prove that a self-referential fixed point generically fails to exist. Paper III uses it as motivation only and does not quantify it.

4. **Ego-rigidity prior** (T-DOME Paper III[4]). The ego-rigidity prior $\mathcal{I}_{\text{ego}}$ is the Fisher information of the observer's prior distribution over its own self-model parameters: $\mathcal{I}_{\text{ego}} := \int(\partial_\sigma \log \pi_{\text{ego}}(\sigma))^2 \pi_{\text{ego}}(\sigma)\, d\sigma$. It may enter a separately specified Bayesian likelihood model. It does not lower-bound reconstruction loss without such an observation model, and no theorem in this paper uses it to do so.

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